Reconstruction-independent moment tests for sample-based spectral functions

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Abstract

Sample-based quantum diagonalization reconstructs spectral functions $A(\omega)$ in sampled determinant subspaces, and checking them against moments read off $A(\omega)$ is circular. We compare a reconstruction's moments with full-operator moments $\langle J^\dagger (H-e_0)^k J\rangle$ on the same subspace ground state. Their difference is exactly the probe's leak out of the subspace at order k: the missed weight at $k=0$, and zero through order $2K+1$ when the subspace contains the order-K Krylov space of the probe state. This residual is deterministic; it flags every support not closed under the operators, and its size does not track the error. Off an eigenstate, a first-moment check in the textbook commutator form passes a truncation whose first moment is 11% low. In a pre-registered blinded simulation on a doped Hubbard ring, with an oracle estimator (exact moments plus shot noise), the m_0–m_2 battery rejected all 56 truncations, 45 of 61 spurious features (20 at fixed total weight) and none of 71 under-converged Krylov reconstructions at a false-positive rate of 1/112; with the 1.96-fold noise the pre-registration specified, the false-positive rate is 40/112 and 23 Krylov reconstructions are rejected, 17 of them exact through m_2. A reweighting that moves 72% of a five-pole spectrum's weight passes any m_0–m_2 check, and full-support Rayleigh–Ritz, whose $k=2$ leak is 1.9% of m_2, passes the default thresholds with a relative L_1 error of 0.93. On IBM Heron samples, with all moments evaluated classically, captured weight rewards support spreading rather than fidelity: a uniform in-sector sampler leaves less weight uncovered than the device at every shot budget.

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1 Introduction

Dynamical response functions such as the spectral function $A(k,\omega)$, the structure factors $S(q,\omega)$ and $S^{zz}(q,\omega)$ and the optical conductivity are measured by photoemission [1], inelastic neutron scattering [2], resonant inelastic x-ray scattering [3] and infrared spectroscopy [4] and computed classically by many-body methods [5, 6]. Quantum algorithms for chemistry and materials began with ground-state energies [7–10], and dynamical responses are now also computed on quantum processors. Trapped ions have produced structure factors and spectral functions [11, 12] and Green's functions built from measured Hamiltonian moments [13]; superconducting processors have produced an impurity Green's function for density-functional plus dynamical mean-field theory (DFT+DMFT) with up to 14 qubits [14], Green's functions

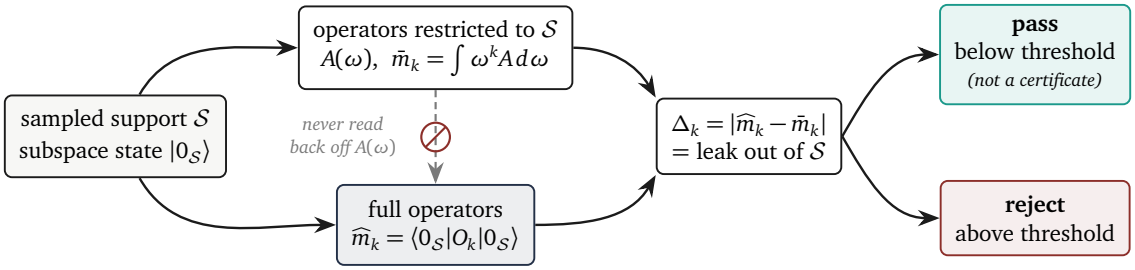


Figure 1: **One subspace state, two operator evaluations.** The sampled support $\mathcal{S}$ fixes the subspace ground state $|0_\mathcal{S}\rangle$. The reconstruction branch diagonalizes H restricted to $\mathcal{S}$, assembles $A(\omega)$ and reads off $\bar{m}_k = \int \omega^k A(\omega)\, d\omega$. The estimator branch evaluates $\widehat{m}_k = \langle 0_\mathcal{S}|O_k|0_\mathcal{S}\rangle$ with the full sparse operators, $O_k \equiv J^\dagger (H - e_0)^k J$ and $e_0 = \langle 0_\mathcal{S}|H|0_\mathcal{S}\rangle$, which reach configurations outside $\mathcal{S}$; it never uses $A(\omega)$ (crossed arrow). Their difference $\Delta_k = |\widehat{m}_k - \bar{m}_k|$ is the leak of $J|0_\mathcal{S}\rangle$ out of $\mathcal{S}$ (Proposition 1), not the distance to the true moment. The leak score $\Delta_k/\widehat{m}_k$ is read against a numerical tolerance; the default thresholds τ_k of the oracle calibration apply to it only as a convention (Sec. 2.2). A pass is not a certificate. Both branches are classical post-processing of the support; on hardware the device supplies $\mathcal{S}$ alone (Sec. 6).

of a four-site Hubbard model by Liouvillian recursion [15], the spin excitation spectrum of the one-dimensional Hubbard model on up to 30 qubits [16], and structure factors with up to 50 qubits, compared quantitatively with neutron data [17]. Sample-based quantum diagonalization (SQD), quantum-selected configuration interaction (QSCI) and quantum Krylov methods for ground states [18–22] have been extended to excited states [23], quasiparticle band structures [24] and spectra [25, 26]: the device samples a subspace, and classical resources diagonalize the Hamiltonian in it and assemble $A(\omega)$, a quantum-centric supercomputing workflow [19, 27].

A reconstructed spectrum can be wrong in a way that a total-weight check misses: spectral weight sits at the wrong frequencies while the total weight and other integrated quantities stay within tolerance. This is the error we target. Critical studies document the ground-state limitations of QSCI and subspace diagonalization [28–30], and error analyses bound how noise and ill-conditioning enter subspace and Krylov ground-state energies [31, 32]. For sample-based spectral functions the companion study shows the misassignment directly: on a subspace that holds the whole probe state, so that no weight is missed, the relative L_1 error is still 0.43 ($L = 6$ ring, $\eta = 0.18\, t$) [25]; the same concern motivates response-targeted determinant selection on the classical side [33]. Such errors matter most where no trusted classical cross-check exists, a regime whose edge moves as classical methods improve and demonstrations are dequantized [34–39] (for sample-based diagonalization see Ref. [40]). A check intended for that regime must first be calibrated where the ground truth is known [41–45], as we do here.

The test compares two sets of moments (Fig. 1). Physics supplies the conditions: every $T = 0$ Lehmann spectrum obeys moment sum rules [46–48] and positivity [49], however it was computed. For a zero-temperature Lehmann function of a probe J, the power moments $m_k = \int \omega^k A(\omega)\, d\omega$ are ground-state expectation values, $m_k = \langle 0|J^\dagger (H - E_0)^k J|0\rangle$; for a Hermitian probe $m_0 = \langle J^2\rangle$ and $m_1 = \frac{1}{2}\langle [J,[H,J]]\rangle$ [46, 49, 50]. We estimate them as $\widehat{m}_k = \langle O_k\rangle$ on the state at hand, never by integrating $A(\omega)$, so $\widehat{m}_k$ can disagree with the reconstruction's moments $\bar{m}_k$; a moment read back off $A(\omega)$ gives $\Delta_k \equiv 0$. These are necessary conditions: failing one rejects the reconstruction, passing does not show that it is correct.

Results. We report five results.

(1) The leak hierarchy (Proposition 1, Sec. 2.3). On the subspace ground state, $\widehat{m}_k - \bar{m}_k$ is exactly the leak of the probe state $J|0_S\rangle$ out of the subspace at order k. It is the missed spectral weight at $k = 0$ and the squared boundary coupling of H at $k = 2$ when no weight is missed, and it vanishes through order $2K + 1$ when the subspace contains the order-K Krylov space of the probe state. The test needs only sparse matrix–vector products from $|0_S\rangle$, not the exact ground state.

(2) The operator form matters off an eigenstate (Sec. 2.3). On a truncated subspace of a doped $L = 6$ ring, $\langle O_1 \rangle = 4.558$ but the textbook double-commutator form gives 3.717; at the threshold $\tau_1 = 0.229$ a first-moment check in the first form rejects a reconstruction whose first moment is 11% low and one in the second form passes it. The full battery rejects this cut on m_0 in either form ($\Delta_0 = 12\,\tau_0$), so the form decides the verdict of checks that test m_1 without m_0, such as checks on spectra normalized to the m_0 sum rule [51] or on $A(k, \omega)$, whose m_0 is pinned to 1. The same warning applies to any sum-rule check evaluated on a variational, subspace or truncated state.

(3) Misplaced weight at fixed total (Secs. 3 and 4). Under determinant truncation the zeroth-order leak, the companion's captured-weight deficit, is the operative channel. When weight moves in frequency at fixed total, m_0 is blind and m_1, m_2 reject it: in the doped $L = 12$ current response from a moved fraction $f \approx 0.057$ and 0.046. The optical f-sum moment m_{-1} shifts most in relative terms, but a 94% Kohn cancellation makes it reject last unless the spurious weight lies at $0.5\,t$ or below. In a pre-registered blinded simulation with an oracle estimator, the battery rejected all 56 truncations, 25 of 27 weight-changing and 20 of 34 weight-preserving spurious features, and none of 71 under-converged Krylov reconstructions (the 56 with two or more Lanczos nodes reproduce m_0–m_2 exactly, the 15 one-node ones miss m_2 by 2.4–4.2%, within τ_2); its false-positive rate is $1/112$ with the simulated noise the threshold assumes and $40/112$ with the 1.96-fold larger noise the pre-registration specified, under which noise also rejects 23 of the Krylov reconstructions, 17 of them exact through m_2.

(4) Limits of low-order sum-rule checks (Sec. 5). With m_0–m_2 the guaranteed power against fixed-pole weight redistributions is zero for every spectrum with four or more distinct poles: on the five-pole $L = 6$ current spectrum a redistribution that moves 72% of the weight passes. The classical Chebyshev–Markov–Stieltjes bracket on a passing reconstruction is vacuous at the spectral mean at this order, and finite tolerances widen it 3.2–7.7-fold just above the band and about tenfold far above it. These two limits hold for every sum-rule validation at this order, not only for this test. The shot cost of exposing a redistribution inside a pole cluster of diameter D grows at least as $D^{-2(s-1)}$, with a bias floor. Rayleigh–Ritz on the full ground-state support of the calibration ring passes the default thresholds τ_k with a relative L_1 error of 0.93: its only residual, the $k = 2$ leak, is 1.9% of m_2, so any tolerance on m_2 above that passes it (the default τ_2 is 5.5%), while its leak score flags it as not closed.

(5) Coverage is not fidelity, on device data (Sec. 6). On supports sampled by IBM Heron (`ibm_fez`) for an $L = 8$ addition sector, the missed weight Δ_0 grows from 0.0044 to 0.37 as coverage falls from 94% to 35%, but it rewards support spreading: at 5×10^4 shots per circuit the device beats all 2000 noiseless replicas of its own circuits, a uniform in-sector sampler beats the device at every budget, and $\Delta_0 = 0.0044$ coexists with a relative L_1 error of 0.35.

Relation to the companion study. The companion study reconstructs these spectra from sampled subspaces and bounds their error at exact-diagonalization scale, given the exact ground state [25]; we adopt its terms (its μ_j is our m_j). It contains the captured weight w, which gives our coverage residual $\Delta_0 = m_0(1 - w)$, the $k = 0$ case of the leak with the probe state built from the exact ground state; the statement and measured reach of the classical moment exactness of Krylov projections [52]; the comparison of reconstruction moments with exact ones on random coordinate subspaces, which miss already m_0; the second-moment deficit $\mu_2 - \mu_2^{\mathcal{S}} = \|Q_{\mathcal{S}} H P_{\mathcal{S}} \varphi\|^2$, the $k = 2$ leak on a subspace of unit captured weight; the obser-

Table 1: **What each approach compares a spectrum against.**

approach	compared against	role
sum-rule checks in DMFT, QMC and continuation [53–56]	classical sum-rule moments or exact results	consistency check
validation framework of Buchs *et al.* [57]	experiment, classical results, sum-rule targets	validation metrics
moment-problem and SDP bounds [71–75, 77, 78]	moments or correlators of the true system	two-sided bounds
state and observable verification [88, 90]	energy variance; validated noise model	certifies a state or validates a number
companion: captured weight [25]	exact ground-state probe $J\|0\rangle$	a posteriori error bound
this work	**full-operator $\langle O_k \rangle$ on the same subspace state (residual = leak), or oracle moments in the calibration**	**necessary-condition test**

vation that exact moments through order $2K+1$ coexist with an $O(1)$ reconstruction error; and the result that captured weight alone cannot certify a reconstruction, of which our $L = 8$ device run is an instance. We add tests computable from the subspace state alone: the full leak hierarchy as a thresholded rejection test, defined for any diagonal $T = 0$ Lehmann $A(\omega)$ (single-particle, current or collective; not only Rayleigh–Ritz reconstructions); its blinded calibration; the exact limits of a pass; and the device evaluation of the coverage channel against reference samplers.

Prior art. Sum-rule checks test spectra against Hamiltonian-derived moments: they are long-standing in dynamical mean-field theory and quantum Monte Carlo, most systematically in the sum-rule program of Turkowski and Freericks [53, 54], and analytic-continuation pipelines use sum rules as prior information [55] or compare with exact results [56]. For quantum-computed spectra, Buchs *et al.* validate against experimental and classical anchors with metrics that include low-order moments and sum-rule residuals, and leave the regime without full classical benchmarks to future work [57]. Moments also enter as inputs: in a quantum linear-response algorithm the zeroth moment restores the normalization of the computed response [51], and frequency moments of spectral densities are obtained exactly from imaginary-time quantum Monte Carlo data [58]. The classical moment problem, with its orthogonal polynomials and Gauss quadrature [52, 59–66], underlies the kernel-polynomial and recursion methods [67–70], gave error bounds on spectroscopic line shapes from their equilibrium moments [71] and Stieltjes–Tchebycheff photoabsorption profiles from dipole spectral moments [72], serves as an a posteriori bound on Lanczos quadrature [73], and now bounds smeared spectral functions from Euclidean lattice correlators [74–76]; semidefinite programs bound ground-state observables [77, 78]. Variational algorithms [8, 79, 80], error mitigation [81–85], randomized measurements [86, 87] and verification methods, from self-verifying variational simulation and state certification to observable estimation without classical verification [88–91], act on states or expectation values, not on a reconstructed function. We claim no novelty for testing a spectrum against sum rules; Table 1 states what each approach compares against.

Scope. The test can reject a reconstruction but cannot certify one. Every moment in this paper is evaluated classically; the device supplies only sampled supports. Its thresholds are calibrated in simulation, for the current probe of one doped $L = 6$ ring, on synthetic errors of classically known truth; all systems are at exact-diagonalization scale ($L \leq 12$ Hubbard

Table 2: **What is demonstrated on hardware, in simulation, analytically, and deferred.**

quantity	source	status
support $\mathcal{S}$ and coverage residual Δ_0, $L = 8$	`ibm_fez` ($\mathcal{S}$ only)	Δ_0 classical on $\mathcal{S}$; rewards spreading (Sec. 6)
off-diagonal moments m_1, m_2 on hardware	simulation	deferred; forecast bias 38–116% unmitigated (App. J)
momentum distribution n_k on hardware	sealed design	not run; no chain met its floors on 2026-09-03 to 09-05 (App. J)
blinded detection and false-positive rates	simulation, oracle	per class in Table 3 (Sec. 4)
shared-state leak score	classical	deterministic; tolerance 10^{-6} (Sec. 2.2)
Hankel positivity of estimated moments	—	not exercised (App. E)
fixed-pole redistribution	exact / analytic	zero guaranteed power with m_0–m_2 (Sec. 5)
same-sample $\widehat{m}_k$ and $\bar{m}_k$	simulation	correlation 0.43 at 84% coverage, →0 at full; no power gain (App. B)
field catch of an unknown-truth error	—	future work

chains and rings, a small lattice-Schwinger model, and a density-matrix renormalization group (DMRG) check of code scalability at $L = 24$). Table 2 separates what is measured on hardware from what is simulated or deferred, and Sec. 7.2 states each limit with its evidence.

Section 2 defines the test and its estimators and proves the leak identity; Sec. 3 shows what the moments detect in exact simulation; Sec. 4 reports the blinded calibration; Sec. 5 derives what a pass implies and what it cannot see; Sec. 6 evaluates the coverage channel on device-sampled supports; Sec. 7 discusses prior work and limitations, and Sec. 8 concludes.

2 Moment tests, their estimators and the leak identity

The test applies to the output of a sample-based-diagonalization pipeline; its residual on the subspace state is an exact operator leak (Proposition 1).

Notation. The hopping t is the energy unit throughout; a frequency rescaled to $[-1, 1]$ is written x (Appendix F). J always denotes the probe operator, and the spin-exchange constant $4t^2/U$ is written J_{ex}. K is the battery order, the highest moment order the test uses; in Theorem 1 it is the highest order a redistribution preserves, and in the moment-exactness statement of Proposition 1 the order of a Krylov space.

2.1 The pipeline and the origin of the samples

In sample-based diagonalization [18–20] the processor does not evaluate $A(\omega)$: each computational-basis shot of a prepared state returns a bitstring (a candidate Slater determinant). In the simplest form, used here and in the companion, the distinct bitstrings surviving post-selection to the physical particle-number and spin sector form the subspace $\mathcal{S}$ [18]; SQD instead repairs the bitstrings by configuration recovery and builds $\mathcal{S}$ from products of the sampled spin-up and spin-down strings [19]. The Hamiltonian is projected onto $\mathcal{S}$ and

diagonalized classically, and the spectrum is the $T = 0$ Lehmann sum

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \, \delta_\eta\big(\omega - (E_n - E_0)\big) \tag{1}$$

over the resulting eigenpairs, with δ_η a narrow broadening [49]. In the companion's formulation the prepared state is a Trotterized real-time evolution of the $(N+1)$-particle seed $c_p^\dagger|\psi_0\rangle$ [25]: a Trotterized Krylov evolution as in Krylov quantum diagonalization [21], sampled as in sample-based Krylov diagonalization [20] and related to time-evolved QSCI [92]. All circuits run on hardware here (the $L = 6$ jobs of 2026-08-21 and the $L = 8$ jobs of 2026-08-29) and in the companion are number- and S_z-conserving samplers of the addition sector, not the Hubbard propagator of the seed; for the companion's run this is stated in Ref. [25]. Neither the companion's run [25] nor ours used the configuration recovery of SQD [19]. The intended role of the processor appears only at scale, where the configurations carrying the weight of the probe state cannot in general be enumerated classically: an ideal sampler of the targeted state would concentrate shots on them and fix $\mathcal{S}$ by measurement rather than by a classical selection heuristic [19]. We do not show that any device sampler does this better than a classical heuristic, and the companion claims no quantum advantage [25].

2.2 Moments, three estimators and the rejection test

Evaluated with the exact eigenpairs of H and $\eta \to 0$, Eq. (1) is the zero-temperature diagonal Lehmann spectral function of the probe J, a positive measure supported on $\omega = E_n - E_0 \geq 0$ [49]. On an exact eigenstate $|0\rangle$ its power moments are ground-state expectation values (Appendix A) [46],

$$m_k = \int \omega^k A(\omega)\, d\omega = \langle 0|J^\dagger (H - E_0)^k J|0\rangle = \langle 0|J^\dagger \, \mathrm{ad}_H^k(J)|0\rangle, \tag{2}$$

with $\mathrm{ad}_H(J) \equiv [H, J]$, so that $m_0 = \langle J^\dagger J\rangle$ and, for a Hermitian probe, $m_0 = \langle J^2\rangle$ and $m_1 = \frac{1}{2}\langle [J,[H,J]]\rangle$. For the current probe m_0 is the total current–current weight and m_1/m_0 the mean frequency; the regular part of the optical f-sum is set by the separate moment m_{-1} (on a ring the Drude weight carries the rest; Appendix A) [47, 48]. On a state $|\psi\rangle$ that need not be an eigenstate we define the moment operator and its commutator form,

$$O_k \equiv J^\dagger (H - e_0)^k J, \qquad C_k \equiv J^\dagger \, \mathrm{ad}_H^k(J), \qquad e_0 = \langle \psi|H|\psi\rangle. \tag{3}$$

On an exact eigenstate $e_0 = E_0$ and $\langle O_k\rangle = \langle C_k\rangle = m_k$; off an eigenstate the two in general differ for $k \geq 1$ and neither equals m_k. The estimator is $\widehat{m}_k = \langle \psi|O_k|\psi\rangle$; C_k is used only on exact eigenstates (Sec. 2.3 gives the reason). The reconstruction moment $\bar{m}_k = \int \omega^k A(\omega)\, d\omega$ is the quantity a circular check would use, and the test statistic is $\Delta_k = |\widehat{m}_k - \bar{m}_k|$. We evaluate $\bar{m}_k$ from the poles and residues of the reconstruction, at $\eta \to 0$ [Eq. (B.1)], never by integrating a broadened $A(\omega)$ on a grid: a Lorentzian has no finite first or second moment, and grid integration biases the moments by an amount that depends on the window. A circular check compares $\bar{m}_k$ with itself, returns $\Delta_k \equiv 0$ and can never reject. We use three estimators, defined once here.

(a) Shared-state estimator (realizable now, classical). $\widehat{m}_k = \langle 0_\mathcal{S}|O_k|0_\mathcal{S}\rangle$ on the reconstruction's own subspace ground state, evaluated with the full sparse operators: with $\varphi_\mathcal{S} = J|0_\mathcal{S}\rangle$, $\widehat{m}_{2j} = \|(H - e_0)^j \varphi_\mathcal{S}\|^2$ and $\widehat{m}_{2j+1} = \langle (H-e_0)^j \varphi_\mathcal{S}|(H-e_0)|(H-e_0)^j \varphi_\mathcal{S}\rangle$, so the battery through $k = 2$ needs $\varphi_\mathcal{S}$ on the configurations that J connects to $\mathcal{S}$ and one further application of H. It needs only the sampled support. It is deterministic, and its residual is exactly the leak of Proposition 1, so it measures whether the reconstruction is closed under the full operators, not how far a moment is from the truth. We report its residual as the leak score $\Delta_k/\widehat{m}_k$, judged

against a numerical tolerance of 10^{-6} (on the 112 faithful controls of the blinded test the score is below 3×10^{-15}). Through $k=2$ the three scores vanish exactly when span $\mathcal{S}$ contains $\varphi_{\mathcal{S}}$ and $H\varphi_{\mathcal{S}}$ (Proposition 1), so at this tolerance the test flags almost every sampled support; its informative output is the vector of leak scores, whose size does not track the reconstruction error (Sec. 5.3). Applying τ_k below to it is an explicit convention calibrated on the oracle estimator; a pass in this paper is a pass under τ_k, and we name the estimator each time.

(b) Oracle estimator (the calibration model: exact moments plus shot noise). The exact moment m_k plus unbiased Gaussian noise of variance $\mathrm{Var}(O_k)/N_s$, with $\mathrm{Var}(O_k)$ the local-estimator variance of Appendix B. It stands for an unbiased finite-shot estimate of the true moment, and the blinded calibration of Sec. 4 uses it. For it the reconstruction is rejected when $\Delta_k > \tau_k$, with $\tau_k = z\,\delta_k$, $z=1.96$ ($\alpha=0.05$ per moment), and $\delta_k^2 = \mathrm{Var}(O_k)/N_s + b_k^2$ the shot variance plus a bias budget b_k. The *default rule* uses $N_s = 5\times10^4$ and $b_k = 0.02\,|\widehat{m}_k|$; for the three-moment battery its nominal family-wise level is $1-0.95^3 = 14.3\%$. The bias budget is assumed, not measured: the level, and the attribution of a rejection to the reconstruction rather than to the estimator, presuppose that the estimator's bias lies within b_k.

(c) Bond-basis device estimator (future). $\langle O_k\rangle$ measured in qubit-wise-commuting Pauli groups, with rotated bases for the hopping terms, on a separately prepared ground state. It is verified in simulation only; its cost and forecast noise bias are in Appendix J.

The battery applies the moment tests for $k=0,1,\dots,K$ (default: $K=2$) and rejects the reconstruction when any of them rejects; a reconstruction that no moment test rejects passes. At $k=0$, where $O_0 = C_0 = J^\dagger J$, the residual is the weight the support misses,

$$\Delta_0 = \langle\varphi|(\mathbb{1}-\Pi_{\mathcal{S}})|\varphi\rangle = \|\varphi\|^2(1-w), \qquad w = \|\Pi_{\mathcal{S}}\varphi\|^2/\|\varphi\|^2,$$

with $\Pi_{\mathcal{S}}$ the projector onto the span of $\mathcal{S}$ and w the companion's captured weight [25]: for the shared-state estimator $|\varphi\rangle = J|0_{\mathcal{S}}\rangle$ and $\|\varphi\|^2 = \widehat{m}_0$; for the device coverage residual $|\varphi\rangle = J|0\rangle$ on the exact ground state, with the addition probe $J = c^\dagger_{0\uparrow}$, and $\|\varphi\|^2 = m_0$ (Sec. 6). Separately, the moments of every positive measure satisfy Hankel positivity $H_n = [\widehat{m}_{i+j}] \succeq 0$ and, for a diagonal $T=0$ Lehmann measure supported on $\omega\geq0$, the shifted Stieltjes condition $H_n^{(1)} = [\widehat{m}_{i+j+1}] \succeq 0$ [59–61]. For a truncated sequence these conditions are not sufficient: $(m_0,m_1,m_2) = (1,0,1)$ satisfies both, yet no measure on $\omega\geq0$ has these moments. A violation rejects the estimate—device, mitigation or estimator—independently of $A(\omega)$; it cannot reject a non-negative reconstruction, whose own moments satisfy these conditions by construction (Appendix E). The battery can also include an exactly known operator identity, such as a conserved charge $[H,G]=0$ with a known physical eigenvalue, whose violation rejects the prepared state, as in symmetry verification [93, 94] and Gauss-law oracles [95]; Appendix H illustrates this with a Gauss law evaluated exactly on the state vector [96]. The construction transports across models and observables because it needs only (a) a positive spectral measure and (b) operators whose ground-state expectations can be estimated without the reconstruction [52, 67].

2.3 The leak identity

Let $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$, $|0_{\mathcal{S}}\rangle$ its ground state, $e_0 = \langle 0_{\mathcal{S}}|H|0_{\mathcal{S}}\rangle$, $J_{\mathcal{S}} = \Pi_{\mathcal{S}} J \Pi_{\mathcal{S}}$ and $\varphi_{\mathcal{S}} \equiv J|0_{\mathcal{S}}\rangle$. A reconstruction built on $|0_{\mathcal{S}}\rangle$ has the moments $\bar{m}_k = \langle 0_{\mathcal{S}}|J_{\mathcal{S}}^\dagger (H_{\mathcal{S}}-e_0)^k J_{\mathcal{S}}|0_{\mathcal{S}}\rangle$ [Eq. (B.1)], with the convention $(H_{\mathcal{S}}-e_0)^0 \equiv [\Pi_{\mathcal{S}}(H-e_0)\Pi_{\mathcal{S}}]^0 \equiv \Pi_{\mathcal{S}}$.

Proposition 1 (Leak identity)**.** *With the shared-state estimator $\widehat{m}_k = \langle 0_{\mathcal{S}}|O_k|0_{\mathcal{S}}\rangle$ and the common reference e_0,*

$$\widehat{m}_k - \bar{m}_k = \langle\varphi_{\mathcal{S}}|(H-e_0)^k|\varphi_{\mathcal{S}}\rangle - \langle\varphi_{\mathcal{S}}|[\Pi_{\mathcal{S}}(H-e_0)\Pi_{\mathcal{S}}]^k|\varphi_{\mathcal{S}}\rangle \tag{4}$$

for every $k \geq 0$. In particular: (i) at $k = 0$ it is the missed weight $\widehat{m}_0(1-w) \geq 0$; (ii) if span$\mathcal{S}$ *contains the Krylov space* span$\{\varphi_\mathcal{S}, H\varphi_\mathcal{S}, \ldots, H^K\varphi_\mathcal{S}\}$*, it vanishes for every $k \leq 2K+1$; (iii) if $w = 1$, it vanishes at $k = 1$, and at $k = 2$ it equals the squared boundary coupling* $\|(\mathbb{1} - \Pi_\mathcal{S})H\varphi_\mathcal{S}\|^2 \geq 0$.

Proof. Appendix B. Statement (ii) is the classical moment exactness of Krylov projections [52, 66], which the companion states for coordinate subspaces and whose failure at order $2K+2$ it measures [25]; the $k = 0$ and $k = 2$ forms are the companion's captured-weight and second-moment results, stated there for the probe on the exact ground state. Their proofs use only the projector $\Pi_\mathcal{S}$ and a common energy shift, so they hold unchanged for $\varphi_\mathcal{S}$ and e_0. What the proposition adds is the whole hierarchy on the subspace state, which a pipeline holds: every order needs only sparse matrix–vector products from $|0_\mathcal{S}\rangle$.

Number-changing probes. For a probe that changes the particle number, such as $J = c_p^\dagger$ of the single-particle Green's function, the ground state comes from an N-particle support $\mathcal{S}_0$ and the reconstruction lives on an $(N\pm1)$-particle support $\mathcal{S}$, built from the eigenpairs of $\Pi_\mathcal{S} H \Pi_\mathcal{S}$ with weights $|\langle n_\mathcal{S}|J|0_{\mathcal{S}_0}\rangle|^2$. Equation (4) and statements (i)–(iii) then hold verbatim with $\varphi_\mathcal{S} = J|0_{\mathcal{S}_0}\rangle$, $e_0 = \langle 0_{\mathcal{S}_0}|H|0_{\mathcal{S}_0}\rangle$ and $\Pi_\mathcal{S}$ the projector onto $\mathcal{S}$, because the proof uses only $\varphi_\mathcal{S}$, $\Pi_\mathcal{S}$ and e_0; for the removal branch take $J = c_p$ and $\omega \to -\omega$.

The proposition fixes the operator form. With O_k the residual is exactly the leak; with C_k it would also contain $\langle 0_\mathcal{S}|(C_k - O_k)|0_\mathcal{S}\rangle$, which at $k = 1$ equals $-\langle 0_\mathcal{S}|J^\dagger J\,(H - e_0)|0_\mathcal{S}\rangle$ and does not vanish off an eigenstate (Appendix A). The difference changes verdicts. Take the current probe of the doped $L = 6$, $U/t = 4$ ring and truncate its sector to the $d = 100$ determinants of largest $|\psi_0|^2$ (44% of the 225-dimensional sector, 54% of the 186-determinant ground-state support). The subspace ground state gives $\langle O_1\rangle = 4.558$ but $\frac{1}{2}\langle[J,[H,J]]\rangle = 3.717$, against the reconstruction moment $\bar{m}_1 = 3.743$ and the exact $m_1 = 4.227$. At the default threshold $\tau_1 = 0.229$ the O_1 form rejects the truncated reconstruction and the commutator form does not; with the truth known here, the rejection is the correct verdict, since $\bar{m}_1$ is 11% below the exact value. At the six other cuts of the same scan the two forms give the same verdict (tie order and the full scan in Appendix A). The battery as a whole rejects the $d = 100$ cut on m_0 in either form, $\Delta_0 = 0.34 = 12\,\tau_0$, because $O_0 = C_0$; the form decides the verdict of a check that tests m_1 without m_0. A first-moment check in the textbook double-commutator form, evaluated on a variational, subspace or truncated state, can thus pass a reconstruction whose first moment is wrong.

3 What the moments detect (exact simulation)

Under determinant truncation the zeroth-order leak responds first; for weight moved in frequency at fixed total, only m_1 and m_2 respond. All results in this section are exact simulations.

3.1 Truncation: the zeroth-order leak is the operative channel

Determinant-subspace truncation is the native failure mode of sample-based diagonalization: the sampled subspace $\mathcal{S}$ omits configurations that carry spectral weight, and the reconstruction is internally consistent with an incomplete basis [18, 19]. We truncate the determinant subspace of the doped $L = 12$ Hubbard ring ($N_\uparrow = N_\downarrow = 4$, $U/t = 8$; full support $n_{\mathrm{sup}} = 2.4 \times 10^5$) to its top d configurations by $|\psi_0|^2$, the noiseless limit of QSCI's selection of the most frequently sampled configurations [18], and evaluate both branches on the same subspace state $|0_\mathcal{S}\rangle$ (Fig. 2). The circular control, $\bar{m}_1$ against itself, is zero by definition; the data show whether, and at what depth, the independent residuals cross a threshold.

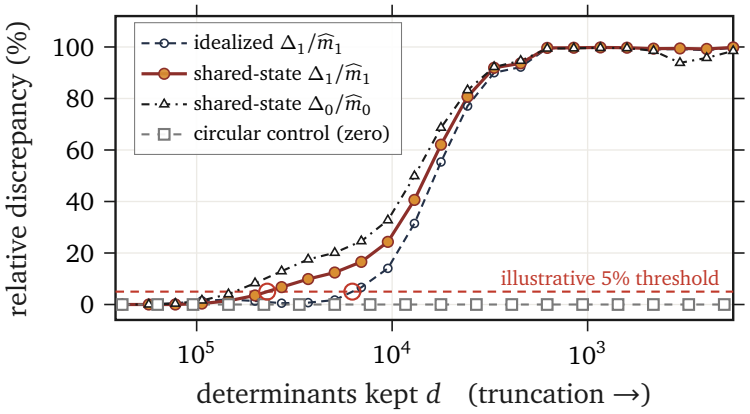


Figure 2: **Independent residuals under determinant truncation.** Doped $L = 12$ ring ($N_\uparrow = N_\downarrow = 4$, 2/3 electrons per site), $U/t = 8$, truncated to its top d configurations by $|\psi_0|^2$ (full support 2.4×10^5). Shared-state curves use the *same* ground state $|0_\mathcal{S}\rangle$ of $H_\mathcal{S} = \Pi_\mathcal{S} H \Pi_\mathcal{S}$, with $e_0 = \langle 0_\mathcal{S}|H|0_\mathcal{S}\rangle$; the evaluation is classical. Dark red, filled dots: $\Delta_1/\widehat{m}_1$, with $\widehat{m}_1 = \langle 0_\mathcal{S}|O_1|0_\mathcal{S}\rangle$ from the *full* operators and $\bar{m}_1$ the within-$\mathcal{S}$ reconstruction moment. Black, dash-dot, open triangles: $\Delta_0/\widehat{m}_0 = 1-w$, the fraction of the weight of $J|0_\mathcal{S}\rangle$ outside $\mathcal{S}$, the companion's captured-weight deficit [25]. Dark blue, dashed, open circles: idealized $\Delta_1/\widehat{m}_1$ with $\widehat{m}_1$ on the exact full-sector ground state; it stays below 2% down to 8% coverage. Grey squares: the circular control, zero by definition. Against the illustrative 5% line, $\Delta_0/\widehat{m}_0$ crosses first (between $d = 6.9\times10^4$ and 5.0×10^4, 29–21% coverage; not marked), then $\Delta_1/\widehat{m}_1$ ($d \approx 4.4\times10^4$, 18%), then the idealized curve (1.6×10^4, 7%); open rings mark these two log-interpolated Δ_1 crossings. At full support, and for $d < 2\times10^3$ where all curves have saturated, $\Delta_0/\widehat{m}_0$ lies below $\Delta_1/\widehat{m}_1$.

The zeroth-order residual $\Delta_0/\widehat{m}_0 = 1-w$, the companion's captured-weight deficit [25] and the $k = 0$ leak of Eq. (4) (here $w = \|\Pi_\mathcal{S} J|0_\mathcal{S}\rangle\|^2/\|J|0_\mathcal{S}\rangle\|^2$), exceeds $\Delta_1/\widehat{m}_1$ at every computed truncation depth from 73% down to 0.9% coverage (for example 1.6% against 0.37% at 39% coverage) and lies below it only at full support, where both are below 10^{-5}, and at the deepest cuts ($d < 2\times10^3$), where both have saturated (93.8% against 99.4% at $d = 337$). It crosses the 5% line between $d = 6.9\times10^4$ (29% coverage, 4.0%) and $d = 5.0\times10^4$ (21%, 8.3%). The shared-state first-moment residual rises by about five orders of magnitude, from 5×10^{-6} at full support to ≈100% at deep truncation (non-monotone near both ends: below 10^{-4} near full support, and at saturation), and crosses the 5% line at $d \approx 4.4\times10^4$ (18% coverage). The idealized curve, with $\widehat{m}_1$ on the exact full-sector ground state, crosses only at $d \approx 1.6\times10^4$ (7%) after a dip at moderate truncation, where the full-state reference and the truncated reconstruction happen to track each other. Between 18% and 7% coverage the shared-state residual therefore exceeds 5% while the reconstruction's first moment is within 5% of the exact one: it registers the leak, not the error (Sec. 5.3). Neither branch stays exact under truncation: $\widehat{m}_1$ rises from 35.1 at full support to 60.1 at the deepest cut, while $\bar{m}_1$ falls toward zero. The residual measures the leak, not the error of either branch against the exact moment.

The blinded simulation of Sec. 4 agrees: in the shared-state rescoring every one of its 56 truncations is rejected on m_0, where the two operator forms coincide, and none is rejected only by m_1 or m_2. For determinant truncation the operative channel is therefore the companion's Δ_0; the moments m_1 and m_2 add reach only for errors that preserve m_0.

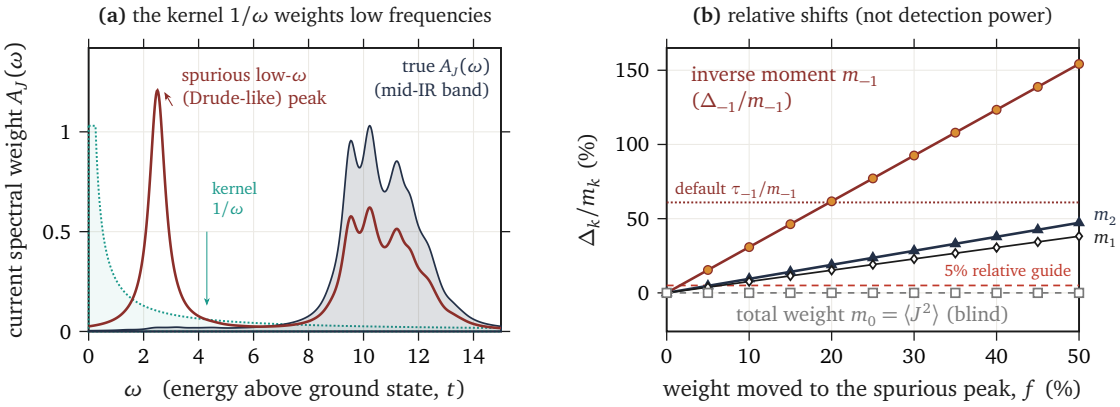


Figure 3: **Current response: misplaced weight at fixed total.** Doped Hubbard ring ($L = 12$, $U/t = 8$, 2/3 electrons per site; $\mathrm{Re}\,\sigma \propto A_J/\omega$). A fraction f of the weight above $6\,t$ moves to a spurious pole at $2.5\,t$ at fixed m_0. **(a)** True spectrum (dark blue; 1.4% of m_0 below $6\,t$) and wrong reconstruction at $f = 0.4$ (dark red: spurious pole plus depleted band); Lorentzian broadening $\eta = 0.35\,t$. Teal: kernel $1/\omega$, rescaled, capped below $0.25\,t$. **(b)** Exact Δ_k/m_k versus f: $\Delta_0/m_0 < 10^{-13}$ (grey, blind); m_1 (black, open diamonds), m_2 (dark blue) and $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D} = 0.324$ (dark red) move by up to 38%, 47% and 154%. Dashed: common 5% guide. Default thresholds: 4.3% of m_1 and m_2 but 61% of m_{-1} (dotted), since the Kohn stiffness $\mathcal{D}$ (from the flux curvature, not the same samples) cancels 94% of $\frac{1}{2}\langle -\hat{T} \rangle$; rejection at $f \approx 0.046$ (m_2), 0.057 (m_1), 0.20 (m_{-1}).

3.2 Weight misplaced at fixed total weight

In the doped one-dimensional Hubbard model the regular ($\omega > 0$) current spectral function $A_J(\omega)$ is dominated by a mid-infrared/Mott band of total weight $m_0 = \langle J^2 \rangle$ [4,48]; on the finite ring the charge stiffness is a separate zero-frequency Drude term [47], absent from $A_J(\omega)$. A wrong reconstruction moves a fraction f of the weight above $6\,t$ into a spurious low-frequency pole at $2.5\,t$, keeping the total fixed, so a total-weight check passes it (Fig. 3; Lanczos ground state, Haydock/Ritz current poles [66, 68], every moment exact, no sampling). The first moment $m_1 = \frac{1}{2}\langle [J,[H,J]] \rangle$ [46, 47] differs from the reconstruction's by up to 38% of m_1 at $f = 0.5$ and, under the default threshold rule, rejects from $f \approx 0.057$. Here the rule takes $z = 1.96$, an assumed 2% bias budget and a shot term at $N_s = 5 \times 10^4$ with the local-estimator variance of each estimated operator on the exact $L = 12$ ground state (Appendix B); its blinded calibration covers only the $L = 6$ catalog of Sec. 4. Integrated weight cannot see this error: m_0 is preserved by construction.

The misplaced weight also moves $m_{-1} = \int_0^\infty A_J(\omega)/\omega\, d\omega = \sum_n |\langle n|J|0\rangle|^2/\omega_n$, whose kernel $1/\omega$ weights low frequencies, unlike the kernel ω^k of the positive moments. With $P = \sum_j j\, n_j$ the polarization of an open chain and $\hat{T}$ the hopping (kinetic) energy, the current of our convention is $J = -i[H,P]$, and this optical f-sum moment is a ground-state kinetic energy: $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle$ on open chains and $\frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D}$ on the ring, with $\mathcal{D}$ the finite-size Drude weight (derived and verified in Appendix A). On this doped ring only 1.4% of m_0 lies below $6\,t$, and $m_{-1} = 0.324$ is the small difference of $\frac{1}{2}\langle -\hat{T} \rangle = 5.025$ and $\mathcal{D} = 4.701$: the stiffness cancels 94% of the kinetic term. On the ring the negative-order moment m_{-1} therefore moves most in relative terms (154% against 38% for m_1), 4.0× the shift of m_1 and 3.3× that of m_2 (47%); on the open chain it moves least, with relative slopes 0.233 for m_{-1} against 0.719 for m_1.

A larger relative shift does not mean earlier detection. Because of the cancellation, a 2%

bias on $\frac{1}{2}\langle -\hat{T}\rangle$ sets $\tau_{-1} = 0.198$, which is 61% of m_{-1}, whereas τ_1 and τ_2 are 4.3% of m_1 and m_2. So m_2 rejects first ($f \approx 0.046$), then m_1 (0.057), and m_{-1} only at $f \approx 0.20$. The open chain gives the same order, and across spurious-pole positions of 0.25–4 t the moment m_{-1} rejects first only for poles at 0.25 t and 0.5 t, on both geometries (Appendix A). It is thus a complementary check for weight very close to $\omega = 0$, not the most sensitive check for this error model. On the ring, $\mathcal{D}$ requires ground states at twisted flux and is not a same-sample quantity. We claim no novelty for the identity, the textbook f-sum [47, 48].

3.3 Single-particle and collective channels

The construction applies to any diagonal $T = 0$ Lehmann measure, and Appendix D illustrates it on three more probes, with injected corruptions and no calibrated threshold. For $A(k, \omega)$ of the half-filled $L = 12$ ring, the anticommutator first moment, the Hartree centroid, exposes a corruption that flattens the band-centroid dispersion while keeping the Mott gap, the band edges and the per-k total weight. On the charge and spin structure factors of a half-filled $L = 6$ ring, a reconstruction that moves band weight into a spurious low-frequency peak at fixed total weight passes the m_0 check (residual $\sim 10^{-15}$), but its first moment is off by 29–36% once 40% of the band weight is moved.

4 A blinded, pre-registered calibration in simulation

On a sealed catalog of 300 instances the battery rejected every truncation and most spurious features, missed every under-converged Krylov reconstruction, and its false-positive rate depends on the scale of the simulated noise (Table 3).

Design. Following blind-analysis and pre-registration practice [97, 98], before any instance existed we sealed by SHA-256 hash (`0e4c3368...`) the battery (m_0, m_1, m_2 plus an interval-Hankel condition), the decision rule and its threshold inputs (N_s, z, b; the formula for τ_k is fixed in the classifier code, not in the hashed text), the error catalog, the primary endpoint, the secondary predictions and the random seed. The rule is the default rule of Sec. 2.2: $N_s = 5 \times 10^4$, the per-circuit budget of the companion's `ibm_fez` run [25] (also the largest of our later $L = 8$ runs), an assumed bias $b = 2\%$, and $\mathrm{Var}(O_k)$ the local-estimator variance on the prepared state (the truncated-subspace ground state for truncations, the exact ground state otherwise). The generator produced $G = 300$ instances: 112 faithful controls and three *algorithmic* error causes, determinant-subspace (SQD) truncation (56), under-converged Krylov reconstruction (71), and an analytic-continuation artifact adding a spurious spectral feature (61); the classifier saw no labels. Every instance uses the current probe J of the doped $L = 6$ Hubbard ring ($N_\uparrow = N_\downarrow = 2$, sector dimension 225) at $U/t \in \{3, 4, 5, 6\}$. Truncations keep the top $d = 29$–138 of the 186 ground-state configurations by $|\psi_0|^2$ (16–74% of the support) and are mild by weight (75.1–98.7% of $|\psi_0|^2$); in four, $J|0_S\rangle$ has no component inside $\mathcal{S}$ and the reconstruction is empty. Krylov instances are n_l-node Lanczos representations, $n_l \in \{1, 2, 3, 4\}$. A spurious feature is a point (δ) atom of weight 2–25% of m_0 at a frequency uniform between the sector's lowest and highest excitation energies; with probability 1/2 the measure is rescaled to restore m_0, and in this m_0-preserving subclass the atom carries 2–20% of the reconstructed weight. The independent estimate is the oracle estimator of Sec. 2.2, drawn independently of $\bar{m}_k$. Generator and classifier re-verify the hash and refuse to run if it is broken. The seal is self-attested: it carries no third-party timestamp, the blinding is enforced in code, and sealing, generation, classification and scoring ran in a single invocation, so the code predates the seal but is not covered by it; because the run is deterministic given the sealed seed, the seal cannot by itself exclude earlier runs. Departures of the implementation from the sealed text are listed in Appendix C.

Table 3: **Blinded calibration by class** (current probe of the doped $L = 6$ ring, oracle estimator, default rule). "As implemented" (the sealed record): simulated noise with the standard error that the threshold's shot term assumes; Wilson 95% intervals in brackets. "1.96-fold noise": the same draws rescaled by 1.96, the noise scale the sealed text specifies, threshold unchanged (post hoc). Pooled primary endpoint as implemented: detection rate $101/188 = 53.7\%$ ($[46.6, 60.7]\%$) at a false-positive rate of $1/112$; with 1.96-fold noise $126/188 = 67.0\%$ at $40/112$.

class	instances	rejected (as implemented)	rejected (1.96-fold noise)
truncation	56	56 $[93.6, 100]\%$	56
spurious feature, weight-changing	27	25 $[76.6, 97.9]\%$	25
spurious feature, m_0-preserving	34	20 $[42.2, 73.6]\%$	22
Krylov, $n_l = 1$	15	0 $[0, 20.4]\%$	6
Krylov, $n_l \geq 2$	56	0 $[0, 6.4]\%$	17
faithful controls (false positives)	112	1 $[0.2, 4.9]\%$	40 $[27.4, 44.9]\%$

Results by class. All 56 truncations were caught, 55 of them on m_0. Of the spurious features, 93% (25/27) of the weight-changing ones were caught, which the diagonal m_0 channel can reach, and 59% (20/34) of the m_0-preserving ones, which only the off-diagonal moments can reach. The truncation result is a property of the sampled cuts, not a guarantee: an exhaustive scan of the $U/t = 4$ ring, with exact moments (the oracle without noise), a deterministic tie order and the same thresholds, finds 12 of 185 truncated reconstructions that pass the full battery, three at mid coverage ($d = 86, 87, 95$ of 186) and nine near-complete ($d \geq 160$), plus three narrow windows in which m_1 alone, at the shot-only threshold, has power only at the false-positive level (Appendix B). The near-complete cuts lie outside the catalog's range ($d \leq 138$), no $U/t = 4$ instance of the catalog fell on the mid-coverage ones, and the other U values were not scanned; the shared-state estimator passes a different set (Sec. 5.3). The pooled detection rate is low because 71 of the 188 corrupted instances are under-converged Krylov reconstructions.

The Krylov misses follow from Gauss quadrature: an n_l-node Lanczos representation reproduces $m_0, \ldots, m_{2n_l-1}$ exactly [52,64], as Proposition 1(ii) states for any subspace that contains the Krylov space. For $n_l \geq 2$ the reconstruction matches m_0–m_2 to below 10^{-15} relative and is indistinguishable from a faithful one at this order. For $n_l = 1$ it misses m_2 by $m_0 \mathrm{Var}_A(\omega)$, the variance of the normalized spectrum times its weight, which is 2.4–4.2% of m_2 on these sharply peaked spectra, and passes only because the threshold budgets a 2% bias (largest residual $0.99\,\tau_2$). At the shot-only threshold (no bias term) 6 of the 15 one-node instances would be rejected on m_2, but 5 of the 56 exact $n_l \geq 2$ instances would also be rejected, on some moment, from noise alone. Two sealed secondary predictions failed: the one-node Krylov case, expected to be caught, was caught 0/15; the m_0-preserving subclass, sealed as designed-blind with low expected detection, was caught 20/34, an outcome the pre-registration did not anticipate, because the rescaling restores only m_0 and still shifts m_1 and m_2. The other secondary predictions held.

Sensitivity to the noise model. The false-positive rate as implemented is low by construction: the simulated estimator is unbiased, its noise has the standard error assumed by the threshold's shot term, and the threshold budgets a 2% bias comparable to the shot term. Redrawing the noise for the 112 faithful instances gives a modeled rate of 1.3%, far below the sealed nominal level of 14.3%; one or fewer false positives occur in more than half of the redraws, so the sealed 1/112 is typical. A realized bias of +2% (−2%) on all three moments, at the

edge of the budget, gives 8.0% (11.5%; the threshold scales with $|\widehat{m}_k|$); no larger bias was modeled. With the 1.96-fold noise of the sealed text the same draws give $40/112 = 35.7\%$ (Wilson $[27.4, 44.9]\%$), above the sealed nominal level and typical of that model (expected rate 34.4% over fresh noise draws), and a detection rate of $126/188 = 67.0\%$ that includes 17 of the 56 Krylov reconstructions with $n_l \geq 2$, rejected by noise alone. The shot term uses the amplitude-based local-estimator variance, which a device cannot realize. Inflating that variance by a factor c, in both the simulated noise and the threshold (sealed draws rescaled by $\sqrt{c}$), the frozen rule still rejects all 56 truncations; at $c = 4$ it rejects 17/34 m_0-preserving and 22/27 weight-changing features at a false-positive rate of 15/112, and at $c = 10$ 15/34 and 19/27 at 19/112. These rates come from one rescaled draw; the expected false-positive rates over fresh noise are 7.1% ($c = 4$) and 10.7% ($c = 10$). The calibration therefore confirms the implementation; robustness to a real estimator bias is untested.

Shared-state rescoring. Post hoc, after unblinding, we rescored the 56 sealed truncations with the shared-state estimator, after a rebuild that reproduces every sealed reconstruction moment, variance and noise draw to rounding error. Read without noise as a leak score, every one lies above the numerical tolerance 10^{-6} on each of m_0, m_1 and m_2 (smallest $\Delta_0/\widehat{m}_0 = 0.25$); with the sealed noise draws and τ_k as a convention, the frozen rule rejects all 56, every one on m_0 (smallest $r_0 = 5.9$, where $r_k = |\widehat{m}_k - \bar{m}_k|/\tau_k$). This count does not depend on the estimator form: at $k = 0$ the two forms coincide, $O_0 = C_0 = J^\dagger J$. For the 112 faithful controls the shared state is the exact ground state (each U has a unique ground state), so the shared-state estimate equals the oracle-centered one and its false-positive rate is the sealed 1/112 (40/112 with 1.96-fold noise). The shared-state estimate is itself far from the exact moment on truncations (for one instance $\widehat{m}_0 = 1.04$ against $m_0 = 0.37$): the test registers the leak, not the distance to the truth.

Figure 4 shows how the two channels divide the work. Determinant truncation moves the total-weight sum rule $m_0 = \langle J^\dagger J \rangle$, so the diagonal channel catches it (55/56 with the oracle estimator, 56/56 shared-state), as it catches most weight-changing spurious features (21 of the 25 caught). The m_0-preserving subclass keeps the total weight and the determinant support fixed by construction, so neither the m_0 residual nor a support-coverage check can see it (cf. Fig. 3); only the off-diagonal moments reject it (20/34), for example id 222 with $(r_0, r_1, r_2) = (0.16, 3.3, 9.1)$. In all, 26 sealed instances have $r_0 < 1$ but an off-diagonal residual above threshold (m_1 on 16, m_2 alone on 10). This off-diagonal reach holds only in simulation. Within this catalog the test detects support truncation and most added spectral features, and misses reconstructions that reproduce the tested moments within the thresholds; the ground truth is classically known and merely withheld from the classifier, so this is a blinded test, not a field catch.

5 What a pass implies and what it cannot see

At the order used here, m_0–m_2, a pass constrains a reconstruction very little: the guaranteed power against weight redistributed among fixed poles is zero and the classical bracket on its cumulative weight is vacuous at the spectral mean, two statements that concern low-order moment checks in general; and the most natural sample-based form of misplaced weight passes the default thresholds.

5.1 Bracketing a passing reconstruction

The moments $m_0, \ldots, m_{2n}$ of a positive measure bracket its cumulative weight $F(x) = \int_{-\infty}^{x} A(\omega)\, d\omega$: every positive measure sharing them lies within the Christoffel function

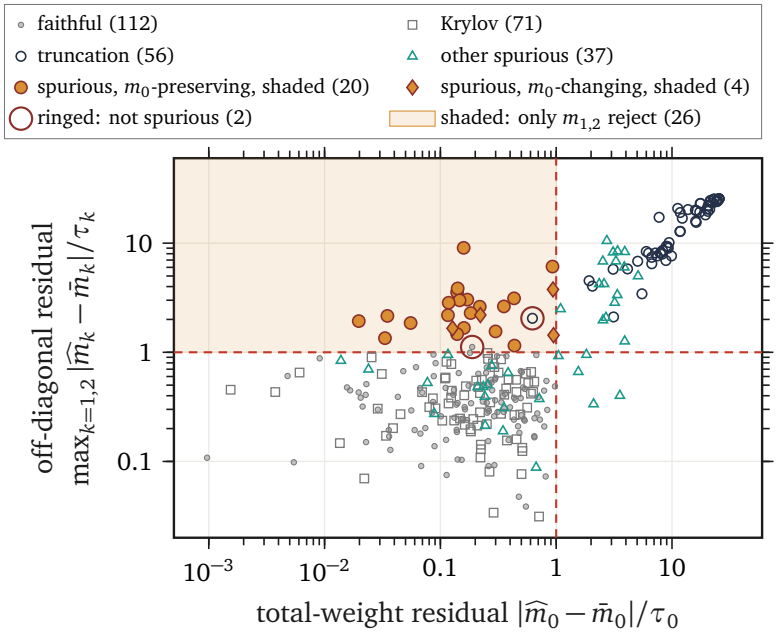


Figure 4: **The diagonal and off-diagonal moments have complementary reach** (sealed blinded simulation, current probe of the doped $L = 6$ ring). $r_k = |\widehat{m}_k - \bar{m}_k|/\tau_k$ with the frozen thresholds ($z = 1.96$, $N_s = 5 \times 10^4$, assumed 2% bias); $r_k > 1$ rejects. All 300 instances are plotted at their sealed r_k; none is clipped. Truncations: 56/56 rejected, $r_0 > 1$ in 55. Shaded: the 26 rejected only off-diagonally: 20 m_0-preserving and 4 m_0-changing spurious features, false positive id 184 and truncation id 162 (both ringed; id 162 has shared-state $r_0 = 15.8$). Krylov (0/71): rejected by neither channel. A structural separation in simulation with an amplitude-based shot variance, not a device-realizable rejection.

$W_n(x) = 1/\sum_{j=0}^{n} p_j(x)^2$ of the orthonormal polynomials, which is also the largest atom a moment-matching measure can place at x, and integrated window weights obey a two-sided Markov–Krein bracket of width at most $W_n(c)+W_n(d)$ (Theorem 2 and Proposition 2, classical and re-derived in Appendix F) [52, 59–61, 99]. Optimal two-sided bounds from equilibrium moments were derived early for a related functional, the response to a damped-harmonic perturbation, which gives the broadened shape of a nuclear-resonance line [71]; Stieltjes–Tchebycheff moment theory built photoabsorption profiles from spectral moments [72]; and matched-moment brackets serve as a posteriori Kolmogorov–Smirnov and Wasserstein bounds for Lanczos spectrum approximation [73]. Here they bound a reconstruction under test. On the doped $L = 6$, $U/t = 8$ current measure (Appendix F), at the default order ($n = 1$) the bracket is the one-sided Chebyshev (Cantelli) bound, whose maximum m_0 lies at the spectral mean, $0.01\,t$ from the dominant atom: passing places no constraint on the cumulative weight at the peak. Finite tolerances loosen it further: the largest atom that a positive measure with $|m_k - m_k^{\text{true}}| \le \tau_k$ ($k \le 2$) can place above the band is 3.2–7.7 times W_1 at $\omega = 14$–$20\,t$, and 3.6–5.7 times on the $U/t = 4$ measure of the blinded catalog. Even with exact moments through m_8 the global maximum falls only to $\max_x W_4 = 0.882$, and at the dominant atom no order can go below its residue 0.778. The companion shows by example that exact moments through order $2K + 1$ coexist with an $O(1)$ reconstruction error [25]; the Christoffel function makes this distribution-free at the default order, for every measure that passes. Rejection does not involve W_n: the looseness limits what passing implies, not the power to reject.

5.2 Detectability of fixed-pole redistributions

The limiting error class is the *within-sector redistribution*: a corruption that keeps the retained determinant support and the reconstructed poles and only reweights those poles. Because the support is unchanged, it contributes no leak term to Eq. (4); it is visible only through the moment shifts characterized below. We study this class because no support-based check can signal it; we know of no mechanism by which a sample-based pipeline produces it. Part (a) of the statement restates the classical Prony/Vandermonde uniqueness of an atomic measure from its moments [59, 100], and part (b) turns it, with the sharp Chebyshev bound, into a detectability-versus-shot-budget statement.

Theorem 1 (Detectability and completeness of the moment battery). *Let*

$$A(\omega) = \sum_{n=1}^{N} w_n\, \delta(\omega - \omega_n), \qquad w_n = \sum_{a:\, E_a - E_0 = \omega_n} |\langle a|J|0\rangle|^2 > 0,$$

be a diagonal $T = 0$ Lehmann measure on N distinct *poles $\{\omega_n\}$ (the frequencies that carry weight), with w_n the total weight of the eigenstates at ω_n, and let a within-sector redistribution $\mathbf{w} \to \mathbf{w} + \boldsymbol{\delta}$ ($\boldsymbol{\delta} \neq 0$, keeping $w_n + \delta_n \geq 0$) act on the* same *poles. Write*

$$\mathbf{m} = V\mathbf{w}, \quad V_{kn} = \omega_n^k, \qquad \Delta_k = |(V\boldsymbol{\delta})_k| = \left|\sum_n \delta_n\, \omega_n^k\right|, \qquad \rho = \|\boldsymbol{\delta}\|_1.$$

Then $\|\boldsymbol{\delta}\|_2 \leq \rho \leq \sqrt{N}\,\|\boldsymbol{\delta}\|_2$, and $\rho/2$ is the weight moved when m_0 is preserved.

(a) (Completeness in exact arithmetic.) *If the redistribution preserves $m_0, \ldots, m_K$ with $K < N-1$, then $\Delta_j > 0$ for at least one j with $K < j \leq N-1$; quantitatively*

$$\max_{K<j\leq N-1} \Delta_j \;\geq\; \frac{\sigma_{\min}(V_{0:N-1})}{\sqrt{N-1-K}}\, \|\boldsymbol{\delta}\|_2, \tag{5}$$

with $V_{0:N-1}$ the $N \times N$ node-Vandermonde on rows $0, \ldots, N-1$ and both sides in one fixed frequency unit. Hence the order-$(N-1)$ battery is complete *for fixed-pole redistributions: no nonzero within-sector redistribution preserves all of $m_0, \ldots, m_{N-1}$. If the poles may also move (two measures of at most N atoms each), completeness holds at order $2N-1$ by Prony counting [59, 100]. The only alteration invisible to every moment is weight traded among degenerate eigenstates at one frequency, which leaves $A(\omega)$ unchanged and is not an error.*

(b) (Finite thresholds.) *At N_s shots, with $\mathrm{Var}(O_k)$ the per-shot variance of the estimator operator O_k and b_k the bias budget (Sec. 2.2), each moment carries the two-sided threshold $\tau_k = z_{1-\alpha/2}\,[\mathrm{Var}(O_k)/N_s + b_k^2]^{1/2}$, that is, $z_{1-\alpha/2}$ times its noise floor (the square root). We call a given $\boldsymbol{\delta}$* exposed *at order j when its noise-free shift exceeds the threshold, $\Delta_j > \tau_j$; whether a finite-shot test then rejects it is a question of power (Corollary 1). Some redistribution preserving $m_0, \ldots, m_K$ is exposed at order $K+1$ iff $\Delta_{K+1}^{\max} > \tau_{K+1}$, with $\Delta_{K+1}^{\max}$ the linear program (G.1). If rows $0, \ldots, K+1$ of V have a nontrivial null space ($N - K - 2 \geq 1$), the smallest Δ_{K+1} over such redistributions is zero for every size ρ up to a limit set by positivity: the guaranteed power at order $K+1$ is zero, however well separated the poles. Two conditions bound exposure: (i) is sufficient for exposure at some order $j \leq N-1$; (ii) is necessary for exposure at order $s-1$.*

(i) Sufficient. *If $\boldsymbol{\delta}$ preserves $m_0, \ldots, m_K$, it is exposed at some order $j \leq N-1$ whenever the right-hand side of (5) exceeds $\max_{K<j\leq N-1} \tau_j$.*

(ii) Necessary at order $s-1$ (cluster bound). *If, for some $s \geq 2$, $\boldsymbol{\delta}$ preserves $m_0, \ldots, m_{s-2}$ and is supported on poles within an interval of diameter D and midpoint ω_c, then $(V\boldsymbol{\delta})_{s-1} =$*

$\sum_n \delta_n\, q(\omega_n)$ *for every monic* q *of degree* $s-1$*, and the Chebyshev choice of* q *gives the sharp bound* $\Delta_{s-1} \le 2\rho\,(D/4)^{s-1}$*, attained by alternating weights on the Chebyshev extrema of the interval (for* $s = 1$*, when nothing is preserved, the sharp bound is* $\Delta_0 \le \rho$*); every higher shift obeys* $\Delta_j \le \rho \sum_{i=s-1}^{j} \binom{j}{i} |\omega_c|^{j-i} (D/2)^i$*, which is also* $O(\rho D^{s-1})$*. Exposure at order* $s-1$ *therefore requires*

$$\rho > \frac{\tau_{s-1}}{2(D/4)^{s-1}}, \qquad \textit{i.e.} \qquad N_s > \frac{z_{1-\alpha/2}^2\, \mathrm{Var}(O_{s-1})}{4\rho^2 (D/4)^{2(s-1)} - z_{1-\alpha/2}^2\, b_{s-1}^2} \tag{6}$$

when the denominator is positive, and no N_s *exposes it at order* $s-1$ *when* $2\rho\,(D/4)^{s-1} \le z_{1-\alpha/2}\, b_{s-1}$*; a higher tested order can still expose it, within the term-by-term bound above. The shot cost to detect a redistribution of fixed size at its first exposable order grows at least as* $D^{-2(s-1)}$ *as the cluster contracts.*

Proof. Appendix G.

Remarks. Super-resolution bounds for clustered nodes [101–103], like the separation threshold of Ref. [104], are proved for Fourier (unit-circle) nodes; for real nodes and power moments, worst-case error bounds of matching order for a near-colliding cluster are given in Ref. [105]. Part (b)(ii) relies on none of them. For real nodes the same $\sigma_{\min} \propto D^{s-1}$ scaling, at fixed relative geometry of the cluster, follows from the lower bound on $\sigma_{\min}$ implied by Gautschi's inverse-norm bound [106, 107] and an elementary null-space upper bound, so for an isolated cluster the sufficient condition (i) weakens with the same power as the necessary bound (ii) (Appendix G, which also gives the shot-cost scaling under uniform cluster contraction). In part (b), K is the highest *preserved* order and $K+1$ the tested one, so a battery through m_{K+1} faces a null space of dimension $N-K-2$; elsewhere in the text K is the highest order of the battery itself (Sec. 2.2), and the same null space has dimension $N-K-1$. The fixed-pole class excludes under-converged Krylov reconstructions, whose poles move; their misses are explained in Sec. 4.

Corollary 1 (Detection power vs. cluster size and shot count)**.** *Under a Gaussian approximation to the estimator (for the* 5×10^4*-shot mean of the local estimator on the* $U/t = 4$ *ring the exact skewness is* -0.014 *to* -0.009 *and the excess kurtosis below* 5×10^{-4} *at orders* $k = 0,\dots,4$*; Appendix B checks* $\widehat{m}_1$ *by Monte Carlo), the test reaches power* $1-\beta$ *at level* α *when* $\Delta_j \gtrsim z_{\alpha,\beta}\,\tau_j/z_{1-\alpha/2}$*, that is, when the shift exceeds* $z_{\alpha,\beta}$ *noise floors, with* $z_{\alpha,\beta} = z_{1-\alpha/2} + z_{1-\beta}$ *and the bias budget counted in the noise floor as in the default rule. By Theorem 1(b)(ii) this requires* $N_s \gtrsim z_{\alpha,\beta}^2\,\mathrm{Var}(O_{s-1})/[4\rho^2(D/4)^{2(s-1)} - z_{\alpha,\beta}^2\, b_{s-1}^2]$ *at the first exposable order* $s-1$ *(*$s \ge 2$*), and no budget suffices at order* $s-1$ *below the bias-limited cluster diameter* $D_\star = 4[z_{\alpha,\beta}\, b_{s-1}/(2\rho)]^{1/(s-1)}$*. Without bias, and when* $\Delta_{s-1} \neq 0$ *(always so if* $\boldsymbol{\delta}$ *is supported on exactly* s *poles), the order-*$(s-1)$ *cost is finite for every* $D > 0$ *and diverges only as* $D \to 0$*.*

For the default $L = 6$, $U/t = 4$ battery the $N = 5$ poles are well separated (minimal separation $\Delta\omega_{\min} = 0.87\,t$; on nodes rescaled to $[-1,1]$, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$), yet its guaranteed power against fixed-pole redistribution is zero. The reason is the null-space dimension, not conditioning: rows $0,1,2$ of V leave a two-dimensional null space on five poles, so every redistribution in it passes, whatever its size; the same holds for any spectrum with four or more distinct poles. Over m_0, m_1-preserving redistributions the linear program (G.1) puts the signed shift of m_2 in $[-0.92, 7.09]$ (Table 5): up to $3.8\,\tau_2$ ($\tau_2 = 1.848$), but the range contains zero. Table 4 lists explicit reconstructions that pass the default battery. Orders 3 and 4 would catch some of them, but not all: with the same rule ($\tau_3 = 15.0$, $\tau_4 = 122$; the exact-state local-estimator variance behind them is expected to underestimate the true per-shot variance, so realistic thresholds would be wider), over all redistributions that keep

Table 4: **Reconstructions of the $L = 6$ current spectrum that pass the default thresholds** τ_k (m_0, m_1, m_2; $\tau_2 = 1.848$, $\tau_3 = 15.0$, $\tau_4 = 122$). CE1–CE3 and the random case reweight the five exact poles ($U/t = 4$; Appendix G); the Rayleigh–Ritz row is the full-support reconstruction of Sec. 5.3; the last two rows are the exhaustive truncation scan, with exact moments (Appendix B) and with the shared-state estimator (Sec. 5.3). Read as leak scores against the numerical tolerance, the Rayleigh–Ritz reconstruction and every truncation are flagged as not closed.

reconstruction	residuals in the battery	error it carries
CE1	m_0, m_1, m_2 exact	moves 72% of the weight ($\rho = 1.43\, m_0$), empties two poles; dominant residue $0.393 \to 0.037$; $\Delta_3 = 0.23\, \tau_3$, $\Delta_4 = 0.70\, \tau_4$
CE2	m_0, m_1 exact, $\Delta_2 = \tau_2$	moves 86%; dominant residue $0.393 \to 0.073$; $\Delta_3 \approx 3.3\, \tau_3$
CE3	$\lvert\Delta_k\rvert \le \tau_k$, $k \le 2$	$\lVert\boldsymbol{\delta}\rVert_1 = 2.03\, m_0$; empties three poles, the dominant one included
random, m_0, m_1 kept	$\Delta_2 = 0.044\, \tau_2$	moves 25% of the weight
Rayleigh–Ritz, full support	$\Delta_{0,1} < 10^{-14}$ relative, $\Delta_2 = 0.34\, \tau_2$ (leak score 0.019)	relative L_1 error 0.93; poles $7.83 \to 7.45\, t$, $2.73 \to 4.43\, t$
truncations, $U/t = 4$, exact moments	all below τ_k	12 of 185 depths pass ($d = 86, 87, 95$; 160–165; 183–185); relative L_1 errors 0.73–1.26
truncations, $U/t = 4$, shared state	all below τ_k (convention)	16 of 185 depths pass ($d = 170$–185); relative L_1 errors 0.94–1.04

m_0, m_1, m_2 exactly $\Delta_3 \le 0.35\, \tau_3$, and Δ_4 reaches $1.19\, \tau_4$ at the one that maximizes Δ_3, a marginal case. Neither m_3 nor m_4 is estimated in this work. The residual blind spot thus has two sources: the battery order, since with $m_0, \ldots, m_K$ every fixed-pole redistribution in the $(N-K-1)$-dimensional null space passes, regardless of pole separation; and pole clustering, which by (6) raises the shot cost of detection at least as $D^{-2(s-1)}$. Injectivity guarantees that some m_j separates any fixed-pole redistribution; the thresholds decide whether that separation is observable at a given budget.

5.3 Full-support Rayleigh–Ritz, and non-closure versus error size

The natural sample-based form of misplaced weight passes the default thresholds. Rayleigh–Ritz on the full ground-state support of the calibration ring (186 determinants at $U/t = 4$, so that $|0_S\rangle = |0\rangle$) captures all of $J|0\rangle$, so by Proposition 1(iii) Δ_0 and Δ_1 vanish (below 10^{-14} of m_0 and m_1) and the first residual is the boundary coupling at $k = 2$: $\Delta_2 = 0.62$, 1.9% of $m_2 = 33.39$, which is $0.34\, \tau_2$. Under the default thresholds τ_k the reconstruction passes. Read as a leak score, $\Delta_2/\widehat{m}_2 = 0.019$ lies far above the numerical tolerance, so the shared-state test flags it as not closed at $k = 2$, but nothing in that score shows that its relative L_1 error at $\eta = 0.15\, t$ is 0.93: Rayleigh–Ritz moves the dominant pole from $7.83\, t$ to $7.45\, t$ and the pole at $2.73\, t$ to $4.43\, t$. The pass under τ_k is set by the assumed bias budget: without it the $k = 2$ rule would reject above $N_s \approx 2.2 \times 10^5$ shots, and any tolerance on m_2 below 1.9% of m_2 rejects it (the default τ_2 is 5.5% of m_2). Over $U/t = 3$–8, Δ_2 is 0.20–0.36 τ_2 (0.95–2.2% of m_2) and the relative L_1 error 0.68–0.99; it passes τ_k at every U/t (`src/full_support_rr.py`,

data/2026-10-02_full_support_rr.json). This is the companion's no-go at unit captured weight [25], now as a reconstruction that passes a thresholded test.

Non-closure is not error size. Over the 176 truncations of the $U/t = 4$ ring with a non-degenerate subspace ground state and a non-empty reconstruction ($7 \le d \le 185$), the shared-state leak scores $\Delta_k/\widehat{m}_k$ correlate only moderately with the relative L_1 error (Spearman 0.44, 0.43 and 0.31 for $k = 0, 1, 2$), and they are large (0.38–0.46 at $k = 0$) at the mid-coverage truncations that pass τ_k with exact moments. With the shared-state estimator and τ_k as a convention, 16 of the 185 truncations pass ($d = 170$–185, 91–99.5% coverage, relative L_1 errors 0.94–1.04), while the nine exact-moment passes at $d = 86$, 87, 95 and 160–165 are rejected; read as leak scores against the 10^{-6} tolerance, all 185 are flagged (the $k = 2$ score is at least 0.014). A large leak score shows that a reconstruction is not closed under the full operators; a small one does not show that it is accurate.

6 Device-sampled supports: coverage is not fidelity

On supports sampled by ibm_fez, the coverage residual Δ_0 rewards support spreading, not fidelity: noise lowers it, a sampler with no quantum content does best, and the reconstruction with the smallest residual still has a relative L_1 error of 0.35. On hardware the device supplies only the sampled support $\mathcal{S}$; every moment in this section is computed classically on it, with exact amplitudes.

Setup. We ran four jobs at $L = 8$ (16 qubits, periodic chain, $U/t = 4$; one job per shot budget, job IDs under Data availability) and deposit their raw counts; at $L = 6$ even noiseless sampling of the companion's seven circuits, pooled at 5×10^4 shots each, covers the whole 300-configuration sector [25]. The probe is $J = c^\dagger_{0\uparrow}$ and the reference state is $|\varphi\rangle = J|0\rangle$, with $|0\rangle$ the exact half-filled ground state ($E_0 = -4.6035\,t$); here φ is built from the exact ground state, whereas Proposition 1 defines it on the subspace state. It lies in the addition sector $N_\uparrow = 5$, $N_\downarrow = 4$ of dimension 3920. Its weight is $m_0 = \langle J^\dagger J\rangle = \langle c_{0\uparrow} c^\dagger_{0\uparrow}\rangle = 1 - \langle n_{0\uparrow}\rangle = 1/2$ exactly, because translation invariance fixes $\langle n_{j\uparrow}\rangle = 4/8$ on every site (operator and Lehmann values agree to 6×10^{-17}). The seven circuits belong to the companion's family [25]: each prepares a computational-basis determinant with X gates (five up and four down electrons) and applies three Trotter steps of on-site R_{zz} and nearest-neighbor $R_{xx}R_{yy}$ hopping, without Jordan–Wigner strings, up to times $0, 0.5, \ldots, 3.0$ in units of $1/t$. They conserve particle number and S_z, so without noise they sample only the addition sector; they do not apply the Hubbard propagator to $c^\dagger_p|0\rangle$ and are not targeted at $|\varphi\rangle$. We ran them at 5×10^4, 3×10^4, 1.6×10^4 and 4×10^3 shots per circuit (labeled 50k, 30k, 16k and 4k below; 3.5×10^5 shots in total at 50k) with the SamplerV2 primitive of Qiskit Runtime, using measurement twirling, Pauli gate twirling and XpXm dynamical decoupling (pairs of π pulses about $+x$ and $-x$ on idle qubits); no readout-error mitigation was applied (SamplerV2 returns twirled raw counts). Each bitstring of the pooled record is mapped to a configuration in Qiskit's bit order (the rightmost bit is qubit 0) and post-selected to the sector, with no configuration recovery, which keeps 23.5–24.0% of the shots (82,697 of 350,000 at 50k: 92% of the shots of the zero-time circuit and 11–13% of each of the six evolved circuits).

The retained bitstrings define $\mathcal{S}$, on which the coverage residual $\Delta_0 = |m_0 - \bar{m}_0| = \langle\varphi|(\mathbb{1} - \Pi_\mathcal{S})|\varphi\rangle$ is evaluated exactly. As the budget falls from 50k to 4k, the configuration coverage $|\mathcal{S}|/3920$ falls from 94% to 35% and the residual grows: $\Delta_0 = 0.0044$, 0.0323, 0.1232 and 0.3726 (Fig. 5). In the companion's notation $\Delta_0 = m_0(1-w)$, with w the captured weight of $\mathcal{S}$ [25]; coverage and captured weight differ, and at 50k 94% of the configurations carry 99.1% of the weight ($1-w = 0.009$). The first-moment residual $\Delta_1 = |\widehat{m}_1 - \bar{m}_1|$ (0.019, 0.057, 0.024, 0.869) is not monotone in the budget.

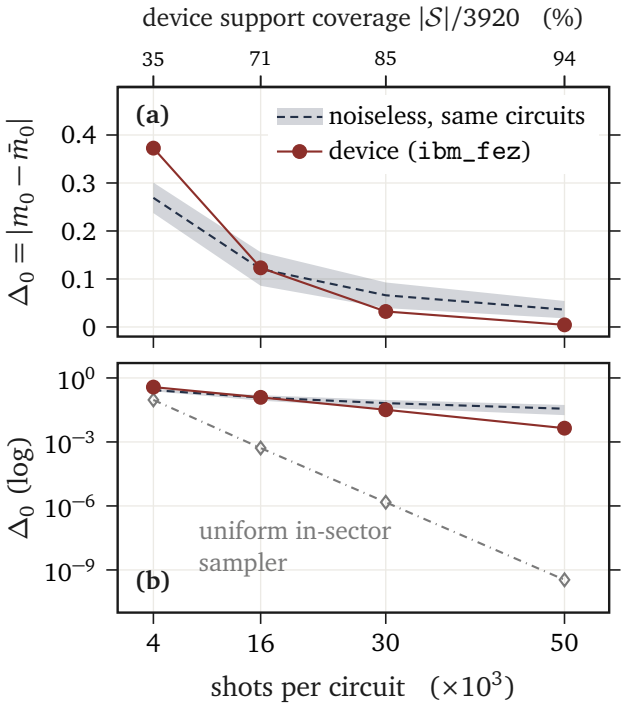


Figure 5: **Coverage residual on device-sampled supports** (`ibm_fez`, $L = 8$, 16 qubits, $U/t = 4$, sector dimension 3920, seven number-conserving sampler circuits). Dark red dots: Δ_0, exact on the device-retained support $\mathcal{S}$; top axis of (a): coverage $|\mathcal{S}|/3920$. Dark blue dashed line and grey-blue band: mean and ±1 sd (right-skewed, not a 68% interval) of 2000 noiseless multinomial replicas of the same circuits at the device's raw shots. Grey dash-dot line with open diamonds, (b) only: uniform in-sector sampler at the device's retained count N_{ret}, $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^{N_{\mathrm{ret}}}$. (a) Linear, (b) log scale. The device lies below all noiseless replicas at 50k and above all at 4k (percentiles in text); the uniform sampler, which uses no device, scores lower at every budget. Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic.

Reference samplers. We report empirical percentile ranks (the fraction of replicas at or below the device value), not Gaussian z-scores, because the weight of $|\varphi\rangle$ is concentrated (1% of the configurations carry 58% of m_0) and the distributions of Δ_0 are skewed. (i) *Noiseless sampling at the device's raw shots.* Two thousand multinomial replicas of the exact output distributions of the same seven circuits give $\Delta_0 = 0.036 \pm 0.018$, 0.066 ± 0.027, 0.121 ± 0.035 and 0.269 ± 0.032 (mean ± sd) at 50/30/16/4k. The device lies below all 2000 replicas at 50k, at the 1.7th percentile at 30k, at the 57.9th at 16k and above all 2000 at 4k. Its support is larger than in every replica at 50k and 30k ($|\mathcal{S}| = 3682$ against 3379 ± 14; 3330 against 3183 ± 16) and smaller than in every replica at 16k and 4k (2798 against 2877 ± 18; 1379 against 2027 ± 21). Noiseless sampling cannot reach $\Delta_0 = 0$: the seven output distributions together cover 3864 of the 3920 configurations, leaving 8.8×10^{-5} at infinite shots. The device's Δ_1 lies at the 35th, 82nd and 27.5th percentiles at 50/30/16k and above all replicas at 4k. (ii) *Uniform noise.* Mixing each circuit's output with a fraction p of uniformly random bitstrings changes the expected Δ_0 at 50k from 0.036 at $p = 0$ to a minimum of 0.0020 near $p \approx 0.93$, and 0.0024 at $p = 1$. At the p that reproduces the device's retention, $p \approx 0.81$, the mixture's mean (0.0023, 0.017, 0.075, 0.295) lies below the device at every budget, while the device's Δ_0 lies inside the mixture's replica distribution at 50–16k (Appendix I). (iii)

Uniform in-sector sampler. Drawing configurations uniformly from the sector at the device's retained count N_{ret}, with no device and no circuit, gives $\mathbb{E}[\Delta_0] = m_0(1-1/3920)^{N_{\mathrm{ret}}}$, which is 3.4×10^{-10}, 1.5×10^{-6}, 5.2×10^{-4} and 0.093, below the device at every budget. At 50k and 30k this sampler covers essentially the whole sector, and in all ten replicas checked its reconstruction is exact to the Lanczos floor (relative L_1 error below 4×10^{-9}). (iv) Noiseless sampling at the device's retained rather than raw counts puts the device below every replica at 50, 30 and 16k (Appendix I).

What the device run shows. The noisy device leaves less weight uncovered than all 2000 noiseless replicas of its own circuits at 50k and than 98% of them at 30k, where its support is the larger one; at 16k and 4k, where its support is smaller, it is typical or worse. Uniform noise lowers Δ_0 at 50k, and a sampler with no quantum content does best, because the 3920-configuration sector is small enough for uniform sampling to cover all or most of it at these budgets (81% of the configurations at 4k); a related effect has been reported for ground-state quantum-selected configuration interaction, where sampling noise can enlarge the configuration space beyond what the ideal ansatz supports [30]. Δ_0 grows as the budget falls for every sampler considered. The coverage residual can be evaluated on a device-sampled support; it cannot distinguish a good device from a noisy one and does not certify the reconstruction. This is a practical caution for any study that cites coverage or captured weight as evidence of sampler quality.

Under the default thresholds, a false negative. The relative L_1 error of each reconstruction on the job's 600-point frequency grid at $\eta = 0.15\,t$, $\int |A_{\mathcal{S}} - A|\,d\omega / \int A\,d\omega$, is 0.35, 0.56, 0.90 and 1.12 at 50/30/16/4k. At 50k both residuals, $\Delta_0 = 0.0044$ and $\Delta_1 = 0.019$, lie below the bias terms that the default rule would set ($\tau_k \geq z\,b\,m_k$ with $z = 1.96$ and $b = 2\%$: $\tau_0 \geq 0.020$, $\tau_1 \geq 0.082$), yet the relative L_1 error is 0.35. These thresholds are counterfactual: the rule was calibrated for the $L = 6$ current probe, and here $\widehat{m}_0, \widehat{m}_1$ are exact. Under the same rule the other three reconstructions, with large L_1 errors (0.56, 0.90, 1.12), would be rejected on Δ_0 (0.0323, 0.1232, 0.3726), since the shot term cannot raise τ_0 that far ($O_0 = J^\dagger J = 1 - n_{0\uparrow}$ is a projector, with variance at most 1/4). On these device data the rule would thus reject three poor reconstructions and pass the 50k one (read at numerical tolerance, as the leak score is, $\Delta_0/m_0 = 0.009$ already flags it as not closed, which says nothing about the size of its error), a false negative and a device-data instance of the companion's result that captured weight alone cannot certify a reconstruction: its bound places the error above $1 - w = 0.009$ and needs the boundary coupling for an upper bound [25] (the companion's own relative L_1 error of 0.35 at $L = 8$ refers to a different setting, the probe's full support at $\eta = 0.18\,t$). In L_1 error the device beats all 80 noiseless replicas at 50k and loses to all of them at 4k, and the uniform in-sector sampler beats the device at every budget (Appendix I).

Complete supports and device-native estimators. The companion's own $L = 6$ `ibm_fez` run retained the whole 300-configuration addition sector, so its reconstruction equals exact diagonalization by coverage; it shows that the pipeline executes, and every determinant it retained arose from a device error after a reversed bit order [25]. Two earlier $L = 6$ jobs of ours (2026-08-21; raw records deposited) likewise retain the whole 300-configuration sector, under either bit order, so a moment test on them returns the same verdict with or without the device data; they enter no result (Appendix I). The discriminating moments m_1, m_2 need a device-native estimator, which we have verified only in simulation; its forecast depolarizing bias without mitigation and the diagonal n_k alternative are in Appendix J.

7 Discussion

7.1 Relation to prior work

Self-verifying variational simulation certifies a prepared *state* through an energy-variance bound measured on the device [88]; observable-estimation validation bounds the accuracy of error-mitigated *expectation values* by validating the noise model and the mitigation [90]. Our test does neither: it tests a reconstructed spectral *function* for weight misassignment, documented by the companion for sample-based spectral functions [25], alongside the limitations documented [28–30] and the noise sensitivity analyzed [31, 32] for the ground-state energies of subspace methods. Cross-checking the classical diagonalizer across independent LAPACK paths [108] catches solver errors but not sampling incompleteness or weight misassignment. The approaches are complementary: a hardware spectrum could be produced under observable-estimation validation from a self-verified input state and then tested with our moments. Pipelines that hold the probe acting on the sampled ground state exactly, such as the Green's-function solver of Patel *et al.* [26], never lose spectral weight; for them the coverage channel carries no signal, and only the higher moments test them.

Moment information can instead be built *into* the reconstruction: Nevanlinna continuation imposes the Pick/positivity structure of a causal response [109], maximum-entropy continuation can take moments and sum rules as prior information [55], stochastic continuation samples positive spectra at a normalization fixed by the data [56], stochastic pole expansion enforces the zeroth-moment sum rule on its pole weights and admits further prior constraints on pole positions and amplitudes [110], and explicit pole representations such as minimal-pole fits [111] could carry further linear constraints pinning low-order moments to their operator values. Where available this is preferable, but such a spectrum is consistent only with the moments *it* was given, within the method that enforced them, for the data it was built from. Irmejs and Santos propose building dynamical correlation functions on hardware without real-time evolution, from weight-$O(k)$ moments measured on a prepared eigenstate, prove exponential accuracy away from the real axis, and demonstrate the method in classical simulations of small lattices [112]; a circuit-differentiation framework builds the retarded Green's function from derivatives of real-time-evolution circuits under realistic noise [113]; classical shadows [86, 114] could supply device-side estimates of $\langle O_k \rangle$, their spectral specialization extracts gaps without forming $A(\omega)$ [115], and their fermionic variant constructs dynamical correlation functions [116]. These programs construct the spectrum. Our test is their diagnostic counterpart for a user who holds the pipeline's sampled support $\mathcal{S}$ (and hence $|0_{\mathcal{S}}\rangle$), or an independent estimate of the moments, even if the reconstruction itself was produced elsewhere, on the real axis, where the constructive guarantees are weakest. Semidefinite moment bounds [77, 78] applied to the operators O_k would bound the true moments, and membership of the read-off moments in such intervals would be a second necessary condition of the same logical type as ours (Appendix E). No finite set of moment conditions certifies $A(\omega)$.

7.2 Limitations

Necessary, not sufficient. Unless the moments lie on the boundary of the moment body (singular Hankel matrix: fewer poles than the moments resolve), infinitely many positive measures share a finite set of moments, and within any nonzero tolerance they always do, so passing can never certify correctness [59, 60]. Section 5 makes this quantitative at the order used here: zero guaranteed power against fixed-pole redistributions, a bracket vacuous at the mean, and reconstructions that pass the default thresholds with relative L_1 errors of 0.68–0.99 (full-support Rayleigh–Ritz, $U/t = 3$–8) and up to 1.26 (truncations that pass with exact moments; `data/2026-10-02_full_support_rr.json`). A passing test does not resolve a

line shape [52, 63, 67].

The estimator. The realizable estimator, $\widehat{m}_k = \langle 0_S|O_k|0_S\rangle$, is deterministic, and its residual is the leak, not the distance to the true moment: at $L = 24$ its $\widehat{m}_0$ lies 10–15.5% below the DMRG m_0 (Appendix K). Its tolerance is numerical, and the default τ_k apply to it only by convention. The rejection rates with shot-noise thresholds come from an oracle estimator, the exact moment plus unbiased Gaussian noise; the 2% bias budget is assumed, not measured, so a rejection is attributed to the reconstruction only under that assumption, and the false-positive rate depends on the simulated noise scale (1/112 or 40/112; Sec. 4). Same-shot estimation of both branches has been realized only in simulation, with a local estimator that needs exact amplitudes (Appendix B).

Error classes. For determinant truncation the operative channel is m_0, the companion's captured-weight deficit (Sec. 3). Under-converged Krylov reconstructions with two or more Lanczos nodes reproduce m_0–m_2 and cannot be detected at this order; the one-node case misses m_2 by $m_0 \mathrm{Var}_A$, below τ_2 on the catalog spectra but not small in general (36% for the current spectrum of the open $L = 12$ chain of Appendix A). That error class calls for a separate instrument, most naturally a Ritz-residual check on the Krylov basis. Neither the moment test nor the Gauss-law check is a general device-noise diagnostic; that is the domain of observable-estimation validation [90].

Synthetic errors and scope. Every error rejected here is synthetic: an injected corruption or a controlled truncation of otherwise exact data, with the truth classically known and withheld only from the classifier. The calibrated scope is the current probe of the doped $L = 6$ ring at $U/t \in \{3, 4, 5, 6\}$, truncations keeping 15.6–74.2% of the support configurations but 75.1–98.7% of the ground-state weight, and spurious point atoms carrying 2–25% of m_0, modeled as analytic-continuation artifacts whose realism for sample-based pipelines is not established. The single-particle and collective cases are illustrations without calibrated thresholds. No error has yet been caught in a spectrum whose correctness was unknown in advance.

Hardware. No moment has been estimated from device counts. For a surrogate bond-basis probe, without error mitigation, the forecast depolarizing bias of m_1 at 295 CZ and two-qubit error 2.5×10^{-3} is 38% of the signal under a static local noise model and 116% under global depolarizing noise; the global part can be removed by noise renormalization once its strength is calibrated [117, 118], and the depth of the ground-state preparation that the estimator needs is not established (Appendix J). The sealed on-device n_k test has not been run: on the calibrations checked (2026-09-03 to 2026-09-05) no 12-qubit chain met its sealed coherence floors, and no job was submitted (Appendix J).

Scale. One-dimensional Hubbard rings at $L \leq 12$ are solved outright by DMRG and exact diagonalization, so no quantum resource is needed for their spectra. This is deliberate: calibration requires exact moments, and the test is characterized against these methods, not in competition with them. Its intended regime, where no controlled classical spectral reference exists, is real-time dynamics of two-dimensional and frustrated lattices and *ab initio* molecular spectra. The only run beyond exact diagonalization is a code-scalability check (Appendix K).

Positivity. The positivity that underwrites the moment bounds is that of a diagonal $T = 0$ Lehmann measure [49]. Off-diagonal spectral functions are not positive measures in general. At finite temperature the diagonal spectral function stays positive, but its support extends below $\omega = 0$ and its moments are thermal averages rather than ground-state expectation values, so the estimator and the bounds used here need separate treatment.

7.3 Outlook

A device-side use of the test needs the mitigation bias measured and bounded rather than assumed, for example from error-model violation [119] or by validating the noise model and the mitigation [90]. Hamiltonian moments of small Hubbard and impurity models have been

measured on trapped ions [13], and related quantities on IBM hardware [15,120]; estimating m_1, m_2 for the probes here with the bond-basis estimator awaits a constructed ground-state preparation and calibrated mitigation or lower gate error. The Gauss-law check, evaluated here exactly rather than from samples, belongs with symmetry verification [93, 94], Gauss-law oracles [95] and sum-rule checks [53, 54]: exactly known identities, each testing a different error class [96]. Extending the test to off-diagonal spectra, where positivity fails, and to finite temperature, where the support extends below $\omega = 0$ and the estimator must target thermal moments, is the main theoretical opening. Built on kernel-polynomial, recursion and quadrature machinery [52,67,68], the test attaches to sample-based diagonalization and Krylov pipelines developed for ground states [18–20,121,122] and extended to the spectra it tests [25,26].

8 Conclusions

On the subspace ground state that a sample-based pipeline holds, the difference between full-operator moments and the moments of the reconstructed $A(\omega)$ is exactly the leak of the probe state out of the subspace, order by order: the missed weight at $k = 0$, the boundary coupling at $k = 2$ when no weight is missed, and zero through order $2K+1$ on a Krylov-adequate subspace. The operator form $J^\dagger(H-e_0)^k J$ is what isolates the leak; off an eigenstate a first-moment check in the textbook commutator form passes, at the default threshold, a reconstruction whose first moment is 11% low. A rejection shows that the reconstruction is not closed under the full operators, not how far a moment is from the truth.

For truncation, the native error of sample-based diagonalization, the zeroth-order leak, the companion's captured-weight deficit, is the operative channel. The first and second moments reach the error that captured weight cannot see, weight moved in frequency at fixed total: in exact simulation they reject it from a moved fraction of a few percent, and in a pre-registered blinded simulation with an oracle estimator the battery rejected all 56 truncations, 74% of the spurious features (45/61) and none of the 71 under-converged Krylov reconstructions, at a false-positive rate of 1/112 with the simulated noise the threshold assumes; with the 1.96-fold noise the pre-registration specified the false-positive rate is 40/112, and noise also rejects 23 Krylov reconstructions, 17 of them exact through m_2.

Low-order moment checks have limits: with m_0–m_2 the guaranteed power against fixed-pole weight redistributions is zero for any spectrum with four or more distinct poles, and the cumulative-weight bracket is vacuous at the spectral mean; these two limits apply to every sum-rule validation at this order. Rayleigh–Ritz on the full ground-state support of the calibration ring passes the default thresholds, and any tolerance on m_2 above 1.9% of m_2, with a relative L_1 error of 0.93; its leak score flags it as not closed but does not measure that error.

On `ibm_fez`-sampled supports the coverage residual grows from 0.004 to 0.37 as coverage falls from 94% to 35%, but it rewards support spreading rather than device fidelity: a uniform in-sector sampler does better at every budget, and the reconstruction with the smallest residual has a relative L_1 error of 0.35. Open problems are estimating m_1 and m_2 on hardware, where the forecast bias for a surrogate probe is 38–116% of the signal without mitigation; calibrating the test with a measured mitigation bias and a same-sample estimator realized on a device; closing the within-sector blind spot with higher moments at an affordable shot cost; applying the semidefinite moment bounds of Refs. [77, 78] to the same moment sequence; and a field test on a spectrum whose correctness is not known in advance.

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Data availability The code and data that support the findings of this study, the code under the MIT licence and the data and documentation under CC BY 4.0, are openly available in the repository https://github.com/nicolasbonilla/reconstruction-independent-moment-tests. The repository contains the exact sector-Lanczos/Haydock engine; the interval-moment, blinded-test, within-sector and hardware-analysis scripts; the pre-registrations with their SHA-256 seals and the sealed records; the per-figure data; and Makefile targets and a notebook that run the generators. The file `docs/REPRODUCE.md` maps each figure and the key quoted numbers to their scripts and data files and lists the known gaps. Among them, two items behind quoted results do not regenerate exactly from source: a re-run of the $L = 12$ shared-state sweep of Fig. 2 reproduces both of its 5% crossings but moves its three smallest-d points, because that sweep left ties in degenerate $|\psi_0|^2$ shells to floating-point order and its subspace eigensolves are not bit-reproducible, so the deposited sweep is the one plotted; and the blinded truncation instances regenerate from the seed only with a stored tie order (below). The raw measurement counts of our four $L = 8$ `ibm_fez` jobs, one per shot budget, are included: `da983qse74ec73ajfr2g` (5×10^4 shots per circuit), `da98k16rbfbs73chckvg` (3×10^4), `da98icsjbipc73fej78g` (1.6×10^4) and `da98cksjbipc73fej1tg` (4×10^3). The raw records of two earlier $L = 6$ `ibm_fez` jobs of 2026-08-21, `da44t8u1vhnc73fkie30` and `da459juaa69c739jcb80`, are included with their reanalysis (`src/gate_tp_analysis.py`); they enter no result (Sec. 6). The sealed n_k test submitted no device job; the record of its two launches is `data/2026-09-05_nk_device_attempt_record.json`, and the transpiled CZ counts of all sampler circuits that were run are in `data/sampler_cz_count.json`. The exporter `src/export_device_dat.py` of Fig. 5 and the reference-distribution generator `src/delta0_reference_mc.py` with its output `data/2026-09-27_delta0_reference_mc.json` are also included. The design history of the sealed tests (the simulated gate versions of the n_k test, its amendment and the re-sealed verdict file) is recorded in `docs/PROCESS_RECORD.md`. The sealed blinded files are the canonical record: regenerating the truncation instances from the seed depends on tie-breaking inside degenerate $|\psi_0|^2$ shells, and the tie order that reproduces them is stored with the blinded-test addenda (`data/2026-09-27_blind_addenda.json`).

A Derivation of the moment identities

We derive the operator form of the spectral moments used throughout. Each m_k is a ground-state expectation value computable without the dynamics, and we state where the derivation needs an exact eigenstate. Let $H = H^\dagger$ have a complete orthonormal eigenbasis $\{|n\rangle\}$, $H|n\rangle = E_n|n\rangle$, and let $|0\rangle$ be an eigenstate of lowest energy E_0. The derivation uses only that $|0\rangle$ is an eigenstate. If the lowest level is degenerate, $A(\omega)$ and its moments refer to the chosen member; for the $L = 6$, $U/t = 8$ current measure, whose lowest level is a spin triplet, the measure is the same for all three members because J is a spin scalar (Appendix F). The probe J (a current or density) is Hermitian unless stated otherwise; we keep $J^\dagger$ on the bra side so that the same steps also cover the non-Hermitian probes treated below. The diagonal zero-temperature Lehmann spectral function

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \, \delta\big(\omega - (E_n - E_0)\big) \tag{A.1}$$

is a non-negative measure supported on $\omega = E_n - E_0 \geq 0$, and its power moments are

$$m_k = \int \omega^k A(\omega)\, d\omega = \sum_n (E_n - E_0)^k \, |\langle n|J|0\rangle|^2 .$$

Writing $|\langle n|J|0\rangle|^2 = \langle 0|J^\dagger|n\rangle\langle n|J|0\rangle$, using $\langle n|(H-E_0)^k = (E_n - E_0)^k\langle n|$, and closing with completeness $\sum_n |n\rangle\langle n| = \mathbb{1}$,

$$\begin{aligned} m_k &= \sum_n \langle 0|J^\dagger|n\rangle (E_n - E_0)^k \langle n|J|0\rangle \\ &= \langle 0|\, J^\dagger (H-E_0)^k J\,|0\rangle = \langle 0|O_k|0\rangle, \end{aligned} \tag{A.2}$$

where

$$O_k \equiv J^\dagger (H - e_0)^k J, \qquad e_0 = \langle H\rangle,$$

is the moment operator of the estimator, with e_0 the energy of the state on which it is evaluated ($e_0 = E_0$ on the exact eigenstate $|0\rangle$). Because $|0\rangle$ is annihilated by $H - E_0$, acting to the right gives $(H-E_0)J|0\rangle = [H,J]\,|0\rangle$, and iterating,

$$(H-E_0)^k J|0\rangle = \mathrm{ad}_H^k(J)\,|0\rangle, \qquad \mathrm{ad}_H(J) \equiv [H,J], \tag{A.3}$$

where ad_H^k denotes the k-fold nested commutator $[H,[H,\cdots[H,J]]]$. Substituting Eq. (A.3) into Eq. (A.2) gives the commutator form

$$m_k = \langle 0|\, J^\dagger \,\mathrm{ad}_H^k(J)\,|0\rangle = \langle 0|C_k|0\rangle, \qquad C_k \equiv J^\dagger \,\mathrm{ad}_H^k(J). \tag{A.4}$$

Both O_k and C_k are built solely from H and J, with no reference to the spectrum or to the reconstructed $A(\omega)$, and $\langle C_k\rangle = \langle O_k\rangle = m_k$ on an exact eigenstate; off an eigenstate they differ (below).

The lowest moments specialize immediately. For $k = 0$, $O_0 = C_0 = J^\dagger J$ gives the total weight $m_0 = \langle J^\dagger J\rangle$, which is $\langle J^2\rangle$ for a Hermitian probe. For $k = 1$ and $J = J^\dagger$, $m_1 = \langle 0|J[H,J]|0\rangle$; expanding the symmetric double commutator $[J,[H,J]]$ $= 2JHJ - J^2H - HJ^2$ and using $H|0\rangle = E_0|0\rangle$,

$$\langle 0|[J,[H,J]]|0\rangle = 2\langle 0|J(H-E_0)J|0\rangle = 2m_1, \tag{A.5}$$

so that

$$m_1 = \tfrac{1}{2}\langle 0|[J,[H,J]]|0\rangle. \tag{A.6}$$

This is the first-moment (double-commutator) sum rule of A_J; for the current operator J it fixes the mean frequency of the spectral weight. It is distinct from the optical-conductivity f-sum rule $\int_0^\infty \sigma_1\, d\omega = \frac{\pi}{2}\langle -\hat{T}\rangle$ (a ground-state kinetic energy), whose regular part is πm_{-1}: all of it on open chains, while on the ring the Drude weight takes the rest (see below) [46–48]. On an exact eigenstate the symmetrized form (A.6) is manifestly real and independent of the additive constant E_0, which the one-sided expression $\langle 0|J[H,J]|0\rangle$ obscures.

Negative-order moment. The negative-order moment $m_{-1} = \int_0^\infty A_J(\omega)/\omega\, d\omega$ of Fig. 3 carries the kernel $1/\omega$ and weights low frequencies; on the ring of that figure it rejects after m_1 and m_2. It carries the regular part of the optical f-sum, $\int_0^\infty \mathrm{Re}\,\sigma_{\mathrm{reg}}\, d\omega = \pi m_{-1}$, and it too is a ground-state expectation: since the current is the polarization velocity, $J = -i[H,P]$ with $P = \sum_j j\, n_j$ and $\hat{T}$ the hopping term of H (the sign is our convention for J and does not enter moments quadratic in J), $m_{-1} = \frac{1}{2}\langle [P,[H,P]]\rangle = \frac{1}{2}\langle -\hat{T}\rangle$, a kinetic energy [46–48]. On open chains we verify this identity against the Lehmann sum over the computed poles, to a relative 2×10^{-15} at $L = 6$ and 8 and 2×10^{-14} at $L = 12$. On the ring the polarization is multivalued and the identity acquires Kohn's stiffness correction, $m_{-1} = \frac{1}{2}\langle -\hat{T}\rangle - \mathcal{D}$ with $\mathcal{D} = \frac{1}{2}\,\partial_\theta^2 E_0|_{\theta=0}$ the finite-size Drude weight, for the Peierls phase $t \to t\, e^{i\theta}$ on every bond (per-bond, not total, flux); the identity is verified at $L = 12$ to a relative 2×10^{-6}, with $\mathcal{D}$ from the flux curvature. The open chain ($\mathcal{D} = 0$) does not restore an advantage for the corruption of Fig. 3, because 41% of its regular weight already lies below $6\,t$ and $m_{-1} = 4.81$ is large: there the thresholds are crossed at $f \approx 0.17$ for m_{-1}, 0.059 for m_1 and 0.045 for m_2. When the spurious pole is moved across 0.25–4 t, m_{-1} rejects first only for poles at $0.25\,t$ and $0.5\,t$, on both geometries (at $0.25\,t$ on the ring, $f \approx 0.015$ against 0.044 for m_1 and 0.043 for m_2). On open chains m_{-1} is also cheap: $\langle -\hat{T}\rangle$ is a one-body operator, measured in two qubit-wise-commuting settings that are a subset of those the m_0, m_1 estimator already requires (Appendix J).

Off an eigenstate the two forms differ, and the choice matters. Equations (A.2), (A.4) and (A.6) agree because $|0\rangle$ is an exact eigenstate of H. On any other state $|\psi\rangle$, with $e_0 = \langle\psi|H|\psi\rangle$, for example the ground state $|0_S\rangle$ of a truncated subspace Hamiltonian (Appendix B), $\langle\psi|O_k|\psi\rangle$ and $\langle\psi|C_k|\psi\rangle$ are no longer equal for $k \geq 1$, and in general neither equals the exact moment m_k. At $k = 1$, $C_1 - O_1 = -J^\dagger J\,(H - e_0)$, so

$$\langle\psi|C_1|\psi\rangle - \langle\psi|O_1|\psi\rangle = -\langle\psi|J^\dagger J\,(H-e_0)|\psi\rangle,$$

a term set by the state's own residual $(H-e_0)|\psi\rangle$, which vanishes on an eigenstate (for Hermitian J the real part of $\langle\psi|C_1|\psi\rangle$ is $\frac{1}{2}\langle\psi|[J,[H,J]]|\psi\rangle$). At $k = 0$ the two forms coincide.

We evaluate O_k, for a structural reason (Appendix B). With O_k and the reference e_0 shared with the reconstruction, the residual $\widehat{m}_k - \tilde{m}_k$ between the estimate and the reconstruction moment is exactly the operator leak of Eq. (4), the part of the probe state that H carries outside the retained subspace: it vanishes whenever that subspace contains enough of the Krylov space of the probe state, and at $k = 0$ it is the companion's captured-weight deficit $\widehat{m}_0(1-w)$ [25]. With C_k the residual would also contain the state-residual term above, which need not vanish in that case, so it would not isolate the leak. The sealed blinded test of Sec. 4 does not depend on this choice: its estimate is the exact moment plus Gaussian noise (its sealed shot widths are local variances of $J^\dagger(H-E_0)^k J$ on the prepared state, `src/blind_generate.py`). The shared-state analysis of Fig. 2 uses O_k (`data/2026-08-26_teeth_shared_state.json`). The post hoc shared-state rescoring of the 56 sealed truncations (Sec. 4) does not depend on the choice either, because every one of them is rejected at $k = 0$, where $O_0 = C_0$.

The following example shows that the choice matters (script `src/estimator_form_offeigenstate.py`, output in `data/2026-09-27_theory_numerics.json`). Take the current probe of the doped $L = 6$, $U/t = 4$ ring and truncate its sector to the $d = 100$ determinants of largest $|\psi_0|^2$ (44% of the 225-dimensional sector, 54% of the 186-determinant ground-state support). With the determinants ordered deterministically (descending $|\psi_0|^2$,

ties broken by sector index, NumPy `lexsort`), the subspace ground state gives $\langle O_1\rangle = 4.558$ but $\frac{1}{2}\langle[J,[H,J]]\rangle = 3.717$, against the reconstruction moment $\bar{m}_1 = 3.743$ and the exact $m_1 = 4.227$. At the default threshold $\tau_1 = 0.229$ (Appendix B) the O_1 form rejects the truncated reconstruction and the commutator form does not; with the truth known here, the rejection is the correct verdict, since $\bar{m}_1$ is 11% below the exact value. At the six other cuts of the same scan ($d = 30$, 50, 75, 150, 186 and 200) the two forms give the same verdict. This cut falls inside a 24-fold degenerate shell of $|\psi_0|^2$, so these values depend on how ties are broken: NumPy `argsort`, which does not fix the order of ties, gives 4.605, 3.707 and 3.810 in the deposited run, with the same two verdicts. The m_0 comparison, common to both forms, rejects this cut ($\Delta_0 = 12\,\tau_0$). In sum, the estimator is $\widehat{m}_k = \langle\psi|O_k|\psi\rangle$ with $e_0 = \langle\psi|H|\psi\rangle$ on the prepared state, and the commutator forms are used only on exact eigenstates.

Two features of these identities make them the operational core of the test. First, O_k is an explicit operator polynomial in H and J; mapped to qubits it is a weighted sum of Pauli strings, whose diagonal terms are read from computational-basis counts and whose hopping terms need rotated bases (Appendix J). Its expectation can therefore be evaluated on a prepared state without forming $A(\omega)$; this is the estimator $\widehat{m}_k$ of Appendix B. Besides the idealized evaluation on the exact ground state, it is evaluated here in three ways, none on hardware: classically on the subspace state that also yields the reconstruction (Fig. 2, a deterministic $|\psi_0|^2$ truncation; the $L = 24$ DMRG test of Appendix K, a shot-sampled subspace); as a local estimator from the same simulated shots, which needs the exact amplitudes (joint sampling, Appendix B); and from simulated Pauli-grouped measurements of a separately prepared ground state (Appendix J). Second, the derivation used only completeness of the eigenbasis and that $|0\rangle$ is an exact eigenstate, so Eqs. (A.2)–(A.4) hold for any probe and Hamiltonian, which is why the test transports across models. Positivity of the measure (A.1) is not needed for the identities; it enters only in the positivity conditions and the bounds built on them.

Because the steps above never used $J = J^\dagger$, Eqs. (A.2)–(A.4) hold unchanged for a non-Hermitian probe, with $m_0 = \langle J^\dagger J\rangle$; only the symmetrized form (A.6) needs a Hermitian J. The single-particle spectral function $A_{k\sigma}(\omega)$ of Fig. 7 is the two-sided (photoemission plus inverse-photoemission) spectrum of the fermionic probe $c_{k\sigma}$, and its moments close under *anticommutators* rather than the double commutator:

$$\begin{aligned} \int A_{k\sigma}\, d\omega &= \left\langle \{c_{k\sigma}, c_{k\sigma}^\dagger\}\right\rangle = 1, \\ \int \omega A_{k\sigma}\, d\omega &= \left\langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\}\right\rangle = \epsilon_k + U\langle n_{-\sigma}\rangle, \end{aligned} \tag{A.7}$$

the Hartree-shifted band centroid [54]. The Hermitian double-commutator map above is the special case $J = J^\dagger$; Eq. (A.7) is the version used for the single-particle test of Fig. 7, and it is likewise a ground-state expectation of explicit operators.

Collective probes. For the non-Hermitian probes ρ_q and S_q^z of Fig. 9 the double commutator is the symmetrized form $\frac{1}{2}[m_1(q) + m_1(-q)]$, equal to $m_1(q)$ by the inversion symmetry of the ring; both forms are evaluated on the exact ground state, where they coincide. Only the kinetic term contributes to m_1, since the Hubbard interaction is diagonal in the occupation basis and commutes with ρ_q and S_q^z.

Derivation of the anticommutator moment identities. Equation (A.7) is not a fresh sum rule but the non-Hermitian specialization of Eqs. (A.2)–(A.4), derived here for completeness. Write $d \equiv c_{k\sigma}$, $d^\dagger \equiv c_{k\sigma}^\dagger$, with $\{|n\rangle\}$ the complete eigenbasis of H over all particle-number sectors, $H|n\rangle = E_n|n\rangle$, ground state $|0\rangle$, energy E_0. The two-sided single-particle spectral function is

the sum of an electron-addition ($N{+}1$) and an electron-removal ($N{-}1$) branch,

$$A_{k\sigma}(\omega) = \sum_n |\langle n|d^\dagger|0\rangle|^2\,\delta\big(\omega-(E_n-E_0)\big) + \sum_n |\langle n|d|0\rangle|^2\,\delta\big(\omega+(E_n-E_0)\big), \tag{A.8}$$

a non-negative measure with addition poles at $\omega = E_n^{N+1}-E_0$ and removal poles at $\omega = E_0-E_n^{N-1}$. Its support *straddles* $\omega=0$ once frequencies are measured from a chemical potential μ in the gap, that is with H replaced by $H-\mu\hat{N}$: the addition poles are then positive and the removal poles negative (in the raw-H frame removal poles can be positive), and the identities below hold unchanged in form, with μ_1 shifted by $-\mu$. Its power moments split accordingly,

$$\mu_p \equiv \int \omega^p A_{k\sigma}\,d\omega = \sum_n (E_n-E_0)^p |\langle n|d^\dagger|0\rangle|^2 + (-1)^p \sum_n (E_n-E_0)^p |\langle n|d|0\rangle|^2. \tag{A.9}$$

Applying to each branch the steps of Eqs. (A.2)–(A.4), namely completeness together with $(E_n-E_0)^p\langle n|X|0\rangle = \langle n|\,\mathrm{ad}_H^p(X)|0\rangle$ for $X=d^\dagger$ and $X=d$, gives the closed operator form, the commutator form C_p of each branch probe:

$$\mu_p = \langle 0|\,d\;\mathrm{ad}_H^p(d^\dagger)\,|0\rangle + (-1)^p\langle 0|\,d^\dagger\;\mathrm{ad}_H^p(d)\,|0\rangle, \tag{A.10}$$

built solely from H, d, and $d^\dagger$. For $p=0$ this is the canonical anticommutator,

$$\mu_0 = \langle 0|\,dd^\dagger + d^\dagger d\,|0\rangle = \langle\{c_{k\sigma}, c_{k\sigma}^\dagger\}\rangle = 1.$$

For $p=1$, using $H|0\rangle = E_0|0\rangle$ and $\langle 0|H = E_0\langle 0|$ on the outer factors one verifies term by term that

$$\mu_1 = \langle 0|\,d[H,d^\dagger]\,|0\rangle - \langle 0|\,d^\dagger[H,d]\,|0\rangle = \big\langle\{[c_{k\sigma},H],c_{k\sigma}^\dagger\}\big\rangle, \tag{A.11}$$

the anticommutator first-moment identity, manifestly real and independent of the additive constant E_0. For the Hubbard Hamiltonian $H=H_0+H_U$ with

$$H_0 = \sum_{k'\sigma'} \epsilon_{k'}\,c_{k'\sigma'}^\dagger c_{k'\sigma'}, \qquad H_U = U\sum_i n_{i\uparrow}n_{i\downarrow},$$

and $c_{k\sigma} = L^{-1/2}\sum_j e^{-ikj}c_{j\sigma}$, the commutators evaluate in closed form,

$$[c_{k\sigma},H_0] = \epsilon_k\,c_{k\sigma}, \qquad [c_{k\sigma},H_U] = \frac{U}{\sqrt{L}}\sum_j e^{-ikj}c_{j\sigma}n_{j,-\sigma},$$

and since $\{c_{j\sigma}n_{j,-\sigma}, c_{j'\sigma}^\dagger\} = \delta_{jj'}n_{j,-\sigma}$,

$$\big\langle\{[c_{k\sigma},H],c_{k\sigma}^\dagger\}\big\rangle = \epsilon_k\,\langle\{c_{k\sigma},c_{k\sigma}^\dagger\}\rangle + \frac{U}{L}\sum_j\langle n_{j,-\sigma}\rangle = \epsilon_k + U\langle n_{-\sigma}\rangle, \tag{A.12}$$

which is the centroid quoted in Eq. (A.7) and evaluated per momentum in Fig. 7.

Scope: which checks transport to the single-particle channel. The zeroth- and first-moment checks carry over intact: μ_0 and μ_1 are ground-state expectation values of explicit operators [Eqs. (A.10)–(A.11)], so they give reconstruction-independent residuals Δ_0, Δ_1 at each k. So

does the two-sided Hamburger–Hankel positivity $H_n = [\mu_{i+j}] \succeq 0$, since a positive measure on $\mathbb{R}$ has a positive semidefinite Hankel matrix whatever the sign of its support; it tests the estimated moments, not a non-negative reconstruction (Appendix B). The shifted-Stieltjes condition $H_n^{(1)} = [\mu_{i+j+1}] \succeq 0$ does *not* transport: it presupposes support on $[0, \infty)$, and $A_{k\sigma}$ is supported on both signs of ω. The Christoffel bracket of Appendix F applies to any positive measure on $\mathbb{R}$, but we have evaluated it only for the one-sided current measure and do not deploy it here. The single-particle test is therefore Δ_0, Δ_1 and Hamburger–Hankel positivity; the shifted-Stieltjes condition belongs to the one-sided current and density responses.

B The reconstruction-independent moment estimator

We give the explicit estimator $\widehat{m}_k$ and show why it is independent of the reconstruction and, unlike the spectral moment $\bar{m}_k$, detects subspace truncation. Sample-based diagonalization returns a set $\mathcal{S}$ of computational-basis determinants $\{|x\rangle\}$ and diagonalizes the projected Hamiltonian $H_\mathcal{S} = \Pi_\mathcal{S} H \Pi_\mathcal{S}$, with $\Pi_\mathcal{S} = \sum_{x\in\mathcal{S}} |x\rangle\langle x|$, on a classical computer [18, 19]. This gives eigenpairs $\{E_n^\mathcal{S}, |n_\mathcal{S}\rangle\}$, among them the subspace ground state $|0_\mathcal{S}\rangle = \sum_{x\in\mathcal{S}} c_x |x\rangle$ with energy $e_0 = \langle 0_\mathcal{S}|H|0_\mathcal{S}\rangle$. Unless $\mathcal{S}$ contains the support of the exact ground state, $|0_\mathcal{S}\rangle$ is an eigenstate of $H_\mathcal{S}$ but in general not of H, so O_k and C_k of Appendix A differ on it; we use O_k, for the reason given there and made explicit by Eq. (4) below. The probe J and the Hamiltonian H are sparse in this basis, with known matrix elements $J_{yx} = \langle y|J|x\rangle$ and $H_{yx} = \langle y|H|x\rangle$.

The moment $\langle 0_\mathcal{S}|O_k|0_\mathcal{S}\rangle$ admits two evaluations, which coincide only when $\mathcal{S}$ is complete for the operators involved. Inserting the *subspace* resolution $\Pi_\mathcal{S}$ in place of the identity yields the reconstruction (spectral) moment,

$$\begin{aligned} \bar{m}_k &= \sum_n (E_n^\mathcal{S} - e_0)^k |\langle n_\mathcal{S}|J|0_\mathcal{S}\rangle|^2 \\ &= \langle 0_\mathcal{S}|J^\dagger [\Pi_\mathcal{S}(H-e_0)\Pi_\mathcal{S}]^k J|0_\mathcal{S}\rangle \\ &= \langle 0_\mathcal{S}|J_\mathcal{S}^\dagger (H_\mathcal{S}-e_0)^k J_\mathcal{S}|0_\mathcal{S}\rangle, \qquad J_\mathcal{S} = \Pi_\mathcal{S} J \Pi_\mathcal{S}, \end{aligned} \tag{B.1}$$

with the sum over the eigenpairs of $H_\mathcal{S}$. It is exactly $\int \omega^k A(\omega)\, d\omega$ and lives entirely inside $\mathcal{S}$. Here $H_\mathcal{S} - e_0$ acts on span $\mathcal{S}$, so the $k = 0$ term carries the convention $[\Pi_\mathcal{S}(H-e_0)\Pi_\mathcal{S}]^0 \equiv (H_\mathcal{S} - e_0)^0 \equiv \Pi_\mathcal{S}$, and $\bar{m}_0 = \langle 0_\mathcal{S}|J^\dagger \Pi_\mathcal{S} J|0_\mathcal{S}\rangle$ is the weight captured by $\mathcal{S}$ rather than the full $\langle J^\dagger J\rangle$. The reconstruction-independent estimator instead applies the sparse operator $O_k = J^\dagger (H-e_0)^k J$ directly,

$$\begin{aligned} \widehat{m}_k &= \langle 0_\mathcal{S}|J^\dagger (H-e_0)^k J|0_\mathcal{S}\rangle \\ &= \sum_{x,x'\in\mathcal{S}} c_x^* c_{x'} \langle x|J^\dagger (H-e_0)^k J|x'\rangle, \end{aligned} \tag{B.2}$$

where each matrix element $\langle x|O_k|x'\rangle$ is a finite product of sparse operators summing over intermediate determinants that need *not* lie in $\mathcal{S}$. This is the operational meaning of independence: $\widehat{m}_k$ probes the action of J and H on $|0_\mathcal{S}\rangle$ beyond the sampled subspace, whereas $\bar{m}_k$ is confined to it. Evaluated on $|0_\mathcal{S}\rangle$, Eq. (B.2) is the shared-state estimator of Fig. 2. Evaluated instead on the exact ground state $|0\rangle$ (with $e_0 \to E_0$), it gives the exact moment m_k, where O_k and C_k coincide; this idealized version, more independent than a device allows, is used in the idealized curve of Fig. 2, in the sealed blinded test (Sec. 4, with Gaussian shot noise added), and in the Monte-Carlo and interval-battery demonstrations below.

Proof of Proposition 1. Equation (4) follows directly from Eqs. (B.1) and (B.2), because both branches are built on the same probe state $\varphi_\mathcal{S} \equiv J|0_\mathcal{S}\rangle$ and the same reference e_0, and $J_\mathcal{S}|0_\mathcal{S}\rangle = \Pi_\mathcal{S}\varphi_\mathcal{S}$. (i) At $k = 0$, with the convention above, it reduces

to $\langle\varphi_S|(\mathbb{1}-\Pi_S)|\varphi_S\rangle = \widehat{m}_0(1-w) \geq 0$, with $w = \|\Pi_S\varphi_S\|^2/\|\varphi_S\|^2$ the companion's captured weight (Π_S is its P_S) [25]. (ii) If $\operatorname{span}\mathcal{S}$ contains $\operatorname{span}\{\varphi_S, H\varphi_S, \ldots, H^K\varphi_S\}$, then $[\Pi_S(H-e_0)\Pi_S]^j\varphi_S = (H-e_0)^j\varphi_S$ for $j \leq K$; writing $k = a+b$ with $a, b \leq K$ for $k \leq 2K$, and $k = 2K+1$ as $\langle(H-e_0)^K\varphi_S|(H-e_0)|(H-e_0)^K\varphi_S\rangle$ with $(H-e_0)^K\varphi_S \in \operatorname{span}\mathcal{S}$, both terms of Eq. (4) agree for every $k \leq 2K+1$. This moment exactness is classical [52, 66]. (iii) If $w = 1$, then $\Pi_S\varphi_S = \varphi_S$, the $k = 1$ terms agree, and at $k = 2$ the difference is $\|(H-e_0)\varphi_S\|^2 - \|\Pi_S(H-e_0)\varphi_S\|^2 = \|(\mathbb{1}-\Pi_S)H\varphi_S\|^2$, the companion's second-moment deficit. □

Statement (ii) explains why the under-converged Krylov reconstructions of Sec. 4 pass the low-order battery: an n_l-step Lanczos reconstruction started from the probe state projects onto its Krylov space with $K = n_l - 1$ (the argument holds for any orthogonal projector, not only a determinant subspace) and so reproduces $m_0, \ldots, m_{2n_l-1}$. The one-step case reproduces only m_0 and m_1; its m_2 is low by $m_0 \operatorname{Var}_A$, 2–4% on the catalog spectra, and passes below τ_2 (Sec. 4). Under determinant-subspace truncation the Krylov space is not contained, and Eq. (4) measures the spectral weight the truncation leaks outside $\mathcal{S}$. Evaluated instead on the exact probe state $\varphi_0 \equiv J|0\rangle$ (for the device, $J = c^\dagger_{0\uparrow}$), with only $\mathcal{S}$ taken from the device, the $k = 0$ expression is the coverage residual of Sec. 6.

The full-support reconstruction of Fig. 6 is an instance: with $\mathcal{S}$ the whole 186-determinant support of the $L = 6$, $U/t = 4$ ground state (so that $|0_S\rangle = |0\rangle$ and $\varphi_S = \varphi_0$), $w = 1$, Δ_0 and Δ_1 vanish to machine precision, and $\Delta_2 = 0.62$ (1.9% of $m_2 = 33.39$); it captures all of φ_S and is still not exact. Against the default thresholds this reconstruction passes the full battery (Δ_0 and Δ_1 below 10^{-14} of m_0 and m_1, $\Delta_2 = 0.34\,\tau_2$; its leak score $\Delta_2/\widehat{m}_2 = 0.019$ flags it as not closed), although its relative L_1 error at $\eta = 0.15\,t$ is 0.93: Rayleigh–Ritz moves the dominant pole from $7.83\,t$ to $7.45\,t$ and the pole at $2.73\,t$ to $4.43\,t$. The same holds at $U/t = 3$, 5, 6 and 8, with $\Delta_2 = 0.20$–$0.36\,\tau_2$ and relative L_1 errors 0.68–0.99 (`src/full_support_rr.py`, `data/2026-10-02_full_support_rr.json`). The companion also bounds the reconstruction error by $1-w$ from below and by the boundary coupling from above and shows that captured weight alone cannot certify; the use of these identities as a thresholded rejection test is this paper's (Sec. 1). For the shared-state estimator, $\Delta_k = |\widehat{m}_k - \bar{m}_k|$ is nonzero only if $\mathcal{S}$ misses part of φ_S or of the states $H^j\varphi_S$ that order k probes; the independent estimator can then reject, while the circular control ($\bar{m}_k$ against itself) returns $\Delta_k \equiv 0$ (Sec. 3). Weight moved *within* $\mathcal{S}$ contributes no leak term; it is visible only through the shift of the reconstruction moments that it causes, which Theorem 1 and Appendix G characterize.

Equation (B.2) is equivalently an importance-weighted average over $\mathcal{S}$,

$$\widehat{m}_k = \sum_{x\in\mathcal{S}} |c_x|^2\, O_k^{\mathrm{loc}}(x), \qquad O_k^{\mathrm{loc}}(x) = \frac{\langle x|O_k|0_S\rangle}{\langle x|0_S\rangle}, \tag{B.3}$$

the local-operator estimator familiar from variational Monte Carlo [125]; the local moment $O_k^{\mathrm{loc}}(x)$ again sums over determinants connected to x by O_k and so reaches outside $\mathcal{S}$. Drawing N_s configurations from $|c_x|^2$ and averaging O_k^{loc} gives an unbiased Monte-Carlo estimate with variance $\operatorname{Var}(O_k)/N_s$, where $\operatorname{Var}(O_k) = \sum_x |c_x|^2 |O_k^{\mathrm{loc}}(x)|^2 - \widehat{m}_k^2$ is the local-estimator variance. This form needs the amplitudes c_x, which a device does not return, so we use it only in simulation (Fig. 6 and the joint-sampling study below). On a device the same expectation would be estimated from Pauli-grouped measurements (Appendix J), with a variance of its own.

Same-sample statistics. Strict same-shot estimation of both branches has been realized only in simulation, in a study sealed before it was run (SHA-256 `8bf87b5a...`, 2026-09-01, self-attested; `data/2026-09-01_lever1_prereg_seal.json`). For the current probe of the $L = 6$, $U/t = 4$ ring it drew 2000 faithful and 2000 corrupted replicas at five coverages of the ground-state support, 84, 90, 95, 99 and 100%, reached with 2000, 3200, 5000, 12000 and 40000 shots per replica; the sealed corruptions are m_0-preserving reweightings of each

replica's reconstruction within its support (a tilt and a satellite, of both signs, at seven target shifts of m_1 from 2% to 24%). The run applied one sign only, and its realized mean shifts of Δ_1 are 0.4–20% of m_1, not the targets. One set of shots per replica gives both $\widehat{m}_k$ and $\bar{m}_k$; here $\widehat{m}_k$ is the local estimator above, which uses the exact ground-state amplitudes, so it is not device-realizable. The sealed primary endpoints were the realized false-positive rate at the nominal z and the detection-power difference at matched false-positive rate. The variance of the residual is $\mathrm{Var}(\Delta_k) = \mathrm{Var}(\widehat{m}_k) + \mathrm{Var}(\bar{m}_k) - 2\,\mathrm{Cov}(\widehat{m}_k, \bar{m}_k)$, with a same-sample correlation coefficient of about 0.43 at 84% coverage, that is at 2000 shots per replica rather than the 5×10^4 of the default rule; the correlation decays towards zero as coverage approaches 100%. Treating the two estimates as independent inputs, as one would when a moment interval comes from separate data, for example from the finite-statistics certificate of Mortimer *et al.* [78], over-estimates this variance, and the test becomes over-conservative in the undersampled regime (realized false-positive rate 1.5% at a nominal 5%). At matched false-positive rate the correlation-aware test gives no detection-power advantage (power differences between −5.6 and +0.05 percentage points across the ladder of corruptions and coverages that was run), so the covariance is a calibration correction, not a power gain. That comparison used the covariance estimated from the Monte Carlo itself, an oracle, and gives point estimates without confidence intervals (`data/2026-09-01_lever1_oracle.json`); no single-sample plug-in stage was run, since the sealed decision gate calls for one only if the oracle stage shows a gain. This structure is a quantum instance of Hausman-type specification testing [126]: $\widehat{m}_k$ is the consistent, model-free estimator and $\bar{m}_k$ the model-implied one. The independent-input variance $\mathrm{Var}(\widehat{m}_k)+\mathrm{Var}(\bar{m}_k)$ is exact only at full coverage, where the correlation vanishes; Hausman's efficiency limit, $\mathrm{Cov}(\widehat{m}_k, \bar{m}_k) = \mathrm{Var}(\bar{m}_k)$, would instead reduce the residual variance to $\mathrm{Var}(\widehat{m}_k) - \mathrm{Var}(\bar{m}_k)$, and our correlation lies between the two limits.

The exact-moment test takes its finite-shot form here. We model the estimator uncertainty as $\delta_k^2 = \mathrm{Var}(O_k)/N_s + b_k^2$, a shot term over N_s samples plus a bias budget b_k for mitigation and device error, and reject the reconstruction at level α when

$$\Delta_k = |\widehat{m}_k - \bar{m}_k| > \tau_k, \qquad \tau_k \equiv z_{1-\alpha/2}\,\delta_k.$$

The default rule (used here only in simulation and on classically computed moments) takes $\alpha = 0.05$ per moment ($z_{0.975} = 1.96$), $N_s = 5 \times 10^4$ and $b_k = 0.02\,|\widehat{m}_k|$; for the three-moment battery the sealed nominal joint level is $1 - 0.95^3 = 14.3\%$ (Sec. 4). For the current probe of the doped $L = 6$, $U/t = 4$ ring, with $\mathrm{Var}(O_k)$ and $\widehat{m}_k$ taken on the exact ground state (`data/2026-09-27_interval_battery.json`), it gives $\tau_0 = 0.029$, $\tau_1 = 0.229$ and $\tau_2 = 1.848$. The level holds only while the estimator's bias stays within b_k, which is assumed, not measured: a rejection shows an inconsistency between estimate and reconstruction, and attributing it to the reconstruction rests on that assumption. A second, separate check concerns the estimate alone: whether some moment vector in the box $[\widehat{m}_\ell - \tau_\ell, \widehat{m}_\ell + \tau_\ell]$ satisfies the positivity conditions obeyed by the moments of every positive measure, $H_n \succeq 0$ and, for one-sided $T = 0$ support, $H_n^{(1)} \succeq 0$. Infeasibility would show that the estimated moments belong to no spectrum, pointing to the device, the mitigation or the estimator; it cannot refute a non-negative reconstruction, whose own moments $\bar{m}_k$ satisfy these conditions by construction. In the implementations reported here (the sealed blinded test and the interval battery below) the positivity condition is evaluated at $\bar{m}_k$ and therefore cannot reject; every rejection comes from a residual Δ_k. The operators, targets and intervals of each demonstration are in the reproduction materials.

We close with a Monte-Carlo calibration of this test (Fig. 6) at $N_s = 5 \times 10^4$, the per-circuit budget of the sealed rule (that of the companion's `ibm_fez` run, and the largest of our later $L = 8$ budgets). It is a simulation: $M = 4 \times 10^3$ replicas of N_s shots are drawn from the exact ground-state distribution $|\langle x|0\rangle|^2$ of the doped $L = 6$, $U/t = 4$ ring, a system not run on hardware; $\widehat{m}_1$ is the local estimator above for its current probe, and each truncated reconstruction

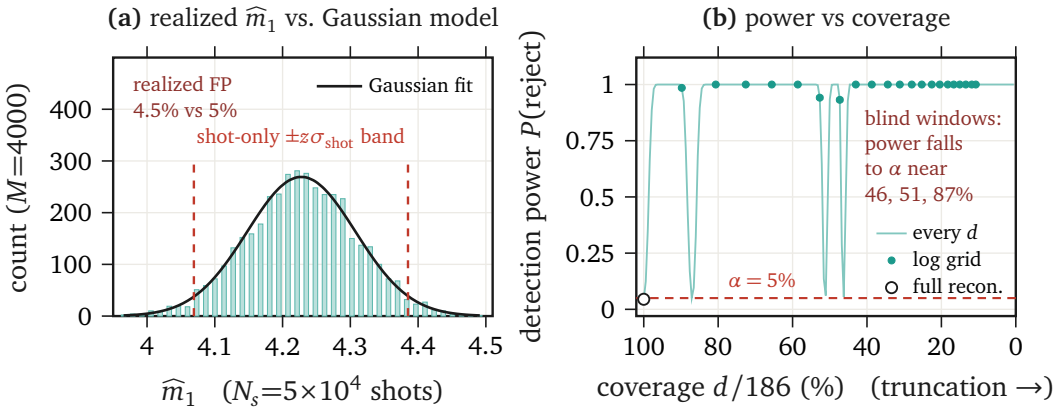


Figure 6: **Monte-Carlo calibration of the m_1 moment test** (simulation only, no device data: current probe, doped $L = 6$, $U/t = 4$ ring; $M = 4 \times 10^3$ replicas of $N_s = 5 \times 10^4$ shots from $|\langle x|0\rangle|^2$). **(a)** $\widehat{m}_1$ is near-Gaussian (curve: Gaussian fit, the draws' mean and s.d.); false-positive rate 4.5% (Wilson 95% interval [3.9, 5.2]%), nominal 5%. Dashed: shot-only band $\pm z\sigma_{\text{shot}} = \pm 0.158$ about the exact m_1, narrower than the default $\tau_1 = 0.229$ (2% bias budget). **(b)** Power of m_1 alone at the shot-only threshold versus coverage $d/186$ of the ground-state support (line: every d; dots: 22-point log grid with Wilson 95% intervals; open marker at $d = 186$: full reconstruction, false-positive rate). Power reaches ≥ 0.95 at 161 of 185 truncations, falling to the false-positive level in windows near 46%, 51% and 87% coverage whose location depends on tie-breaking in degenerate $|\psi_0|^2$ shells. The shot term (exact-state variance) likely understates a device estimator's per-shot uncertainty.

is built from the ground state of the truncated Hamiltonian. The Monte Carlo checks the analytic interval rather than replacing it. The realized estimator is close to Gaussian (sample skewness −0.04 and excess kurtosis −0.05 over the replicas, consistent within sampling error with the exact −0.011 and 3.5×10^{-4} of the 5×10^4-shot mean), its spread matches the analytic shot term to 1% (0.0799 against 0.0806), and its false-positive rate, 4.5% (Wilson 95% interval [3.9, 5.2]%), is consistent with the nominal 5% (Fig. 6a). Two inputs are modeled: the shot term uses the exact-state local-estimator variance, which assumes known amplitudes and omits ratio-estimator and Pauli-grouping overhead, so it likely understates the per-shot uncertainty of a device estimator; the bias term b_k is assumed and swept, not measured. For this probe $\mathrm{Var}(O_1) \approx 3.2 \times 10^2$ gives a shot standard error $\sqrt{\mathrm{Var}(O_1)/N_s} \approx 0.081$ (1.9% of m_1); with the 2% bias budget $\tau_1 = 0.229$ (5.4% of m_1), close to the 5% relative line of Fig. 2.

Power is high but not uniform in the truncation depth. At the shot-only threshold m_1 alone reaches power ≥ 0.95 at 161 of the 185 truncations, already at $d = 181$ (97% coverage, power 0.98). The exceptions are the four mildest truncations ($d = 182$–185, one to four determinants dropped) and three narrow windows, $d = 84$–88, 93–98 and 158–166 (coverage 45–47%, 50–53%, 85–89%), at whose centers power falls to the false-positive level (5.5%, 6.1%, 5.0% at $d = 86$, 95, 162). The 22-point grid of Fig. 6b touches the first two windows only at their edges ($d = 88$, 98; power 0.93, 0.94) and misses the third. The noise-free residuals agree: at the default 2% bias budget $|\widehat{m}_1 - \bar{m}_1| < \tau_1$ for 17 of the 185 truncations ($d = 183$–185, 159–165, 94–97, 85–87), and at a 4% budget for 27.

Where the cut falls matters. The probabilities $|\psi_0|^2$ of this sector come in exactly degenerate shells of 6, 12 or 24 determinants, and which members of a shell a truncation keeps is a convention. All the window cuts, and 19 of the 21 truncated grid points, fall inside a shell. Our convention is deterministic (descending $|\psi_0|^2$, ties broken by sector index), but

verdicts near the windows depend on it. At $d = 98$ it gives $|\widehat{m}_1 - \bar{m}_1| = 0.284 > \tau_1$, while 300 random choices of the 8 kept members of the 24-fold shell give 0.172–0.295 and rejection by m_1 alone in 57% of cases. At the ten cuts that fall on a shell boundary ($d = 6, 18, 30, 42, 66, 90, 114, 138, 150, 174$), and so do not depend on the convention, the noise-free residual of m_1 alone exceeds τ_1 at the 2% budget.

Detection sets in at much higher coverage here than in Fig. 2, whose idealized full-state and shared-state curves cross their 5% line near 7% and 18% retained determinants ($d \approx 1.6$–4.4×10^4 of 2.4×10^5; Fig. 2). This reflects a different test, not a sharper one: those curves score the large $L = 12$ sector by a *relative* discrepancy $|\widehat{m}_1 - \bar{m}_1|/\widehat{m}_1$ against a fixed line, this Monte Carlo the small $L = 6$ sector by an *absolute* threshold set by the shot budget. The coverage percentages thus come from different rulers on different-sized sectors (and, at $L = 12$, two reference states), not from one test giving several answers.

The direction of the residual uncertainty matters for a test that can only reject. Underestimating δ_k shrinks τ_k: safe against missed errors (false negatives, acceptable for a necessary condition) but anti-conservative against false rejections (false positives, the costly error for such a test). The false-positive rate above relies on $\bar{m}_1$ being exact at full support ($|\widehat{m}_1 - \bar{m}_1| \sim 10^{-15}$), so that the null hypothesis holds exactly there. For m_2 it does not: the 186-determinant support contains all of φ_S but not $H\varphi_S$, so the reconstruction on it is wrong at second order, $\Delta_2 = 0.62$ ($0.48\,\tau_2$ at the shot-only threshold, $0.34\,\tau_2$ at the default one), the boundary-leakage term above. An m_2 rejection of it would be a correct detection, not a false positive; the Monte Carlo tests m_1 only. Away from full coverage the threshold should be set on $\mathrm{Var}(\widehat{m}_k - \bar{m}_k)$, for which $\mathrm{Var}(O_k)$ is a proxy valid only when the reference is trusted (same-sample statistics above; Sec. 7).

The second moment catches some lone-m_1 misses, not all. In the noise-free scan at the 2% budget, m_2 rejects 3 of the 17 truncations that m_1 alone passes ($d = 97, 96, 85$), and m_0 rejects two more ($d = 94$ and 159, residuals 0.044 and 0.037 against $\tau_0 = 0.029$). The clearest is $d = 97$ (52% coverage): in the deterministic order $|\widehat{m}_1 - \bar{m}_1| = 0.186 < \tau_1 = 0.229$ but $|\widehat{m}_2 - \bar{m}_2| = 2.73 > \tau_2 = 1.85$, whatever the tie convention (m_1 alone rejects in none and the battery in all of 300 random shell choices). With exact moments the full (m_0, m_1, m_2) battery nonetheless passes the remaining 12 truncated reconstructions at the 2% budget: $d = 183$–185 (98–99.5% coverage), 160–165 (86–89%), and 86, 87, 95 (46–51%). Some of these passes are tie-convention effects (the battery rejects in 44%, 40% and 11% of random shell choices at $d = 87$, 160 and 95), but not those at $d = 86$ (1.3%), 161–165 (at most 0.3%) or 183–185 (none). Without a bias budget the battery catches the mid-coverage windows but still passes $d = 161$–164, 184 and 185; with a 4% budget it passes 22 truncations. The joint battery is thus a partial remedy for the single-moment windows: with finite thresholds no battery of necessary conditions is free of blind spots, and at the default order this one has blind spots that we can list. The shared-state estimator passes a different set, 16 near-complete truncations (Sec. 5.3).

In this demonstration the discrepancy is measured against the exact moments, which serve as an oracle: it calibrates the interval where classical ground truth exists, and a device-side version that propagates measured rather than assumed bias remains to be done.

C Blinded test: protocol details

Decision rule. The sealed rule rejects when any $|\widehat{m}_k - \bar{m}_k| > \tau_k$, $k \le 2$ (the thresholds of Sec. 2.2, $z = 1.96$), when the interval-Hankel condition on the estimates fails, or when the m_1 interval lies entirely below zero (tolerance 10^{-6}). The implementation evaluates $\mathrm{Var}(O_k)$ with H shifted by the exact E_0 (Appendix B).

Deviations from the sealed protocol. Three departures of the implementation from the sealed text, and one post hoc analysis, qualify the result. (i) The sealed catalog calls the spurious feature a Gaussian atom and gives no width; the generator adds a point atom. (ii) The sealed rule applies the interval-Hankel condition and the m_1-sign clause to the estimates. The classifier evaluated both on the reconstruction moments, as positivity of the 2×2 Hankel matrix H_1 of $\bar{m}_0, \bar{m}_1, \bar{m}_2$ and as $\bar{m}_1 \geq 0$, which cannot fail on a catalog of positive measures on $\omega \geq 0$ (smallest eigenvalue of H_1 over the 300 instances -3.6×10^{-15}, rounding error). That branch rejected 0/300 by construction: every rejection came from the moment comparisons, and the sealed battery acted as the three comparisons alone. (iii) The sealed shot model sets the noise standard deviation to z times the local-estimator standard error; the generator draws it with the standard error itself, $\sqrt{\mathrm{Var}(O_k)/N_s}$, the shot term inside τ_k; the reported 1/112 rests on this scale, and the rescoring at the sealed scale ($\widehat{m}_k - m_k \to 1.96\,(\widehat{m}_k - m_k)$, threshold unchanged) is the right-hand column of Table 3. (iv) The shared-state rescoring of Sec. 4 was not pre-registered and covers the 56 truncations. The sealed files were not modified, and the reported endpoint is the sealed one.

Shared-state rescoring. The rebuild of the sealed truncations depends on the stored eigensolver tie order inside degenerate $|\psi_0|^2$ shells. The rescoring uses $\widehat{m}_k = \langle 0_S | J^\dagger (H - e_0)^k J | 0_S \rangle$ plus the sealed noise draw and variances; before noise, $\widehat{m}_0 - \bar{m}_0 = \widehat{m}_0(1 - w)$ is exactly the weight of $J|0_S\rangle$ outside $\mathcal{S}$, and the m_0 comparison alone rejects all 56. The m_1, m_2 thresholds reuse the sealed variances, which are centered on E_0 rather than e_0; this leaves the m_0 verdicts unchanged. The oracle-centered sealed estimator rejected on m_0 in 55 of the 56; the exception, id 162 ($r_0 = 0.62$, rejected only off-diagonally), is an artifact of the oracle-centered estimate and has $r_0 = 15.8$ under the shared-state estimator; its shared-state estimate $\widehat{m}_0 = 1.04$ is far from the exact $m_0 = 0.37$.

Bookkeeping of Fig. 4. The 26 instances with $r_0 < 1$ but an off-diagonal residual above threshold are the 20 m_0-preserving spurious features, 4 weight-changing artifacts whose weight shift fell below τ_0, truncation id 162 and false positive id 184. The other 14 preserving and 23 changing artifacts form the legend entry "other spurious" (37). Reading the 20/34 m_0-preserving catches as a capability is post hoc; low detection was pre-registered. Theorem 1 does not apply to these artifacts: they add a pole, leaving its fixed-pole class.

D Illustrations: single-particle and collective channels

The figures of this appendix apply the construction to three more probes with injected corruptions; no threshold is calibrated for them.

Figure 7 shows $A(k, \omega)$ of the half-filled Hubbard ring, a one-dimensional correlated insulator with Hubbard bands separated by the Mott gap [127–129]. The single-particle gap approaches U only in the atomic limit $t \to 0$, and at the size ($L = 12$) and broadening of the figure the spinon and holon branches are only partly resolved. An injected corruption flattens the band-centroid dispersion by moving weight between the Hubbard bands at each momentum, leaving the Mott gap and the per-k total weight intact. For this probe m_0 is pinned to 1, so the first informative moment is m_1 [49,54]: from the anticommutator sum rule for the non-Hermitian probe $c_{k\sigma}$, measured from $U/2$, it is the Hartree centroid $\epsilon_k + U(\langle n_{-\sigma} \rangle - \frac{1}{2}) = \epsilon_k$ at half filling (Appendix A), and it exposes the flattening (right panel). The corruption moves the centroid by a fixed fraction of ϵ_k and the moment is closed-form, so the residual is set by construction and no threshold is calibrated for this probe: the panel illustrates the check rather than testing it.

Figure 8 shows the collective response: the charge structure factor $S(q, \omega)$, gapped by a particle–hole onset Δ_c, and the spin structure factor $S^{zz}(q, \omega)$, whose gapless spinon contin-

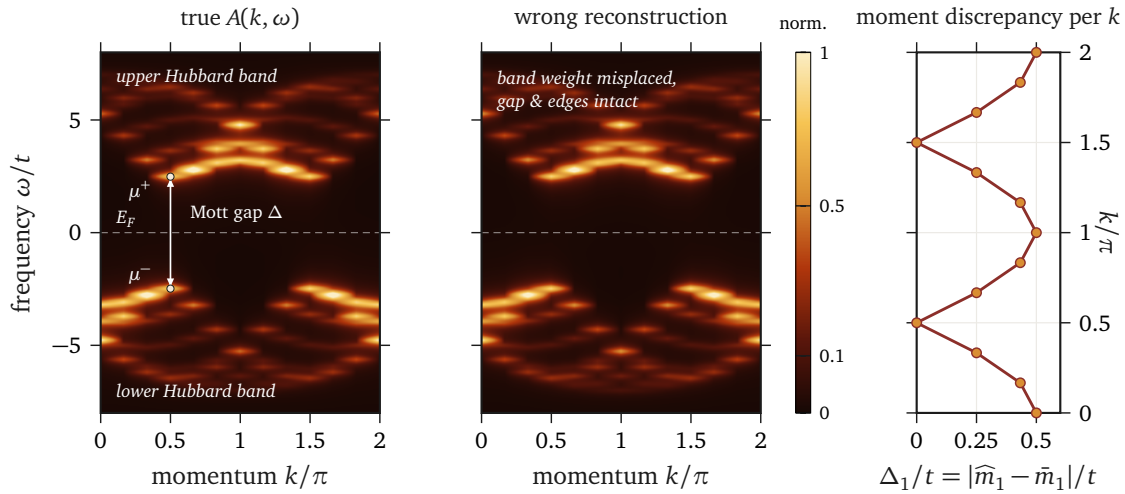


Figure 7: **First-moment check on the single-particle spectral function $A(k,\omega)$ of the Hubbard ring.** Left: exact $A(k,\omega)$ at half filling, frequencies measured from $E_F = U/2$ (dashed); the band edges $\mu^{\pm} = \pm 2.484\,t$ at $k_F = \pi/2$ (dots, arrow) give the Mott gap $\Delta = \mu^+ - \mu^- = 4.97\,t$. Center: an injected corruption rebalances intensity between the Hubbard bands so that the band centroid becomes $(1-\alpha)\epsilon_k$, $\alpha = 0.25$. The gap stays free of poles (only the Lorentzian tail reaches E_F), $\mu^{\pm}$ are unchanged and $m_0 = 1$ per k, so a total-weight check accepts it. Right: $\Delta_1 = |\widehat{m}_1 - \bar{m}_1|$, with $\widehat{m}_1(k) = \langle\{[c_{k\sigma}, H], c^\dagger_{k\sigma}\}\rangle$ the closed-form Hartree centroid, not a sampled estimate; $\Delta_1 = \alpha|\epsilon_k|$ by construction ($0.50\,t$ at $k = 0, \pi$; zero at $k = \pi/2, 3\pi/2$). $L = 12$ (12 momenta; the $k = 2\pi$ column repeats $k = 0$), $U/t = 8$, $\eta = 0.16\,t$; color $\propto (A/A_c)^{0.8}$, saturated at A_c, the 99.6th percentile (colorbar ticked at true A/A_c); Lanczos with Haydock Green's functions [66, 68].

uum [130] signals spin–charge separation [5]. The onset Δ_c is a finite-size quantity distinct from the single-particle Mott gap ($\approx 4.97\,t$); both are set by U and coincide only in the thermodynamic limit. The charge (spin) data, on the scale $\sim U$ ($\sim J_{\rm ex}$, with $J_{\rm ex} = 4t^2/U$ the exchange constant), are computed to $11\,t$ ($2.6\,t$), displayed to $10\,t$ ($2.4\,t$) and linearly interpolated in q. Each channel is a positive $T = 0$ Lehmann measure with its own moment sum rules [46], so the test applies to each; we apply it at $L = 6$ (Fig. 9), not to these $L = 12$ spectra. Such spectra are standard output of classical many-body methods [55, 131–135]; at these sizes we compute them exactly.

On the collective channels (Fig. 9), for each probe $J \in \{\rho_q, S^z_q\}$, with $\rho_q = \sum_j e^{-iqj} n_j$ and $S^z_q = \sum_j e^{-iqj} S^z_j$, both moments are ground-state expectations, $m_0(q) = \langle J^\dagger J\rangle_c$ and $m_1(q) = \langle 0|J^\dagger(H - E_0)J|0\rangle = \frac{1}{2}\langle[J^\dagger,[H,J]]\rangle$. On a half-filled $L = 6$ ring at $U/t = 8$ the operator moments reproduce the exact Lehmann moments of $S(q,\omega)$ and $S^{zz}(q,\omega)$ to better than 10^{-14} at every momentum, and the two forms of m_1, equal on the exact ground state (Appendix A), agree to the same precision. A reconstruction that moves band weight into a spurious low-frequency peak at fixed total weight passes the m_0 check (residual $\sim 10^{-15}$), but its first moment is off by 29–36% once 40% of the band weight is moved, for both observables at $q/\pi = 1/3, 2/3, 1$; this is read against a fixed 5% guide, not a calibrated threshold. The mean excitation frequency m_1/m_0 registers the two energy scales, set by U for charge ($\approx 8.5\,t$ at $q = 2\pi/3$) and by $J_{\rm ex} = 4t^2/U$ for spin ($\approx 1.0\,t$). These are exact simulations at $L = 6$, not the $L = 12$ spectra of Fig. 8, and not a hardware result.

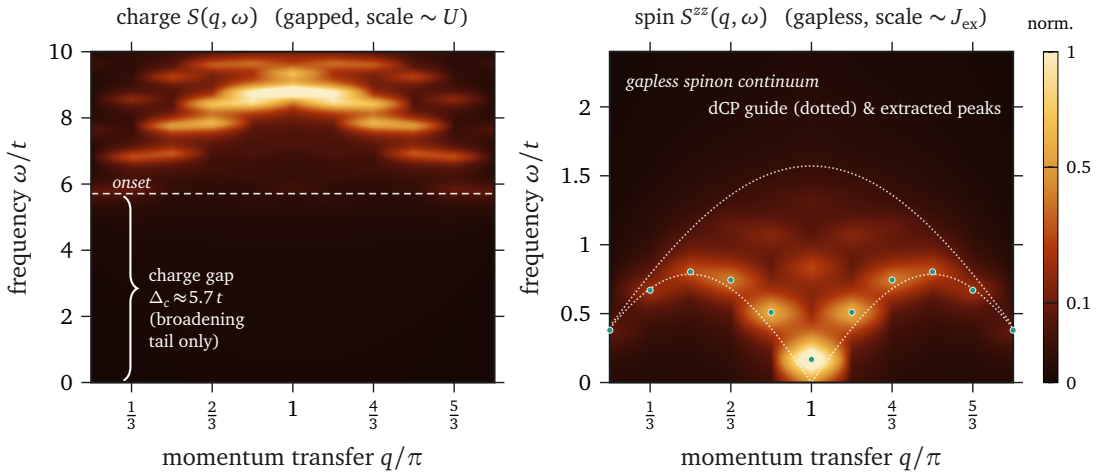


Figure 8: **Charge and spin structure factors of the half-filled Hubbard ring.** Left: $S(q,\omega)$ is gapped by a particle–hole onset $\Delta_c \approx 5.7\,t$ (dashed: lowest frequency above 5% of the peak intensity); below it only the broadening tail appears. Right: $S^{zz}(q,\omega)$ carries gapless spinon weight. White dotted: des Cloizeaux–Pearson (dCP) two-spinon boundaries $\omega_L = (\pi J_{ex}/2)|\sin q|$ [136] and $\omega_U = \pi J_{ex}|\sin(q/2)|$ [137], $J_{ex} = 4t^2/U = 0.5\,t$, an ideal large-U guide, not a strict bound (finite U, size and broadening put some weight below ω_L). Teal points: dominant spinon peak at $q/\pi \in \{1/6,\ldots,11/6\}$, within $0.17\,t$ of ω_L (largest offset at $q=\pi$). No moment test is applied here; Fig. 9 applies it at $L=6$. $L=12$, $U/t=8$, $\eta = 0.20\,t$ (charge), $0.12\,t$ (spin); color $\propto (S/S_c)^{0.62}$, saturated at S_c, the 99.5th percentile of each panel, with one colorbar for both panels, ticked at true S/S_c; Lanczos, Haydock evaluation [66, 68].

E Moment-body conditions and the semidefinite layer

The moment vector of every positive measure lies in the moment body: its Hankel matrices, and its localizing matrices on any interval known to contain the support, are positive semidefinite [59, 60]. A violated Hankel condition on the estimated moments rejects the estimate without reference to $A(\omega)$ (Fig. 10); it cannot reject a non-negative reconstruction. In the blinded test this branch was applied to the reconstruction moments, where it cannot fail, and rejected none of the 300 instances; it has not been applied to estimated moments anywhere in this paper.

Wang *et al.* turn the same machinery into certified two-sided bounds on many-body ground-state properties [77], and Mortimer *et al.* extend it to finite measurement statistics [78]. Applied to the operators O_k on the ground state, these semidefinite programs (SDPs) would bound the true moments, $m_k \in [\ell_k, u_k]$; this is possible in principle but not done in the cited works or here. Membership of the read-off moments $\bar{m}_k$ in such intervals would be a second necessary condition, of the same logical type as ours; their conjunction is at least as strict as either alone and is not a composition with a sufficient condition.

The classical moment-body conditions (not the Wang/Mortimer programs) illustrate such bounds at exact moments on the $L=6$, $U/t=8$ current measure of Appendix F (outside the blinded catalog's $U/t \in \{3,4,5,6\}$), normalized to $m_0 = 1$ (true $m_2 = 125.4$). Positivity gives $m_2 \geq m_1^2/m_0 = 123.3$. For a support contained in $[a,b]$, the Hausdorff condition $\int (\omega-a)(b-\omega)\,d\mu \geq 0$ gives $m_2 \leq (a+b)m_1 - ab\,m_0$: 147.6 for the weighted support $[2.49, 13.92]\,t$, which is oracle knowledge requiring the exact spectrum; a priori, 243.3 for the excitation range $[0, 21.91]\,t$ of the probe's four-electron, $S_z = 0$ sector and 579.8 for the

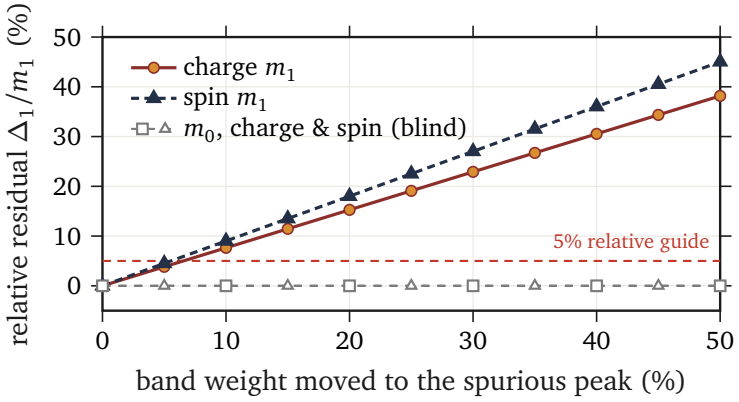


Figure 9: **Moment test on the charge and spin channels.** The comparison of Fig. 3 applied to $S(q,\omega)$ and $S^{zz}(q,\omega)$ of a half-filled Hubbard ring ($L=6$, $U/t=8$; exact simulation). A wrong reconstruction moves band weight into a spurious peak at $\omega = 2.0\,t$ (inside the charge gap) or $0.10\,t$ (spin), conserving m_0 exactly. At $q/\pi = 2/3$ the total-weight residuals (grey; squares charge, triangles spin, both $\sim 10^{-15}$) stay blind, while the relative first-moment residual of the charge (ρ_q, dark red) and spin (S^z_q, dark blue) probes grows linearly and crosses a fixed 5% guide, not a calibrated τ_1. Operator and exact Lehmann moments agree to better than 10^{-14} at every $q/\pi \in \{1/3, 2/3, 1\}$. $L=12$ spectra: Fig. 8.

full Fock range $[0, 52.21]\,t$ (573.7 with the lower edge at the smallest positive Fock excitation, $0.148\,t$). An under-dispersed moment vector ($m_2 = 122.1$) violates the positivity bound; an over-dispersed one ($m_2 = 148.8$) violates only the oracle Hausdorff bound and passes both a priori bounds. Neither condition has been applied to device-estimated moments.

The two layers could meet on integrated window weight: over all positive measures that share given moments, the window mass $\mu([c,d]) = \int_c^d A\,d\omega$ ranges over an interval, given in closed form at exact moments by Proposition 2 (the Markov–Krein envelope, of width at most $W_n(c) + W_n(d)$), and an SDP fed with interval-valued moments [78] would give a finite-statistics counterpart against which a reconstruction's window mass could be checked. We do not implement this. Nor do we attempt frequency-resolved weights, a limitation rather than a stylistic choice: a weight $|\langle n|J|0\rangle|^2$ at a resolved frequency is an excited-state matrix element, outside the ground-state moment hierarchy that makes these SDPs tractable; the resolved line shape almost certainly remains beyond this machinery.

F The Christoffel bracket and the window bound: statements, proof and evaluation

This appendix states what a *passing* test implies for cumulative and window weight. The bound below is classical (moments and orthogonal polynomials); our contribution is its evaluation as a diagnostic for a reconstructed spectrum, not the mathematics, and the Christoffel–Darboux kernel is an established data-analysis tool [52, 63, 138, 139]. Let $F(x) = \int_{-\infty}^{x} A(\omega)\,d\omega$ be the cumulative weight of a diagonal $T=0$ Lehmann measure whose moments $m_0, \ldots, m_{2n}$ are known exactly. Here x is a frequency; Fig. 11 plots it after an affine map onto $[-1,1]$, which does not change W_n (Appendix F), and the hopping t remains the energy unit. The moments fix the orthonormal polynomials $p_0, \ldots, p_n$ via the Hankel system [59], and the Christoffel function is $W_n(x) \equiv \lambda_n(x) = 1/K_n(x,x) = 1/\sum_{j=0}^{n} p_j(x)^2$, with $K_n(x,y) = \sum_{j=0}^{n} p_j(x)p_j(y)$

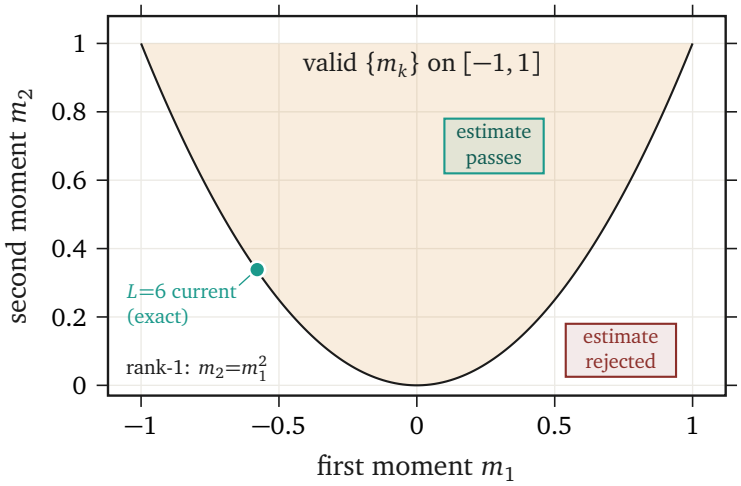


Figure 10: **Hankel-positivity region in the (m_1, m_2) plane.** With $m_0 = 1$ and frequencies rescaled to $x \in [-1, 1]$ as in Fig. 11 (Appendix F), moments lie in the shaded body $m_1^2 \le m_2 \le 1$, bounded below by the Hankel (rank-one) edge of a single δ and above by the interval (Hausdorff) localizing condition. The shifted (Stieltjes) condition refers to the unrescaled $\omega \ge 0$ frame and is not drawn. Schematic shot-budget boxes: the dark-red one ("estimate rejected"), wholly Hankel-negative, would reject the estimate without reference to $A(\omega)$; the teal one ("estimate passes") would pass. A non-negative reconstruction's moments always lie inside; the check was not applied to estimated moments here. Teal dot: exact moments of the doped $L = 6$, $U/t = 8$ current measure; one pole carries 78% of its weight, and the dot lies just above the rank-one edge, at variance $m_2 - m_1^2 = 0.003$ in this frame [59, 60].

the Christoffel–Darboux kernel (construction from the moments in Appendix F) [52, 63].

Theorem 2 (Cumulative-weight miss-distance bound; classical, re-derived)**.** *Let $A(\omega)$ be a diagonal $T = 0$ Lehmann spectrum, so that $d\mu = A(\omega)\, d\omega$ is a positive measure; let it have compact support and a positive-definite Hankel matrix $H_n = [m_{i+j}]_{i,j=0}^{n}$, and suppose the moments $m_0, \ldots, m_{2n}$ are known exactly. For $x \in \mathbb{R}$ define the cumulative-weight miss-distance as the extremal spread $\sup |F_{\nu_1}(x) - F_{\nu_2}(x)|$ over all positive measures ν_1, ν_2 on $\mathbb{R}$ sharing those moments. Then for every $x \in \mathbb{R}$ this spread equals the Christoffel function $W_n(x)$, which is also the supremum of the mass a moment-matching measure can place at x (the Gauss–Radau weight $\lambda_n(x)$, attained except at the n zeros of the orthonormal polynomial p_n). In particular, any non-negative reconstruction (a positive measure, as every Lehmann reconstruction is) whose moments $m_0, \ldots, m_{2n}$ equal the true ones obeys $|F_{\mathrm{rec}}(x) - F_{\mathrm{true}}(x)| \le W_n(x)$ at every $x \in \mathbb{R}$ (Appendix F) [99, 142].*

Remarks. No convex-hull hypothesis is needed: for $H_n \succ 0$ the Gauss–Radau rule with a node at x has positive weights for every real x except the n zeros of p_n, and its atom at x is $W_n(x)$ (Appendix F). The bound is monotone, $W_{n+1} \le W_n$. For N distinct poles, $H_n \succ 0$ only for $n \le N - 1$, and at a pole $W_n(\omega_i) \ge w_i > 0$ (the residue), with equality at $n = N - 1$: W_n does not vanish on the support. In the thermodynamic limit, where the discrete spectrum densifies, $W_n(x) \to \mu(\{x\})$, which vanishes away from atoms for any compactly supported measure [59, 65]; the $1/n$ rate, $n\, W_n(x) \to A(x)/\omega_S(x)$, is established for Stahl–Totik regular measures obeying a local Szegő condition (Appendix F) [140, 141].

The equality is the classical content of the Chebyshev–Markov–Stieltjes inequalities and Gauss–Christoffel moment theory [52, 59, 60]. We say "bound", not "certify": the test never certifies that $A(\omega)$ is correct. The bounded object is cumulative weight, not the pointwise value of $A(\omega)$ or a peak position [52], and the contraction *rate* is established under that regu-

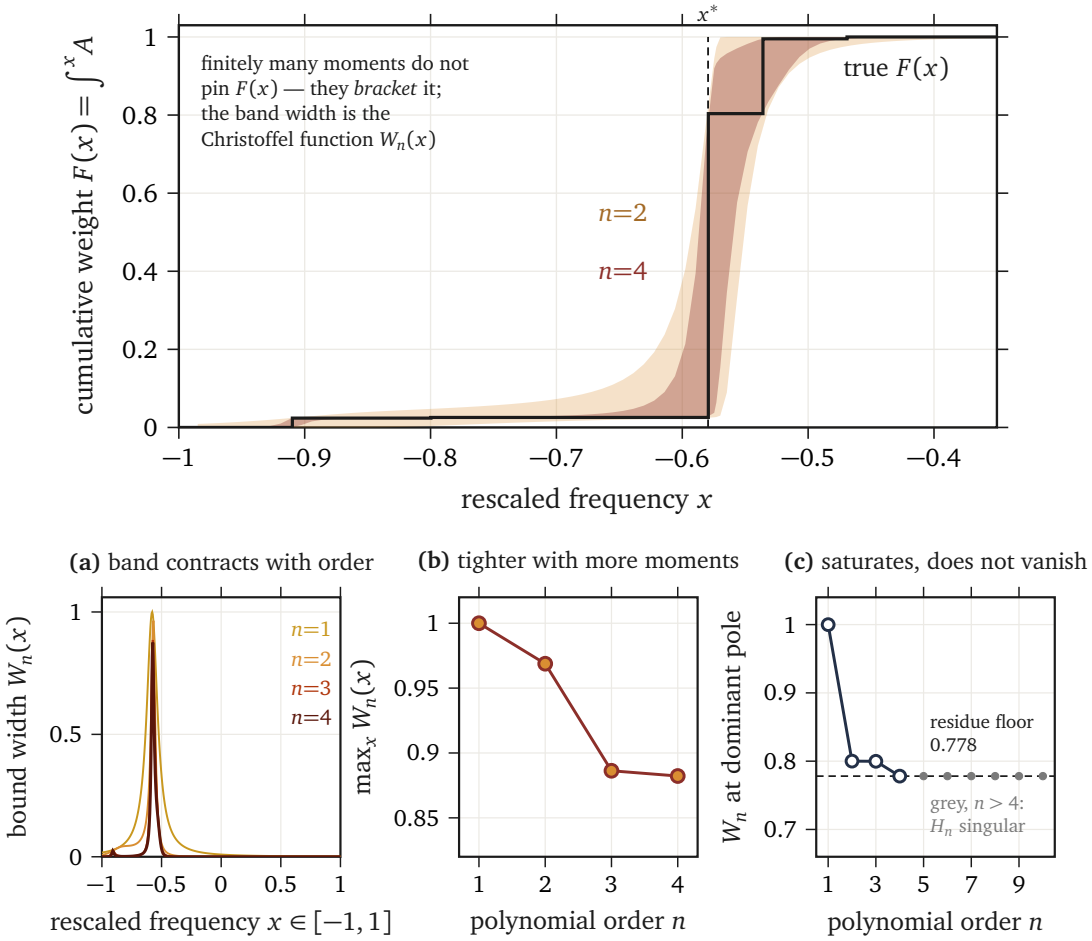


Figure 11: **The Christoffel miss-distance bound at exact moments.** *Top.* The moments $m_0, \ldots, m_{2n}$ *bracket* the cumulative weight $F(x)$ between extremal Gauss–Radau measures, in a band of Christoffel width $W_n(x)$; the n=4 band (dark red) nests in the n=2 band (amber); x^* (dashed) marks the dominant atom. **(a)** $W_n(x)$ contracts as n grows. **(b)** $\max_x W_n = 1.000, 0.969, 0.886, 0.882$ for $n = 1, \ldots, 4$; at the default order $n = 1$ it is $W_1 = m_0$, at the mean: the bracket is vacuous. **(c)** At the dominant atom ($\omega = 11.09\,t$, residue 0.778) W_n reaches the residue (dashed) at $n = N - 1 = 4$, the last order with $H_n \succ 0$ (five weighted atoms); grey points ($n > 4$, H_n singular) lie outside the theorem's hypothesis. All curves assume exact moments; a reconstruction passing within tolerances is bracketed less tightly. Doped $L = 6$, $U/t = 8$ Hubbard current measure, $m_0 = 1$, frequency rescaled to $x \in [-1, 1]$ (Appendix F) [52, 59, 60, 63, 140, 141].

larity [140, 141]. The top panel of Fig. 11 shows the mechanism: the true $F(x)$ lies between the extremal Gauss–Radau envelopes, a bracket of width $W_n(x)$.

The theorem assumes exact moments; the default test passes a reconstruction whose residuals lie within tolerances, $|\Delta_k| \leq \tau_k$, and only through order $2n = 2$. We evaluate the consequences on the measure of Fig. 11, the doped $L = 6$, $U/t = 8$ current measure. It lies outside the blinded catalog ($U/t \in \{3, 4, 5, 6\}$, Sec. 4) and is not the $U/t = 4$ measure of Appendix G; we use it as an illustrative measure (the $U/t = 4$ check below gives the same conclusion), and the thresholds used below are those the default rule would give on this state. There are two consequences. First, at $n = 1$ the exact-moment bracket is the one-sided Chebyshev (Cantelli) bound [99], $W_1(\omega) = m_0\,\sigma^2/[\sigma^2 + (\omega - \bar{\omega})^2]$ with mean $\bar{\omega} = 11.10\,t$ and

variance $\sigma^2 = 2.10\,t^2$; its maximum m_0 lies at the mean, $0.01\,t$ from the dominant atom ($11.09\,t$), so the bracket is vacuous at the peak, and just above the top pole, at $\omega = 14\,t$, it is still 0.20. Second, tolerances loosen it further: the largest atom that a positive measure with $|m_k - m_k^{\text{true}}| \le \tau_k$ ($k \le 2$, the default threshold rule evaluated on this state) can place at ω is 3.2–7.7 times W_1 at $\omega = 14$–$20\,t$ (0.63 against $W_1 = 0.20$ at $14\,t$), 10–11 times W_1 far outside the band ($\omega \approx 34$–$50\,t$), and more than m_0 at the mean ($1.045\,m_0$). On the $U/t = 4$ measure of Appendix G the same computation gives a similar inflation, 3.6–5.7 times W_1 at $\omega = 12$–$20\,t$, above its top pole at $10.66\,t$. At exact-moment orders $n \le 4$ ($m_0, \ldots, m_8$) the global maximum falls only to $\max_x W_4 = 0.882$, and at $n = N - 1 = 4$, W_4 at the dominant atom equals its residue 0.778, below which no order can go (panel c; Appendix F). No positive-order moment beyond m_2 is estimated in this work. At the default order, passing the test therefore places no constraint on the cumulative weight at the peak. The companion shows by example that exact moments through order $2K + 1$ coexist with an $O(1)$ reconstruction error [25]; the Christoffel function makes this distribution-free at the default order, for every measure that passes the test.

Refutation and bracketing are separate statements. Rejection ($|\Delta_k| > \tau_k$ for some k) does not involve W_n, and its reach is set by the thresholds (Sec. 5); a Hankel/Stieltjes violation of the *estimated* moments rejects the estimate, not a non-negative reconstruction. W_n governs only how tightly a *passing* reconstruction's cumulative weight is bracketed. The looseness above therefore limits what passing implies; it does not limit the power to reject.

At a frequency the measure leaves empty, however, $W_n(x)$ bounds the weight any moment-matching reconstruction can place there; far above the band ($\omega \approx 34$–$50\,t$), $W_1 = 1.4$–4×10^{-3} and the higher orders are far smaller (Appendix F). These small values reflect the distance from the band, not resolution: at $n = 1$ they are the Cantelli tail. Nearer the band the bracket is wider ($W_1 = 0.20$ at $\omega = 14\,t$), and the default tolerances raise the far-out-of-band $n = 1$ values about tenfold. This is a computed property of this measure, not a theorem, and we do not extrapolate it to the two-sided single-particle Mott gap, a distinct measure not tested here.

Proposition 2 (Two-sided Markov–Krein window-mass bound). *Let $d\mu = A(\omega)\,d\omega$ be a positive measure (a diagonal $T = 0$ Lehmann spectrum) with $H_n = [m_{i+j}]_{i,j=0}^{n} \succ 0$, and suppose $m_0, \ldots, m_{2n}$ are known exactly. For real x with $p_n(x) \neq 0$, let $F_-(x)$ be the mass that the $(n+1)$-point Gauss–Radau rule with a node fixed at x places strictly left of x, and let $F_+(x) = F_-(x) + W_n(x)$, where $\lambda_n(x) = W_n(x)$ is that rule's atom at x; at the n zeros of p_n, where the rule does not exist, both are taken as limits. These are the lower and upper ends of the Chebyshev–Markov–Stieltjes bracket on $F(x)$ (the ordinate-gap Lemma 3 of Appendix F). For any frequency window $[c, d]$ with $c \le d$ real, the mass $\mu([c,d]) = \int_c^d A(\omega)\,d\omega$ of* every *positive measure sharing those moments obeys*

$$\max\{0,\, F_-(d) - F_+(c)\} \;\le\; \mu([c,d]) \;\le\; F_+(d) - F_-(c) \;\le\; m_0. \tag{F.1}$$

The bracket is two-sided; its upper end never exceeds m_0, since $F_+(d) \le m_0$ and $F_-(c) \ge 0$. Before the lower end is clipped at 0 its width is exactly $W_n(c) + W_n(d)$; after it, the width is still at least $\max\{W_n(c), W_n(d)\}$, because $F_\pm$ are non-decreasing. The bracket is therefore tight exactly when W_n is small at both window edges, and it inherits the monotone contraction $W_{n+1} \le W_n$. The sharp bounds, the infimum and supremum of $\mu([c,d])$ over all positive measures sharing $m_0, \ldots, m_{2n}$ (the Markov–Krein extremal problem [61]), are no wider. Equation (F.1) follows directly from the Chebyshev–Markov–Stieltjes inequalities and extends the point bound on the cumulative weight, $|F_{\text{rec}} - F_{\text{true}}| \le W_n$, to a two-sided bound on integrated weight [52, 61, 99, 142].

Evaluated on the same $L = 6$, $U/t = 8$ current measure, again at exact moments (Appendix F, evaluation of the Markov–Krein window bound), the bracket (F.1) is narrow at

$n = 4$ for a window whose two edges both lie where this measure has no weight: a peak-containing window gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744; at the default order $n = 1$ the same window gives the wider $[0.848, 1]$. It is as loose as W_n when an edge sits on the peak, and the default tolerances loosen it further. The narrowness at weight-free edges is a computed property of this measure, on which W_n is small at those edges; it is not part of Proposition 2. In general W_n need not be small where a measure has no weight: $W_1 = m_0$ at the spectral mean, which for two equal atoms is the empty midpoint between them. Optimal moment bounds on a broadened line shape go back to Gordon [71]; the bound here is also the real-axis, ground-state-moment counterpart of the Euclidean-time two-sided spectral bounds obtained from the moment problem analytically, via Kovalishina's solution [74], or by semidefinite programming [75, 76]; its moments are ground-state operator expectations, not lattice correlators. As with the one-sided bound, we do not extrapolate it to the two-sided single-particle Mott gap.

Proof of Theorem 2

We re-derive the equality of the extremal cumulative-weight spread with the Christoffel function from three classical lemmas [52, 60, 61]; nothing in this derivation is new. Here x is a real frequency point in any affine frame (Fig. 11 uses the rescaled frame $x \in [-1, 1]$); the hopping t remains the energy unit. Write $\mathcal{P}_n$ for the real polynomials of degree at most n, $K_n(x, y) = \sum_{j=0}^{n} p_j(x) p_j(y)$ for the Christoffel–Darboux kernel and $\lambda_n(x) = 1/K_n(x, x)$ for the Christoffel function, and assume the moment vector is interior to the moment cone, $H_n = [m_{i+j}]_{i,j=0}^{n} \succ 0$ (for a finite N-atom Lehmann spectrum this holds for $n \le N - 1$). The measures ν compared below are positive measures on $\mathbb{R}$. Restricting them to the Lehmann half-line $\omega \ge 0$ can only narrow the bracket, so W_n remains an upper bound there.

Lemma 1 (Variational identity). *$\lambda_n(x) = \min\{\int |P|^2 \, d\mu : P \in \mathcal{P}_n,\ P(x) = 1\}$, with minimizer $P^\star = K_n(\cdot, x)/K_n(x, x)$.*

Proof. Expanding $P = \sum_{j=0}^{n} c_j p_j$ gives $\int |P|^2 \, d\mu = \sum_j c_j^2$, while by Cauchy–Schwarz

$$1 = P(x)^2 = \Big(\sum_{j=0}^{n} c_j p_j(x) \Big)^2 \le \Big(\sum_{j=0}^{n} c_j^2 \Big) K_n(x, x),$$

with equality at $c_j = p_j(x)/K_n(x, x)$; the minimum is $1/K_n(x, x) = \lambda_n(x)$. □

Lemma 2 (Maximal atom). *Among positive measures ν with $\int \omega^k \, d\nu = m_k$ for $k \le 2n$, the supremum of the mass at a point x is $\sup_\nu \nu(\{x\}) = \lambda_n(x)$. It is attained if $p_n(x) \ne 0$, and approached but not attained at the n zeros of p_n.*

Proof. The minimizer $P^\star$ of Lemma 1 has $P^\star(x) = 1$, so

$$\nu(\{x\}) = P^\star(x)^2 \, \nu(\{x\}) \le \int (P^\star)^2 \, d\nu = \int (P^\star)^2 \, d\mu = \lambda_n(x),$$

the middle equality using $\deg(P^\star)^2 = 2n$ and that ν shares those moments. If $p_n(x) \ne 0$ the bound is attained by the $(n+1)$-point Gauss–Radau rule with a node prescribed at x: a positive measure matching $m_0, \dots, m_{2n}$ whose weight at x is $\lambda_n(x)$. At the n zeros of p_n this rule does not exist (one node escapes to infinity), and the supremum is approached but not attained. □

Lemma 3 (Ordinate gap). *Let $p_n(x) \ne 0$, and let $\xi_0 < \cdots < \xi_n$ be the nodes of the Gauss–Radau rule of Lemma 2, one of them equal to x, with weights $\lambda_n(\xi_i) > 0$. For every positive ν sharing*

$m_0, \dots, m_{2n}$, with $F_\nu(x) = \nu((-\infty, x])$, the Chebyshev–Markov–Stieltjes inequalities [60, 61, 99, 142] hold:

$$\sum_{\xi_i < x} \lambda_n(\xi_i) \;\le\; F_\nu(x^-) \;\le\; F_\nu(x) \;\le\; \sum_{\xi_i \le x} \lambda_n(\xi_i).$$

Consequently $\sup_{\nu_1,\nu_2} |F_{\nu_1}(x) - F_{\nu_2}(x)| = \lambda_n(x)$, and the same supremum holds at the zeros of p_n, so it holds for every real x.

Proof. The inequalities follow from polynomials $Q_\pm \in \mathcal{P}_{2n}$ with $Q_- \le \mathbf{1}_{(-\infty,x)}$ and $\mathbf{1}_{(-\infty,x]} \le Q_+$ that coincide with these indicators at the nodes, so that $\int Q_\pm \, d\nu = \int Q_\pm \, d\mu$ equal the two sums. The bracket has width $\lambda_n(x)$. Its upper end is attained by the Radau rule itself; its lower end is approached by Radau rules whose prescribed node x' decreases to x. At the zeros of p_n the same supremum follows from the continuity of λ_n and the right-continuity of F. □

Combining the three lemmas, the extremal spread equals $\lambda_n(x) = W_n(x)$ for every real x, and any single reconstruction consistent with the moments obeys $|F_{\mathrm{rec}}(x) - F_{\mathrm{true}}(x)| \le W_n(x)$. No convex-hull hypothesis is needed: the Radau rule has positive weights for every x with $p_n(x) \neq 0$, inside or outside the convex hull of the support. We checked this on the $U/t = 8$ current measure of Fig. 11: 24 Radau rules ($n = 1, \dots, 4$ at six values of x, 20 of them outside the convex hull of the weighted support) all have positive weights, place mass $W_n(x)$ at x to a relative error of 2×10^{-16}, and match $m_0, \dots, m_{2n}$ to 1.4×10^{-17}. Monotonicity $W_{n+1} \le W_n$ follows from $K_{n+1} = K_n + p_{n+1}^2 \ge K_n$.

For any compactly supported measure, $W_n(x) \to \mu(\{x\})$ as $n \to \infty$ [59, 65], so in the thermodynamic limit, where the discrete spectrum densifies, W_n vanishes away from atoms; regularity is needed only for the rate. If the discrete spectrum densifies to a measure that is Stahl–Totik regular with compact support S and absolutely continuous with a density A obeying the local Szegő condition $\int_I \log A > -\infty$ on an interval $I \subset S$, then $n\,W_n(x) \to A(x)/\omega_S(x)$ for almost every $x \in I$, with ω_S the equilibrium density of S [140]; on a single interval this reduces to Máté–Nevai–Totik [141].

Evaluation of the bound from the moments. The Christoffel function is computed from the moments $\{m_k\}_{k=0}^{2n}$ by a short pipeline [52, 64]. With the inner product $\langle f, g\rangle_\mu = \int f(\omega) g(\omega)\, d\mu(\omega)$, so that $\langle \omega^a, \omega^b\rangle_\mu = m_{a+b}$, Gram–Schmidt on $\{1, \omega, \dots, \omega^n\}$ gives the orthonormal polynomials, with $p_0 = 1/\sqrt{m_0}$; they obey the three-term recurrence

$$\beta_{j+1}\, p_{j+1}(\omega) = (\omega - \alpha_j)\, p_j(\omega) - \beta_j\, p_{j-1}(\omega), \tag{F.2}$$

with $\alpha_j = \langle \omega p_j, p_j\rangle_\mu$ and $\beta_j = \langle \omega p_j, p_{j-1}\rangle_\mu$, all fixed by $\{m_k\}$. The map from power moments to (α_j, β_j) becomes ill-conditioned as n grows, so high orders need high-precision arithmetic. The nodes and weights of the $(n+1)$-point Gauss rule are the eigenpairs of the Jacobi matrix $\mathbf{J}_{n+1} = \mathrm{tridiag}(\beta_j, \alpha_j, \beta_j)$ (Golub–Welsch), and $K_n(x,x) = \sum_{j=0}^{n} p_j(x)^2$, evaluated by the same recurrence, gives $W_n(x) = 1/K_n(x,x)$. Every quantity in the bound is thus an explicit function of the moments, readable at any frequency.

We evaluate it on the $L = 6$, $U/t = 8$ current measure of Fig. 11, normalized to $m_0 = 1$; five of its atoms carry weight. (Its ground level is a spin triplet; the current measure is the same for all three states, because J is a spin scalar.) This measure lies outside the blinded catalog ($U/t \in \{3, 4, 5, 6\}$, Sec. 4) and differs from the $U/t = 4$ spectrum of Appendix G; we use it as an illustrative measure, and the in-catalog $U/t = 4$ measure gives the same tolerance inflation (below). The numbers below come from `src/export_christoffel_dat.py` and `src/christoffel_tolerance_lp.py` (`data/2026-09-27_theory_numerics.json`). We run the recurrence in 60-digit arithmetic on the atoms and weights of the measure, which fix the same coefficients as its moments, and locate the maxima by root-finding: $\max_x W_n = 1,\ 0.969,\ 0.886,\ 0.882$ for $n = 1, \dots, 4$. The $n = 1$ maximum equals m_0 and lies at the mean frequency $\bar{\omega} = 11.10\, t$, the zero of p_1, so at the default order (m_0, m_1, m_2) the

bracket is vacuous at the spectral peak. At the dominant atom ($\omega = 11.09\,t$, residue 0.778), W_n reaches the residue at $n = N-1 = 4$, the highest order at which H_n is positive definite, and for $n \le 4$ it cannot go lower, because Lemma 2 with $\nu = \mu$ gives $W_n(x) \ge \mu(\{x\})$: within the range of the theorem no number of moments narrows the bracket at the peak below 0.778. For $n > 4$, H_n is singular and the theorem's hypothesis fails.

The rescaled abscissa $x \in [-1,1]$ of Fig. 11 is the affine image of the frequency range $\omega \in [0.148,\, 52.21]\,t$, the positive excitation energies of the full Fock space, zero-weight states included. The weighted support $[2.49,\, 13.92]\,t$ maps to $[-0.910, -0.471]$, and 73.5% of $[-1,1]$ lies above the last weighted pole. The affine map moves points along the axis but does not change W_n, which is a mass. The rescaled points $x = 0.3,\, 0.6,\, 0.9$ correspond to $\omega = 34.0,\, 41.8,\, 49.6\,t$, far above the band. There $W_1 = 4.0\times10^{-3},\, 2.2\times10^{-3},\, 1.4\times10^{-3}$, and at $x = 0.6$ the higher orders give $W_2 = 1.4\times10^{-5}$, $W_3 = 2.7\times10^{-8}$, $W_4 = 3.6\times10^{-10}$. For $n = 1$ these small values are Cantelli's one-sided Chebyshev bound, $W_1(\omega) = m_0\sigma^2/[\sigma^2 + (\omega-\bar\omega)^2]$ with σ^2 the variance of the normalized measure (it agrees with the orthogonal-polynomial evaluation to 1.6×10^{-15}). They reflect the distance from the mean, not a feature of this spectrum. Just above the top pole, at $\omega = 14\,t$, $W_1 = 0.20$.

All of this assumes exact moments. If the moments are matched only within the default tolerances, $|\Delta_k| \le \tau_k$ for $k \le 2$, the largest atom a positive measure can place at ω is a linear program. With thresholds from the default rule applied to this state ($\tau_0 = 0.051$, $\tau_1 = 0.57$, $\tau_2 = 6.5$ before normalization; an illustration, since the rule was calibrated only on the catalog), it is $3.2\,W_1$ (0.63 of m_0) at $\omega = 14\,t$, $7.7\,W_1$ at $20\,t$, and 10.1–$10.7\,W_1$ at the rescaled points $x = 0.3$–0.9. On the $U/t = 4$ measure of Appendix G, which is in the catalog, the same construction gives 3.6–$5.7\,W_1$ at $\omega = 12$–$20\,t$, so the loosening is not specific to $U/t = 8$. These τ_k use the exact-state local-estimator shot variance of Appendix B, which is expected to underestimate a realistic per-shot variance; wider tolerances would inflate the atom further. At the default order a passing reconstruction is therefore bracketed much more loosely than W_1 alone suggests.

Evaluation of the Markov–Krein window bound (Proposition 2). On the same $L = 6$, $U/t = 8$ current measure (committed script `src/markov_krein_window.py`), at exact moments: at $n = 4$ $(m_0, \ldots, m_8)$, a window with one edge at the 400-point-grid maximizer of W_4, just above the dominant atom, gives $\mu \in [0.070, 0.945]$, width 0.875; the exact maximum of W_4, at a nearby frequency, is 0.882. A window that contains the peak with both edges in weight-free frequency (rescaled $[-0.75, -0.35]$) gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744, and the empty window rescaled $[0.3, 0.9]$ ($\omega \approx 34$–$50\,t$) gives $\mu \in [0, 3.2\times10^{-9}]$. The peak window is tight here because W_4 is small at both of its edges (before the lower end is clipped at 0 the width is $W_4(c) + W_4(d)$); weight-free edges alone do not ensure this, as $W_1 = 0.20$ at $14\,t$, just above the band, shows, so this is a property of this measure. At the default order $n = 1$ (m_0, m_1, m_2) the peak-containing window gives $\mu \in [0.848, 1]$ and the empty window $\mu \in [0, 0.004]$, while a window with an edge on the peak is uninformative (its bracket runs from 0 to within 2×10^{-4} of m_0). These are exact-moment brackets. Under the default tolerances the upper end for the empty window is at least 0.040, the largest atom that a measure with m_0, m_1, m_2 within tolerance can place at its lower edge ($\omega \approx 34\,t$).

G Within-sector redistributions and the default battery

This appendix proves Theorem 1 and evaluates it on one spectrum. The independent estimator detects weight that leaks *outside* the retained determinant support. A redistribution strictly *within* it contributes no leak term; it is visible only through the moment shift $(V\boldsymbol{\delta})_k$, and not at all if it preserves the moments used (Sec. 3). We treat that residual in three steps:

Table 5: Within-sector redistributions of the $L = 6$, $U/t = 4$ current spectrum ($N = 5$ distinct poles). Each row keeps $m_0, \ldots, m_K$ exactly and asks how far m_{K+1} can move. "LP range" is the range of the signed shift $\sum_n \delta_n \omega_n^{K+1}$ over all such redistributions with non-negative corrupted weights [Eq. (G.1)]; every range contains 0. The thresholds τ_{K+1} follow the default rule ($N_s = 5 \times 10^4$, an assumed 2% bias budget, $z_{0.975} = 1.96$; Appendix B); for m_3 and m_4, unused in the default battery, they are the values the same rule would give. When $\dim\ker V_{0:K+1} > 0$, some finite redistributions leave m_{K+1} exactly unchanged, so the guaranteed power at that order is zero and the maximum describes only the best case.

preserved exactly	next moment	LP range of shift [Eq. (G.1)]	τ_{K+1}	max shift/τ_{K+1}	$\dim\ker V_{0:K+1}$	in default battery?
m_0, m_1	m_2	$[-0.92, 7.09]$	1.848	3.8	2	yes
$m_0, \ldots, m_2$	m_3	$[-0.14, 5.26]$	14.98	0.35	1	no
$m_0, \ldots, m_3$	m_4	$[-0.30, 0.39]$	122.0	0.003	0	no

injectivity (part (a): in exact arithmetic, $m_0, \ldots, m_{N-1}$ fix the weights on N distinct poles); *conditioning* (part (b) and the Corollary: a sufficient and a necessary condition for exposing a redistribution at finite shots); and *reach* (a linear program and explicit counterexamples on the doped $L = 6$, $U/t = 4$ current spectrum, with thresholds from Appendix B; Table 5). The result at the default order is negative: the battery (m_0, m_1, m_2) has no guaranteed power against within-sector redistributions of this spectrum, although its poles are well separated. We characterize this residual; we do not close it. As in Theorem 1, K denotes the highest order a redistribution preserves; the default battery tests orders 0, 1 and 2. Unless stated otherwise, the numbers below are in `data/2026-09-27_theory_numerics.json` (script `src/within_sector_lp.py`).

Injectivity (proof of Theorem 1(a)). A $T = 0$ Lehmann current spectrum on a finite sector is a discrete measure on N distinct poles $\{\omega_n\}$,

$$A(\omega) = \sum_{n=1}^{N} w_n \, \delta(\omega - \omega_n), \qquad w_n = \sum_{a:\, E_a - E_0 = \omega_n} |\langle a|J|0\rangle|^2 \geq 0,$$

where w_n is the total weight of the eigenstates at ω_n. Its moments are $m_k = \sum_n w_n \omega_n^k$, i.e. $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$ a Vandermonde matrix. Because the poles are distinct, the block $V_{0:N-1}$ of rows $0, \ldots, N-1$ is invertible, so the weight-to-moment map is injective through order $N-1$: a redistribution $\mathbf{w} \to \mathbf{w} + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \neq 0$ cannot preserve all of $m_0, \ldots, m_{N-1}$. A redistribution that preserves $m_0, \ldots, m_K$ ($K < N-1$) zeroes the first $K+1$ entries of $V_{0:N-1}\boldsymbol{\delta}$, so $\|(\Delta_{K+1}, \ldots, \Delta_{N-1})\|_2 = \|V_{0:N-1}\boldsymbol{\delta}\|_2 \geq \sigma_{\min}(V_{0:N-1})\|\boldsymbol{\delta}\|_2$; the largest of these $N-1-K$ shifts is at least this norm divided by $\sqrt{N-1-K}$, which is Eq. (5). If the poles may also move, an N-atom measure has $2N$ parameters and is fixed by $m_0, \ldots, m_{2N-1}$ among measures with at most N atoms (Prony), and by $m_0, \ldots, m_{2N}$ among all positive measures, since its Hankel matrix H_N is singular [59, 100]. Weight traded among degenerate eigenstates at a single frequency changes no moment and leaves $A(\omega)$ unchanged; it is not an error.

For the $L = 6$, $U/t = 4$ spectrum the sixteen eigenstates with nonzero weight fall on $N = 5$ distinct frequencies, $\omega_n/t = 2.73, 5.57, 7.83, 8.70, 10.66$, with weights 0.040, 0.004, 0.393, 0.116, 0.001 ($m_0 = 0.554$; the pole at $7.83\,t$ carries 71% of it). The map is therefore injective through m_4: a redistribution among these poles that preserves $m_0, \ldots, m_K$ changes some m_j with $K < j \leq 4$ (at most m_9 if the poles also move and the altered spectrum has at most five poles; m_{10} against any positive measure). In exact arithmetic the residual is thus resolved by m_4. That is two orders beyond the default battery, and it says nothing about detection at finite

shots. Equation (5) is valid here but loose: for the redistribution CE1 below, its right-hand side is 3.8×10^{-3}, while the larger of the shifts in m_3 and m_4 is 85. The sufficient condition of Theorem 1(b)(i) is far from met here: for CE1 and CE2 the right-hand side of (5) is below 4×10^{-3}, against $\max_j \tau_j = \tau_4 = 122$ in the same units.

Conditioning, and when detection is possible (proof of Theorem 1(b)). Invertibility of $V_{0:N-1}$ holds in exact arithmetic. At finite shots a shift counts only if it exceeds the threshold τ_j. Two bounds, in opposite directions, turn detectability into a statement about the size of $\boldsymbol{\delta}$ and the geometry of the poles it touches.

Sufficient condition. By Eq. (5), a redistribution preserving $m_0,\dots,m_K$ is exposed (its noise-free shift exceeds the threshold) at some order $j \le N-1$ if

$$\frac{\sigma_{\min}(V_{0:N-1})\,\|\boldsymbol{\delta}\|_2}{\sqrt{N-1-K}} > \max_{K<j\le N-1} \tau_j .$$

Since $\|\boldsymbol{\delta}\|_1/\sqrt{N} \le \|\boldsymbol{\delta}\|_2 \le \|\boldsymbol{\delta}\|_1$, this is a condition on $\rho = \|\boldsymbol{\delta}\|_1$ up to a factor $\sqrt{N}$. It is informative only when $\sigma_{\min}$ is not small, and it uses moments up to order $N-1$.

Necessary condition. Let $s \ge 2$, and let $\boldsymbol{\delta}$ preserve $m_0,\dots,m_{s-2}$ and be supported on poles inside an interval of diameter D and midpoint ω_c. (Among s poles, preserving $m_0,\dots,m_{s-2}$ leaves a one-parameter family, and m_{s-1} is the first moment it changes.) Since $\sum_n \delta_n\, q(\omega_n) = 0$ for every polynomial q of degree at most $s-2$, the signed shift is $\sum_n \delta_n\,\omega_n^{s-1} = \sum_n \delta_n\, q(\omega_n)$ for every monic q of degree $s-1$. The choice $q(\omega) = 2^{2-s}(D/2)^{s-1}\,T_{s-1}\big(2(\omega-\omega_c)/D\big)$, the monic Chebyshev polynomial of the interval, has modulus at most $2(D/4)^{s-1}$ on it, hence

$$|\Delta_{s-1}| \le 2\rho\,(D/4)^{s-1}.$$

The bound is sharp: weights of alternating sign on the s Chebyshev extrema of the interval, with the discrete orthogonality weights that preserve $m_0,\dots,m_{s-2}$, attain it. The cruder choice $q = (\omega-\omega_c)^{s-1}$ gives $\rho\,(D/2)^{s-1}$, larger by 2^{s-2}. Subtracting from ω^j its Taylor polynomial of degree $s-2$ at ω_c gives, for every $j \ge s-1$, the Lagrange-remainder bound

$$|\Delta_j| \le \rho \binom{j}{s-1} \big(|\omega_c| + D/2\big)^{j-s+1} (D/2)^{s-1};$$

expanding instead $\omega^j = \sum_i \binom{j}{i}\omega_c^{j-i}(\omega-\omega_c)^i$, whose terms with $i \le s-2$ are combinations of preserved moments and vanish, and bounding the rest by $|\omega_n-\omega_c| \le D/2$ gives the term-by-term form of Theorem 1(b)(ii). Either way every shift vanishes as D^{s-1}. Exposure at order $s-1$ therefore requires $2\rho\,(D/4)^{s-1} > \tau_{s-1}$. With $\tau_{s-1} = z_{1-\alpha/2}\,[\mathrm{Var}(O_{s-1})/N_s + b_{s-1}^2]^{1/2}$ this reads

$$N_s > \frac{z_{1-\alpha/2}^2\,\mathrm{Var}(O_{s-1})}{4\rho^2(D/4)^{2(s-1)} - z_{1-\alpha/2}^2\, b_{s-1}^2},$$

and no budget exposes it at order $s-1$ once $2\rho\,(D/4)^{s-1} \le z_{1-\alpha/2}\, b_{s-1}$. The shot cost of detecting a redistribution of fixed size inside a collapsing cluster thus grows at least as $D^{-2(s-1)}$. Numerically, the identity holds to a relative error of 1.5×10^{-12}, and the bound holds in 11 structured cases (CE1 and CE2 below, the six optima of Eq. (G.1) and three committed random cases) and in 20 000 random null-space redistributions on random pole subsets (seed 20260927, no violation). The largest ratio $|\Delta_{s-1}|/[2\rho(D/4)^{s-1}]$ is 1.000, 0.998 and 0.924 at $s = 2,3,4$ (against the cruder $\rho(D/2)^{s-1}$: 1.000, 0.499 and 0.231), the sharp form also holds in every case, and alternating weights on the Chebyshev extrema attain it to a ratio of 1.000000. If a cluster contracts uniformly, so that its minimal pole separation falls from $\Delta\omega_{\min}$ to $\Delta\omega$, D shrinks by the same factor and the lower bound (6) on the shot cost grows by at least $(\Delta\omega_{\min}/\Delta\omega)^{2(s-1)}$. A finite separation below which no budget suffices exists only with a bias floor, $b_{s-1} > 0$.

Conditioning numbers. The same collapse appears in $\sigma_{\min}$: for s nodes of fixed relative geometry and diameter $D \to 0$, $\sigma_{\min}$ of the $s \times s$ Vandermonde block scales as D^{s-1}. The sharp super-resolution results for this regime [101–103], like the separation threshold of Ref. [104], are proved for Fourier (unit-circle) nodes; for real nodes and power moments, worst-case error bounds of matching order for a near-colliding cluster are given in Ref. [105]. For the fixed-pole case needed here both directions follow from elementary bounds on the $s \times s$ block V of the cluster. Row n of V^{-1} holds the coefficients of the Lagrange polynomial $\prod_{j\neq n}(\omega-\omega_j)/(\omega_n-\omega_j)$, which gives Gautschi's bound [106, 107] and hence

$$\|V^{-1}\|_\infty \le \max_n \prod_{j\neq n} \frac{1+|\omega_j|}{|\omega_n-\omega_j|}, \qquad \sigma_{\min}(V) \ge \Big[\sqrt{s}\,\max_n \prod_{j\neq n} \frac{1+|\omega_j|}{|\omega_n-\omega_j|}\Big]^{-1},$$

a lower bound that scales as D^{s-1} at fixed relative geometry. Conversely, the redistribution that spans the null space of rows $0,\dots,s-2$ has $\|V\boldsymbol{\delta}\|_2 = |\Delta_{s-1}| \le \rho\,(D/2)^{s-1} \le \sqrt{s}\,\|\boldsymbol{\delta}\|_2\,(D/2)^{s-1}$, so $\sigma_{\min}(V) \le \sqrt{s}\,(D/2)^{s-1}$. A direct check with $s = 2,3,4$ equally spaced nodes near $\omega = 7.83\,t$ (spacing 0.02–0.2 t) gives fitted exponents 0.995, 1.98 and 2.96 (`data/2026-08-24_within_sector_control.json`). Condition numbers depend on the frequency unit, so we quote them only on nodes mapped affinely onto $[-1,1]$ (outermost poles at ±1): for the five poles above, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$. With a minimal separation of $0.87\,t$ these poles are well separated, yet good conditioning does not give the default battery guaranteed power: that is set by the dimension of a null space (next paragraph).

Reach of the default battery (instantiation of Theorem 1(b) and Corollary). The zero-power clause of part (b) is immediate: a vector in the null space of rows $0,\dots,K+1$ of V, scaled so that $w_n + \delta_n \ge 0$ (possible because every $w_n > 0$), preserves $m_0,\dots,m_{K+1}$; every smaller rescaling does too, so the smallest shift at order $K+1$ is zero. The default battery uses m_0, m_1, m_2 with the thresholds of Appendix B, $\tau_0 = 0.0287$, $\tau_1 = 0.229$ and $\tau_2 = 1.848$, at $N_s = 5\times10^4$ with an assumed, unmeasured 2% bias budget $b_k = 0.02\,m_k$; each τ_k includes the factor $z_{0.975} = 1.96$ and is not a 1σ noise floor. The shot term uses the exact-state local-estimator variance, which is expected to underestimate the true per-shot variance, so realistic thresholds would be wider. For a redistribution that preserves $m_0,\dots,m_K$, the largest shift it can impose on the next moment is the linear program

$$\begin{aligned} \Delta_{K+1}^{\max} = \max_{\boldsymbol{\delta}} \Big|\textstyle\sum_n \delta_n\,\omega_n^{K+1}\Big| \quad \text{s.t.} \\ \textstyle\sum_n \delta_n\,\omega_n^{j} = 0\ (j \le K), \quad \delta_n \ge -w_n, \end{aligned} \tag{G.1}$$

the constraint $\delta_n \ge -w_n$ keeping the corrupted weights non-negative; together with $\sum_n \delta_n = 0$ it also bounds $\|\boldsymbol{\delta}\|_1 \le 2m_0$. Extremal problems of this kind over the moment space are classical [61, 99]; here they are evaluated on one measured spectrum.

Table 5 gives the full range of the signed shift. A redistribution that preserves m_0, m_1 can move m_2 by up to $7.09 = 3.8\,\tau_2$, so some such redistributions are exposed at m_2. The guaranteed power, however, is set by the smallest shift a redistribution of a given size can produce, and that is zero. Rows $0,1,2$ of V have a null space of dimension $N-3=2$: some redistributions preserve m_0, m_1 and m_2 exactly, and scaling one of them down gives an exactly invisible redistribution of every smaller size. Three explicit cases, all positive measures that pass the default battery: (i) CE1 preserves m_0, m_1, m_2 exactly and moves 72% of the weight ($\|\boldsymbol{\delta}\|_1/2 = 0.72\,m_0$); it empties two of the five poles and lowers the dominant residue from 0.393 to 0.037. (ii) CE2 preserves m_0, m_1 exactly and shifts m_2 by exactly τ_2, so it sits on the decision boundary and passes only because the rule rejects on the strict inequality $\Delta_k > \tau_k$; it moves 86% of the weight and lowers the dominant residue to 0.073. (iii) CE3 keeps

$|\Delta_k| \le \tau_k$ for $k \le 2$, has $\|\boldsymbol{\delta}\|_1 = 2.03\, m_0$ (above $2m_0$ because m_0 itself moves by τ_0), and empties three poles, the dominant one included. The random $K = 1$ case committed in `src/within_sector_control.py` (seed 0) moves 25% of the weight ($\|\boldsymbol{\delta}\|_1/2 = 0.25\, m_0$; its output file lists $\|\boldsymbol{\delta}\|_1/m_0 = 0.49$ as the moved fraction) and shifts m_2 by $0.044\,\tau_2$; it passes as well. The guaranteed power of the default battery against within-sector redistributions of this spectrum is therefore zero. The reason is the dimension of the null space, not conditioning, and the same holds for any spectrum with four or more distinct poles.

Adding m_3 and m_4 would remove the null space (rows $0, \dots, 4$ have $\dim\ker = 0$), but at this shot budget the extended battery would still miss a redistribution that moves 72% of the weight. Under the same threshold rule, $\tau_3 = 14.98$ and $\tau_4 = 122.0$. Among redistributions that preserve m_0, m_1, m_2 exactly, the largest m_3 shift is $5.26 = 0.35\,\tau_3$ (Table 5), and CE1, like every scaled-down copy of it, stays below both thresholds ($0.23\,\tau_3$ and $0.70\,\tau_4$). Some such redistributions would be caught at m_4: the one that maximizes the m_3 shift moves m_4 by $145 = 1.19\,\tau_4$, a marginal case, since the variance behind τ_4 is expected to underestimate a realistic one. CE2 and CE3, which use the tolerance on m_2, would be caught ($|\Delta_3| = 3.3\,\tau_3$; $|\Delta_4| = 7.1$ and $7.2\,\tau_4$).

The joint battery does catch some errors that m_1 alone misses, but the example is a truncation, not a fixed-pole redistribution: truncating the determinant basis of the same system to the $d = 97$ determinants of largest $|\psi_0|^2$ (52% of the 186-determinant support; Appendix B) gives a reconstruction that m_1 alone passes and m_2 rejects. The cut falls inside a 24-fold degenerate shell, so the gaps depend on how ties are ordered. With the deterministic `lexsort` order used by both the smoke test `src/verify.py` and `src/interval_battery.py` (descending $|\psi_0|^2$, ties by ascending determinant index), $|\bar{m}_1 - \widehat{m}_1| = 0.186 < \tau_1$ and $|\bar{m}_2 - \widehat{m}_2| = 2.729 > \tau_2$; and for all 300 random tie-breaks within the shell m_1 alone rejects none and the battery rejects all (`data/2026-09-27_interval_battery.json`). This is one catch, not a guarantee: in the exhaustive scan of Appendix B ($d = 1, \dots, 185$, same thresholds, `lexsort` order) m_1 alone passes 17 truncated reconstructions, m_2 rejects three of them ($d = 97$, 96, 85) and m_0 two more ($d = 94$, 159), so with exact moments the full (m_0, m_1, m_2)+Hankel battery passes 12 ($d = 86$, 87, 95, 160–165 and 183–185). Under random tie-breaks within the degenerate shells, ten of these pass in at least 89% of 300 draws, and $d = 87$ and $d = 160$ in 56% and 60%.

Positivity cannot help against within-sector redistributions: a redistribution with nonnegative weights is itself a positive measure, so the Hankel and Stieltjes conditions hold by construction. Only independent moments of high enough order, estimated precisely enough, can expose it, and at the default order the guaranteed power is zero. We report this residual; we do not claim to close it.

H Complementary state-level check: the Gauss law

The Gauss-law check mentioned in Secs. 2.2 and 7 is not a moment test; we record its construction and scope here. In the U(1) lattice Schwinger model the generators G_x of gauge transformations commute with the Hamiltonian, $[H, G_x] = 0$, and every physical state satisfies $G_x|\psi\rangle = 0$ at every site, so $\langle \sum_x G_x^2 \rangle = 0$ exactly. The operator $\sum_x G_x^2$ is diagonal in the computational basis, so on hardware it could be estimated from computational-basis samples of the prepared state, with no reconstruction and no moment. It tests the prepared state, not a reconstructed spectrum: it responds to leakage out of the physical sector, an error class the moment test does not target.

Figure 12 illustrates the identity on a small instance: the quantum-link form of the model with $N = 4$ staggered matter sites and 3 spin-$\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$

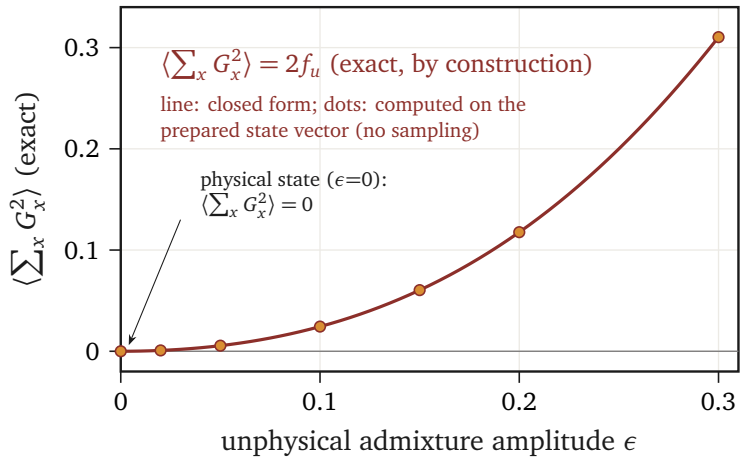


Figure 12: **A state-level check: the Gauss-law identity.** In the U(1) lattice Schwinger model physical states obey $G_x|\psi\rangle = 0$, so $\langle\sum_x G_x^2\rangle = 0$; $[H, G_x] = 0$ keeps the dynamics in that sector. Plotted: $|\psi(\epsilon)\rangle \propto (1-\epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ the lowest-energy charge-neutral state with the smallest violation, $\sum_x G_x^2 = 2$. The amplitude ϵ gives unphysical weight $f_u = \epsilon^2/[(1-\epsilon)^2+\epsilon^2]$, only 0.155 at $\epsilon = 0.3$. Line: the closed form $2f_u(\epsilon)$, which $\langle\sum_x G_x^2\rangle$ equals by construction; dots: values computed exactly on the state vector (no sampling is simulated). It does not measure detection power. The check tests the prepared state, not a reconstructed spectrum, and targets gauge-symmetry breaking. $N = 4$ staggered sites, 3 spin-$\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$.

($[H, G_x] = 0$ to machine precision). The plotted state is $|\psi(\epsilon)\rangle \propto (1-\epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ an unphysical state. The parameter ϵ is an amplitude, not a weight: the unphysical weight is $f_u(\epsilon) = \epsilon^2/[(1-\epsilon)^2 + \epsilon^2]$, 0.059 at $\epsilon = 0.2$ and 0.155 at $\epsilon = 0.3$. Because $\sum_x G_x^2|\psi_0\rangle = 0$ and $\langle\psi_0|\phi_u\rangle = 0$, the violation is $\langle\sum_x G_x^2\rangle = g_u f_u(\epsilon)$ with $g_u = \langle\phi_u|\sum_x G_x^2|\phi_u\rangle$. The curve is therefore fixed by construction: zero for the physical state, rising monotonically with ϵ, with its scale set by g_u alone. It illustrates the identity and does not measure detection power. The values are computed exactly on the state vector; no sampling is simulated.

The admixed state is chosen by an explicit rule: $|\phi_u\rangle$ is the lowest-energy eigenstate of H among charge-neutral states with the smallest nonzero violation, $\sum_x G_x^2 = 2$ (one site with $G = +1$ and one with $G = -1$; a 27-dimensional subspace; generator `src/export_gauss_state.py`). Every state in that subspace has $g_u = 2$, so the plotted curve is $2f_u(\epsilon)$. The plotted values follow from this rule (`data/2026-09-27_gauss_state.json`). Over the 124 unphysical eigenstates of this model g_u ranges from about 1 to 10, so the slope of the curve reflects this choice. Under an independent bit-flip channel on every qubit the violation is positive at every nonzero flip rate (0.25 at $p = 0.05$; 40 sampled flip patterns per rate, too few for a monotone curve: 3.05 at $p = 0.35$, 2.52 at $p = 0.40$; `data/2026-08-18_gausslaw_falsifier.json`).

Gauss-law violation has been measured in a 71-site cold-atom quantum simulator [96] and analyzed theoretically under gauge-breaking errors [143]. The lattice Schwinger model is also the model of self-verifying variational simulation on a trapped-ion analog quantum simulator [88] and of early digital simulations of Schwinger-model dynamics [144, 145].

I Device run: reference samplers, line shapes and complete supports

This appendix records the remaining reference comparisons and the line-shape comparisons of the $L = 8$ device run (Sec. 6), and the complete-support runs at $L = 6$. As in that section, Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic.

(iv) Noiseless sampling at the device's retained counts. This comparison depends on how the retained counts are allocated. Matched circuit by circuit, the comparator puts 46,216 of its 82,697 shots at 50k on the $t = 0$ circuit, which is a single determinant without noise; its mean Δ_0 is 0.224, 0.279, 0.343 and 0.443 at 50/30/16/4k. With the same total split equally over the seven circuits it is 0.150, 0.205, 0.273 and 0.404. Under both allocations the device lies below every replica at 50, 30 and 16k; at 4k it lies at the 0.6th (per circuit) or 5.8th (equal split) percentile. The margin shrinks with the budget: the reference mean exceeds the device value 51-, 8.6-, 2.8- and 1.19-fold per circuit, and 34-, 6.3-, 2.2- and 1.08-fold with the equal split.

(ii) Uniform noise (details). Mixing each circuit's output distribution P_k with a fraction p of uniformly random bitstrings, $(1-p)P_k + p/2^{16}$, before post-selection, changes the expected Δ_0 without being monotone in p: at every budget the exact $\mathbb{E}[\Delta_0](p)$ falls from $p = 0$ to a single minimum and rises again toward $p = 1$, with the minimum at $p \approx 0.93$, 0.88, 0.77 and 0.09 at 50/30/16/4k ($\mathbb{E}[\Delta_0] = 0.0020$, 0.017, 0.075, 0.269). At 50k it falls from 0.036 at $p = 0$ to 0.0056 at $p = 0.5$ and is 0.0024 at $p = 1$. At 4k it is flat to 0.3% up to $p = 0.25$ and then rises to 0.326 at $p = 1$, because post-selection keeps only 6% of uniform shots. At the p that reproduces the device's retention ($p \approx 0.81$), the mixture's mean (0.0023, 0.017, 0.075, 0.295) is below the device at every budget. The device nevertheless lies inside the mixture distribution at 50–16k (85.7th, 91.6th and 88.7th percentiles of 1000 replicas) and at its far upper edge only at 4k (99.9th; one replica lies higher).

Line shape. On the line shape (the relative L_1 error of Sec. 6), too, noiseless sampling (comparator (i)) does worse at the two higher budgets. Its reconstructions give 0.566 ± 0.032 at 50k (all 80 replicas above the device) and 0.630 ± 0.043 at 30k (device at the 1.25th percentile). At 16k and 4k the device is worse (96th percentile, and above all 80 replicas). The uniform in-sector sampler gives 0.014 at 16k and 0.90 at 4k, below the device at every budget.

Complete supports. The companion study ran its pipeline at $L = 6$ (`ibm_fez`, 12 qubits, 3.5×10^5 shots (5×10^4 on each of seven circuits; job `d9s16avpemts73ct6g8g`); $L = 6$ Hubbard model, $U/t = 4$, addition sector $4\uparrow, 3\downarrow$ (300 configurations), $\eta = 0.15\,t$ in its spectrum; measurement twirling, Pauli gate twirling and dynamical decoupling, no readout-error mitigation) [25]. Its retained support is the whole addition sector ($|\mathcal{S}| = 300/300$), so the reconstruction equals exact diagonalization by coverage. The bitstrings were post-selected after being read in reversed qubit order, which swaps the spin labels and moves the circuits' own output ($4\uparrow, 3\downarrow$, read as $3\uparrow, 4\downarrow$) into the discarded $S_z = -\frac{1}{2}$ sector, so all 300 retained determinants came from device errors [25]. Its raw counts are not deposited. The run shows only that the pipeline executes; no moment residual can be nonzero on a complete support. Two earlier $L = 6$ `ibm_fez` jobs of ours (2026-08-21, seven circuits of the same family at $U/t = 4$ and 3.5, 3.5×10^4 shots each; raw records deposited) likewise retain the whole 300-configuration sector under either bit order, so a reconstruction on it is exact diagonalization of the assumed Hamiltonian, and a moment test on it returns the same verdict with or without the device data. They enter no result.

J Cost of a device-native estimator

This appendix estimates the resources and the noise bias of measuring m_1, m_2 on hardware, and describes a diagonal alternative; none of it was run on a device.

The estimator also admits a device-native evaluation. We verify it in simulation because it settles how the off-diagonal (hopping) content of $\widehat{m}_1, \widehat{m}_2$ would be obtained on hardware. For a probe diagonal in the computational basis, the density probe $\rho_q = \sum_j \cos(qj)\, n_j$ at $q = \pi$, the moment operator is O_k of Appendix A with $J = \delta\rho_\pi \equiv \rho_\pi - \langle\rho_\pi\rangle$ and $e_0 = E_0$. It is a sum of Pauli strings, and $\widehat{m}_k = \sum_P c_P \langle P\rangle$, the expectation $\langle 0|O_k|0\rangle$ written as a linear combination of Pauli expectations, is estimated from counts. The m_0 term is a pure computational-basis estimator. The hopping terms in m_1, m_2 need rotated-basis measurements, one circuit per qubit-wise-commuting group of Pauli strings, the standard grouping for Pauli observables; classical shadows are a randomized alternative [86]. This path measures a prepared ground state in rotated bases, with circuits separate from those whose samples define $\mathcal{S}$: it shares the device with the reconstruction but not the samples.

We verify it by finite-shot sampling of the exact ground state of an open $L = 6$ Hubbard chain at $U/t = 4$ (12 qubits), with 5×10^4 shots in each of the 357 qubit-wise-commuting bases that (m_0, m_1, m_2) require (1.8×10^7 shots per repetition). Over 8 repetitions the mean estimates $(\widehat{m}_0, \widehat{m}_1, \widehat{m}_2) = (2.730, 10.55, 47.01)$ lie 0.87, 0.72 and 0.66 standard errors (spread over repetitions divided by $\sqrt{8}$) from the exact $(2.737, 10.61, 47.64)$; for the $L = 4$ chain (8 qubits, 126 bases) every moment is recovered within 0.06 standard errors. The number of bases is set by the Pauli support of O_k, not by the moment values, so it can be counted without the statevector once the probe shift $\langle\rho_\pi\rangle$ is known. That shift vanishes by the reflection symmetry of the open chain: for even L, reflection maps ρ_π to $-\rho_\pi$, so $\langle\rho_\pi\rangle = 0$ in a reflection-symmetric ground state. The ground states are not half filled (2, 4 and 5 electrons at $L = 4$, 6 and 8, where $\langle\rho_\pi\rangle = 0$ was checked numerically; it is assumed at $L = 10$ and 12). The committed script `src/estimator_scaling.py` counts the bases this way and reproduces the count from the full eigendecomposition where affordable ($L \le 8$). The full (m_0, m_1, m_2) battery needs $126, 357, 721, 1191, 1837$ bases at $L = 4, 6, 8, 10, 12$ (8 to 24 qubits); the (m_0, m_1) pair alone needs $19, 35, 51, 67, 83$, growing roughly linearly. The number of settings thus grows polynomially in this range, not exponentially. We do not turn this into a deployability claim: the counts-based recovery was validated only up to 12 qubits and the basis count against full diagonalization only up to $L = 8$; the shot budget and accuracy at 24 qubits remain to be characterized; and a device would have to prepare the ground state that the classical check takes as given.

The estimator, including its off-diagonal content, is therefore measurable from a fixed shot budget in principle; it has not been run on a device here, and without mitigation the forecast bias at the reference depth is a large fraction of the signal. A job script that builds the rotated-basis circuits and applies this estimator (`src/hardware_blind_job.py`) is in the repository; its state preparation is a placeholder, it has not been run on a quantum processor, and an off-diagonal test on hardware requires a calibrated mitigation of this bias or lower gate error.

For the bond-basis estimator of Fig. 13, no off-diagonal moment has been measured on a device in this work. Under global depolarizing noise of strength p the estimate of $m_1 = \langle 0|O_1|0\rangle$, with $O_1 = J^\dagger(H - E_0)J$, becomes $(1-p)\, m_1 + p\, \mathrm{Tr}(O_1)/\dim\mathcal{H}$, a bias of $p\,[\mathrm{Tr}(O_1)/\dim\mathcal{H} - m_1]$. For the $L = 8$ device Hamiltonian and $J = c^\dagger_{0\uparrow}$, $\mathrm{Tr}(O_1)/\dim\mathcal{H} = 6.80$ against $m_1 = 2.10$ (ratio 3.24), so the bias is upward. The forecast uses as a surrogate a bond-basis probe with nearly the same ratio, the $q = \pi$ density probe of the open $L = 6$ chain at $U/t = 4$ ($m_1 = 10.6$, $\mathrm{Tr}(O_1)/\dim\mathcal{H} = 34.3$, ratio 3.23), with no error mitigation modeled. An exact density-matrix simulation with local two-qubit depolarizing noise at $\varepsilon = 2.5 \times 10^{-3}$ puts its first-moment bias at 13% of the signal at 100 controlled-Z (CZ) gates

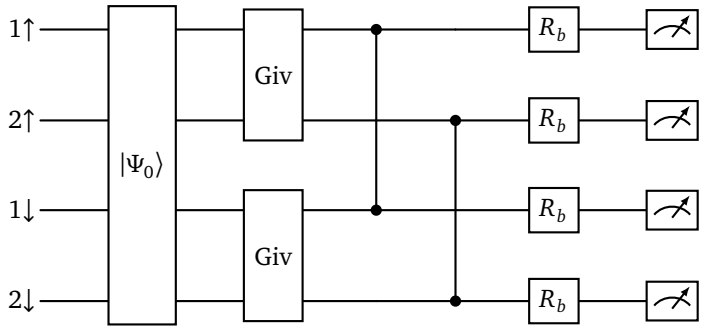


Figure 13: **Measurement circuit of the bond-basis moment estimator (schematic; not run on hardware).** One circuit, drawn for a two-site unit cell (4 qubits: two orbitals per spin). The first blocks stand for state preparation: reference state $|\Psi_0\rangle$, number-conserving hopping per spin species (Giv), on-site interaction $U\hat{n}_\uparrow\hat{n}_\downarrow$ (controlled up–down pairs). A per-qubit rotation R_b (H or $S^\dagger H$ for X or Y, identity for Z) precedes a computational-basis measurement; each qubit-wise-commuting Pauli group needs one circuit. The all-Z setting gives m_0; the hopping content of m_1, m_2 needs rotated settings. The verification above samples the exact ground state in each basis instead of running a preparation. Without mitigation the off-diagonal estimate would carry the bias forecast below.

and 38% at 295 CZ; global depolarizing noise at the same depths gives 49% and 116%, and a coherent-ZZ model 47% at 295 CZ. The local value is exact for a static noise model that omits the operator spreading of a real circuit, leakage and readout error, so it is expected to underestimate, but is not proven to bound, the bias of a real circuit. The depth of 295 CZ is the median transpiled CZ count of our $L = 8$ sampler circuits on `ibm_fez` (range 278–313; the 12-qubit circuits of the companion's family need 213, as deployed in our 2026-08-21 jobs and in a transpiler reproduction) and serves as a reference: the bond-basis estimator needs a ground-state preparation, whose depth is not established here. The global-depolarizing part of this bias is affine in p with a classically computable offset $\mathrm{Tr}(O_1)/\dim\mathcal{H}$, so noise renormalization removes it once p is calibrated [117, 118]; the local, coherent and leakage parts are not removed this way. Related quantities have been estimated on IBM hardware: Hamiltonian moments in circuits of up to about 500 CNOT gates [120] and, with error mitigation, the Liouvillian recursion behind the Green's functions of a four-site Hubbard model [15]. The particle-number post-selection that protects the diagonal channel does not survive the bond-basis rotations. A device-side off-diagonal test therefore requires a calibrated mitigation of this bias or lower gate error; no device datum here bears on off-diagonal content (Table 2).

The off-diagonal limitation concerns the bond-basis estimator of m_1, not every discriminating on-device channel. A diagonal one exists: the momentum distribution n_k, the occupied spectral weight at momentum k, with the sum rules

$$n_k = \int_{\omega<0} A(k,\omega)\,d\omega = \langle c^\dagger_{k\sigma} c_{k\sigma}\rangle, \qquad \int A(k,\omega)\,d\omega = \langle\{c_{k\sigma}, c^\dagger_{k\sigma}\}\rangle = 1,$$

with ω measured from a chemical potential in the gap. The total weight is pinned at every k and is blind to how the weight splits between the occupied and empty sides; n_k is not.

The n_k channel. For the momentum distribution n_k above: in the doped $L = 6$ ground state ($N_\uparrow = N_\downarrow = 2$: density 2/3 per site, i.e. 1/3 per spin orbital, the value a fully flattened n_k takes; committed script `src/nk_falsifier.py`) n_k has a finite-size step $\max_k n_k - \min_k n_k = 0.91$

at $U/t = 4$ and 0.80 at $U/t = 8$; in the thermodynamic limit the one-dimensional step softens to a Luttinger power-law cusp. A reconstruction that flattens the dispersion moves $\bar{n}_k$ toward this filling value, while the true n_k keeps the step. (At half filling the chain is a Mott insulator and n_k is analytic, so this discriminant targets doped sectors, not the half-filled spectrum of Fig. 7.) The occupation n_k is a number operator in the momentum basis, read out diagonally after a number-conserving fermionic Fourier transform (a Givens network of $L(L-1)/2$ rotations per spin sector). The particle-number and spin post-selection that protects the coverage channel therefore survives the basis change. The loss of post-selection and the full-space trace-to-value bias that affect the bond-basis rotations do not arise; depolarization acts within the sector instead, as described next. In an in-sector depolarizing model with post-selection at $\varepsilon = 2.5 \times 10^{-3}$, the ∼60-CZ transform adds a flattening bias of ≈9% of the step; this assumes all-to-all connectivity, before the fermionic-SWAP routing a real device needs, which adds gates, so we expect it to underestimate the bias on hardware. The device estimate $\widehat{n}_k$ is not immune to noise. Depolarization flattens it toward the same filling value that a dispersion corruption does, so it discriminates only corruptions whose flattening exceeds the device's own, and it inherits the depolarization of the state preparation. Readout error and coherent or leakage noise, which need not flatten toward filling, are not modeled, and post-selection costs shots.

The sealed n_k test. A pre-registered on-device n_k test (doped $L = 6$ ring, $U/t = 8$, 12 qubits) was sealed on 2026-09-02 (manifest SHA-256 `4962...ae80`). Its day-of rule runs the test only on a 12-qubit chain whose every qubit has calibrated $T_1 \geq 150\,\mu s$ and $T_2 \geq 100\,\mu s$, with no chain CZ error above twice the device median, and otherwise forfeits the window. Launched twice on 2026-09-03 against `ibm_fez`, it submitted nothing: the first launch stopped in chain selection on a qubit without a calibrated T_2, and the second forfeited (longest compliant chain 7 qubits; the T_1 and T_2 floors bind). Calibrations of `ibm_fez`, `ibm_marrakesh` and `ibm_kingston` re-queried through 2026-09-05 forfeit under the sealed rule as well, and a post-seal amendment with looser floors failed its pre-committed simulated gate and was not adopted. No n_k job was submitted (`data/2026-09-05_nk_device_attempt_record.json`, which also records that no job on the account was created after the seal; history in `docs/PROCESS_RECORD.md`).

K A code-scalability check beyond exact diagonalization

On an open Hubbard chain at $L = 24$, half filling and $U/t = 8$ ($S_z = 0$ sector of dimension $\approx 7.31 \times 10^{12}$), with the total current J as probe, $\mathcal{S}$ is taken from Born-rule samples of a TeNPy DMRG ground state [124]. The subspace ground state $|0_\mathcal{S}\rangle$ of $H_\mathcal{S} = \Pi_\mathcal{S} H \Pi_\mathcal{S}$ on the sampled configurations is found by sparse diagonalization, and both branches are evaluated on it. From 1000 to 15 000 shots Δ_0 falls from 40.6 to 24.5 (DMRG $m_0 = 55.0$); at 500 shots the subspace ground level is two-fold degenerate, and Δ_0 ranges over 40.1–44.5 within it. The relative error of the reconstruction's total weight $\bar{m}_0$ against DMRG falls from 0.89 to 0.55 (0.95 at 500 shots); since $\widehat{m}_0$ varies little, both Δ_0 and this error are dominated by the falling $\bar{m}_0$ and move together by construction. The estimate $\widehat{m}_0 = 46.5$–49.5 itself lies 10–15.5% below the DMRG m_0: the residual measures the leak, not the distance to the truth. Even at 15 000 shots the subspace energy $e_0 = -2.90\,t$ is far above the DMRG $E_0 = -7.66\,t$, so a large residual is expected. There $|\mathcal{S}| = 13\,235$ and $\varphi_\mathcal{S}$ lives on 4.4×10^5 configurations; both branches are evaluated at $k = 0$ and 1 ($\Delta_1 = 121$ against $\widehat{m}_1 = 393$), and $\widehat{m}_2$, which needs $(H - e_0)\varphi_\mathcal{S}$ on the configurations that H connects to that set, was not evaluated beyond exact diagonalization. The check shows that the code runs at this scale; it is not a measure of power and not a beyond-classical claim.

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